Let 1,000 Flowers Bloom (or Wilt): Causal Machine Learning Analysis of Heterogeneity in Charter School Effects *

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Abstract

More than 1,000 U.S. school districts nationally have had at least one charter school and prior research has found that their average market-level treatment effects, including participant, competitive, and matching effects, have been positive (though arguably small). The present study analyses the heterogeneity of these effects, using both staggered difference-in-differences (DD) and a novel method that combines the staggered DD with machine learning (causal forests or CF). We focus on the net or "market-level" effects of all the above market mechanisms and find considerable effect variation across states and districts. Student demographic measures explain more of this variation compared with other moderators related to market setting (e.g., teacher supply) or charter-related policies (e.g., school funding), at least when test scores are the dependent variable. The results are not robust, however, when we switch between test scores

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to high school graduation rates as the dependent variable, consistent with prior research finding that different educational inputs and policies drive different educational outcomes. These general conclusions hold for both the standard DD and DD-CF, informing the role of methodology in understanding effect heterogeneity.

Keywords: Charter schools; competition; high school graduation; student achievement
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1 Introduction

Effect heterogeneity is important for understanding optimal policy design, in part, because the results of social science research often vary considerably across studies and samples. In education research, in particular, we often see variation in treatment effects across units (e.g., some teachers and schools are much better than others) but less attention is paid to explaining this variation, which can be informative for policy design.

Charter schools are a case in point. Research consistently shows that charter schools have positive effects on the achievement of students who attend them (e.g., Sass, 2006, Booker, Gilpatric, et al., 2007, Hoxby and Rockoff, 2004, Abdulkadiroğlu, Angrist, et al., 2011, Angrist, Cohodes, et al., 2016) as well as perhaps high school graduation (Chen and Harris, 2023).[1] However, these charter school participant effects also vary across states (CREDO, Center for Research on Education Outcomes, 2023) and across cities (Angrist, Cohodes, et al., 2016; Harris and Larsen, 2023). The overall distribution of school-level charter effects is also wider than traditional public schools, i.e., a large share of charter schools perform worse than the lowest-performing traditional public schools—and, conversely, the best charter schools are better than the highest-performing traditional public schools (Baude et al., 2020).

To better understand the sources of this variation, We apply and extend prior methods for understanding effect heterogeneity. More than a thousand school districts, across 46 states, have had at least one charter school open in the last 20 years. If charter effects depend, for example, on what types of students they serve, then it might be possible to target charter schools to particular subgroups. Indeed, researchers have commonly found that the participant effects of charter schools on test scores are more positive in urban areas and for disadvantaged students. But what we attribute to student demographics could reflect differences in settings and policy that happen to be correlated with student disadvantage. Given the strong correlation between student race/income and student achievement levels, it may be that the apparent role of the former reflects a role for the latter. If state policy differences – regarding funding, charter accountability, regulations, and more – do explain variation in treatment effects then this would have direct implications for charter

[1] See Cohodes and Roy, 2024 for a review of lottery studies that include both test scores and high school graduation. Our conclusion here is very similar to their review. However, Cohodes and Roy, 2024 report four lottery studies that have examined effects on high school graduation. Three of these have negative point estimates (Angrist, Cohodes, et al., 2016; Cohodes and Feigenbaum, 2023; Angrist et al. (2023)) and one shows positive effects (Dobbie and Fryer, 2013). However, this is a selected sub-sample of over-subscribed schools, whereas Chen and Harris, 2023 find positive market-level effects when looking at the entire country.

policy design. While subgroup heterogeneity could be useful regardless of the reasons behind it,[2] it is still useful to identify those reasons and understand the underlying mechanisms.

We begin by following Chen and Harris, 2023 and apply various standard difference-in-differences (DD) methods using national data on most charter schools that have ever been opened in the United States. We also share a focus on "market-level" or net effects, which combine participant effects and various competitive and match effects. Using this approach, Chen and Harris, 2023 find that a 10-percentage point increase in charter market share increases student test scores by 0.004 standard deviations and increases high school graduation by 1-2 percentage points.

Given the positive, but small, average treatment effects, estimates combined with the growth and relevance of the charter school sector, it is important to understand if this average is masking a potentially wide distribution of effects which may range from strongly positive to strongly negative. Attempting to capture this distribution calls for novel non-parametric methods for estimating heterogeneous treatment effects. Firstly, we find evidence of significant treatment effect heterogeneity at the school district level for both achievement and graduation rates. For graduation rates, districts range in expected charter school impact from over 5pp for an additional 10pp charter school share to as low as -2.5pp for the same additional 10pp charter share, with the majority of districts being unaffected by additional charter schools. For math test scores, we find a normal distribution of charter school impacts ranging from about -0.1sd to about 0.1sd across district-grades.

Despite our very large sample and broad set of potential moderating variables, we still find that student demographic measures explain more of the variation in market-level charter effects than measures related to policy. (One reason is that the role of school funding is complex in this market-level analysis.) Student demographics also explain more than the market setting variables, including baseline outcome of traditional public school, availability of young, educated teachers, market size, and urbanicity.

The results are not robust, however, when we switch from test scores to high school graduation rates as the dependent variable. If we view these outcomes as equally important, this calls into question whether charter schools really are better for disadvantaged students in an overall sense. Others, too, have noted the discrepancies in charter results across outcomes.

[2]Targeting based on student subgroup may improve policy effects even if the underlying cause is some unobservable that is correlated with subgroup status.

Another important policy topic is whether there is some optimal charter market share. These shares range from 0 percent in most districts, to a national average of 7 percent, and to an extreme of essentially 100 percent in New Orleans and elsewhere. The number of high-charter-share districts is small, which makes this analysis somewhat speculative, but our findings provide little evidence of diminishing marginal returns to charter market share.

In addition to our contributions to understanding charter schooling, we advance understanding of the methods of causal inference for effect heterogeneity. Understanding variation in treatment effects poses greater empirical challenges than estimating average treatment effects. Each subgroup could have a different counterfactual and selection process. Moderator variables may be subject to measurement error, making treatment effects appear more similar across subgroups than they really are. When many potential dimensions of effect heterogeneity arise, two additional problems arise: (a) hypothesis testing is subject to multiple comparisons and fishing; and (b) subgroups are more likely to be in sparse cells where a plausible, and sufficiently sized, comparison group is difficult to come by. Conventional econometric methods also typically impose functional form assumptions that are convenient and minimize dimensionality but are often tenuous. Causal effect heterogeneity therefore inherits all the challenges of causal inference and then some.

To address these challenges, we drew the above conclusions using and extending the generalized random forest (GRF) method (Athey, Tibshirani, and Wager, 2019), an extension of the machine learning causal forest (CF) (Wager and Athey, 2018), and we will henceforth refer to these interchangeably. Briefly, CF takes a training sample of the data and splits observations according to the moderator variable that maximizes the sum of squared differences in outcomes between the treated and untreated units. This process repeats and identifies important dimensions of effect heterogeneity, based in part on how many splits are driven by a given moderator. To avoid overfitting, the results are obtained by splitting the data in the same way across the remaining, estimation sample. This general CF method has many advantages over conventional interaction terms; it is non-parametric, not susceptible to multiple comparisons, less susceptible to fishing, and can handle high-dimensional data (large k relative to n).

The CF method, however, was designed for randomized control trials and premised on the idea that the differences between treated and untreated units can be interpreted as "effects." A few studies have recently started marrying CF with difference-in-differences

(DD) (Gavrilova, Langørgen, and Zoutman, 2025 provides what they call the "Difference-in-Differences Causal Forest –DiDCF; Kattenberg, Scheer, and Thiel, 2023 provide what they call the CFFE – "Causal Forest Fixed Effects"). Briefly, these methods demean outcomes, in a way similar to two-way fixed effects, prior to the data splitting. To our knowledge, prior studies have not applied these methods to stacked DD settings, as we have do in our application to charter schools.

In both the DD and DD-CF methods, we identified the correlates market-level effect heterogeneity by using nationwide data from the National Longitudinal School Database (NLSD), which includes nearly all districts in the U.S. from school years 1995 to 2018. To capture the market-level effect, the dependent variable is a weighted average of outcomes of all traditional public schools (TPS) and charter schools located within each local market, i.e., the geographic school district. Prior research on the average market-level treatment effects suggests small positive effects on both achievement and high school graduation nationally (Chen and Harris, 2023). The authors' analyses pass various endogeneity tests (e.g., placebos, parallel trends, and changes in observable demographics), providing empirical support for the idea that we are capturing heterogeneity in causal effects.

Since the DD-CF methods are new, we test their practical import by comparing the above results to those from a DD using the usual linear interactions. While the signs on many of the variables change, both methods point in the same general direction—that demographics explain more than policy and that the few moderator variables associated with positive test score effects are not present, or even reverse signs, for graduation effects. This suggests that while the DD-CF has many advantages and imposes fewer assumptions, it does, at least in some cases yield similar conclusions to standard econometric techniques.

The paper proceeds as follows. Section II starts by describing the data and especially the market-level structure of the data and subsequent analysis. The end of this section reviews research on potential moderator variables that are used in the subsequent analysis. Section III introduces the discusses the methods and identification, for both the GDD and DD-CF methods. The results are shown in section IV. Section V concludes. Many of the details and proofs related to the DD-CF methods are in the appendix.

2 Data: Outcomes and Moderator Variables

2.1 Outcome Data and Structure

Most treatment effect studies, especially in the charter school arena, focus on participant effects, i.e., the effect of attending a charter school compared with counterfactual (usually nearby) TPS. This is less useful in the charter school arena, however, where the rationale for the policy relies not only on participant effects but a range of other market mechanisms whereby charter schools may affect non-participants, too. The entry of a charter school in a given location has been shown to have: *near competitive effects* on the closest public schools (Hoxby, 2003, Bettinger, 2005, Bifulco and Ladd, 2006, Sass, 2006, Zimmer and Buddin, 2009, Imberman, 2011, Linick, 2014, Ridley and Terrier, 2022, Griffith, 2019, Slungaard Mumma, 2022), *far competitive effects* on other traditional public schools located within geographic school district (Hess, 2004, Holley, Egalite, and Lueken, 2013; Figlio, Hart, and Karbownik, 2022),[3] *closure effects* on low-performing traditional public schools (Engberg et al., 2012, Bross, Harris, and Liu, 2023, Carlson and Lavertu, 2016), and *match effects* (Abdulkadiroğlu, Agarwal, and Pathak, 2017; Campos and Kearns, 2022). Given the evidence on all of these mechanisms, it is important to capture all of them – what we call the "market-level" effect – to understand charter school policy.

Our focus on market-level effects also shapes our data structure. The unit of analysis is the geographic district, which generally follows the boundaries of the local administrative school districts. This is for two reasons: first, distance affects the choice sets of families and therefore the geographic scope of competitive processes; second, most schools are traditional public schools (TPSs) that are governed and led by elected school boards whose authority resides only within the district's geographic boundaries.[4]

The shape files and other data for these geographic districts come from the National Longitudinal School Database (NLSD) and contain a near-census of all TPSs, charter schools, and private schools in the U.S. from 1991 to the most recent academic year.[5] The NLSD merges school and district-level data from Common Core of Data (CCD), Stanford Education Data Archive (SEDA), Census Small Area Income and Poverty Estimates (SAIPE), and other sources. We are most interested in the student outcomes, test scores and graduation

[3]There is very little evidence on far competitive effects. These citations are mainly case studies of individual districts. Also, the Figlio et al. citation refers to district responses to private schools.

[4]We recognize that the size of these districts varies regionally, e.g., the geographic districts in the U.S. South are much larger geographically than those in the North and some other states.

[5]All the school years mentioned in this paper are spring school years unless specifically stated otherwise.

rates, but also include a vector of district-level covariates in some models. The street addresses of charter school also allow us to place the location of each charter school within its geographic school district. These districts define the scope of the market, i.e., which schools are assumed to compete with one another.[6]

The specific dependent variables are the average freshmen graduation rate (AFGR) (hereafter, graduation rate) and student math and English Language Arts (ELA) test scores. The AFGR is defined by the U.S. Department of Education and is calculated based on uses aggregate student enrollment data to estimate the size of an incoming freshman class and aggregate counts of the number of diplomas awarded four years later.[7] For example, the AFGR for a school year in 2006 is the total number of diploma recipients in 2006 divided by the average enrollment of the 8th-grade class in 2002, the 9th-grade class in 2003, and the 10th-grade class in 2004. The graduation rate sample includes schools covering grades 9-12 annually from 1995 to 2010.[8]

The test scores are available in 3rd through 8th grade in math and English Language Arts (ELA) over the 2009-2018 school years from the Stanford Education Data Archive (SEDA). The SEDA provides nationally comparable, publicly available test score data for U.S. public school districts (Ho, 2020).[9] For interpretation purposes, we normalize the test scores to have means of zero and standard deviations of unity within the grade, year, and subject.

[6]The SEDA team uses the 2019 Elementary and Unified School District Boundaries https://nces.ed.gov/programs/edge/Geographic/DistrictBoundaries) to define the Geographic School District for test score sample (except for special education or virtual schools). We use the same boundaries to define the Geographic School District for the graduation sample.

[7]The AFGR was the standard federal metric in the years before student-level data systems were prevalent. It was not possible to see where students moved to when they exited district in these earlier years that are important to our analysis.

[8]We noticed some clear errors in the AFGR data, e.g., wild swings from year to year. In these cases, we imputed the errant observations using linear interpolation. (When the errant observation was in the first or last year of the panel, we imputed using the adjacent year.)

[9]To make estimates are comparable across states, grades, and years. The SEDA research team took the following steps: (1) estimate the location of each state's proficiency "thresholds" in the distribution of scores; (2) place the proficiency thresholds on the same scale using the National Assessment of Educational Progress (NAEP), a test taken by a representative sample of students in each state; (3) estimate the mean test scores in each school, district, county, metropolitan statistical area, commuting zone, and state from the raw data and the threshold estimates, and (4) create estimates of average scale scores and achievement growth measures. See details in SEDA website https://edopportunity.org/methods/. While this method is of course based on assumptions (e.g., that the NAEP sample is truly state representative and the distribution of state scores is the same as the NAEP distribution despite the differing content of these various tests), this could only result in bias if test error in charter-entering districts differs from never-charter districts (within grades and states and conditional on other included covariates) and changes over time in conjunction with charter school entry. We could not come up with potential examples where this might be the case.

Treatment status is defined as the charter enrollment share, or the percentage of students attending publicly funded schools who are enrolled in charter schools. These market share measures are created separately by grade level under the theory that charter schools only affect the geographic district outcomes in the specific grades being served (Jinnai, 2014, Slungaard Mumma, 2022).

For both outcomes, data are aggregated to the geographic district level by averaging TPS outcomes with those of charter schools located within their boundaries[10] (weighted by enrollment share[11]). This is crucial to the analysis as it allows us to capture charter effects operating through all the various market mechanisms. In contrast, prior studies of charter schools have focused on the outcomes of individual schools in the examination of school participant effects or the competitive effects of individual charter schools on nearby TPS. Here, we are interested in the total effect of all the mechanisms, which means we need to account for the outcomes of all traditional and charter schools. Aggregating outcomes to the geographic district level also allows us to sidestep the main problem with prior studies, i.e., selection bias from students moving across schools (within districts).

The test score (graduation) data are available for 11,399 (10,658) districts, including 1,597 (1,073) districts that have at least one charter school. The final samples for the test scores (graduation) analyses include 97 (98) percent of the nation's publicly funded schools and 99 (97) percent of nation's charter schools.[12] Table 1 presents the summary statistics for the outcome and control variables. Compared with TPSs, charter schools nationwide tend to enroll a larger proportion of African American and Hispanic students. Charter schools are also more likely to be in urban districts where achievement is relatively low.

We are identifying total charter effects leveraging the timing of charter entry. One key factor affecting this is the timing of the passage of charter laws that allow them to enter. The first law allowing the establishment of public charter schools was passed in Minnesota in 1991. As of fall 2022, charter school legislation had been passed in 45 states and the

[10]SEDA provides test score data at the geographic district level. We converted the graduation rate data to the geographic district level ourselves. Please refer to Appendix B for the description that how the sample was created.

[11]Weighting is necessary to obtain a nationally representative effect (e.g., to avoid counting New York City the same as small rural districts). However, as Solon, Haider, and Wooldridge (2015) note, weighting does not necessarily yield a consistent estimate of the population average treatment effect if the covariate variance depends on the weight. They therefore advise reporting results weighted and unweighted; and testing for effect heterogeneity by weight. We provide these additional results and discussion later as well.

[12]Since geographic districts are the unit of analysis, an observation is only missing if all the schools in the unit-by-school-type are missing. There are likely some missing schools within districts.

District of Columbia.[13] As a result of the growing number of states allowing charter schools, the percentage of publicly funded schools that are charter schools increased from 0 to 7.3 percent, and their percentage of enrollment increased from 0 to 6.2 percent.[14]

To highlight the variation in charter market share we are exploiting, Figure **XXXX** lists the trajectory of charter school growth from the point of the first charter entrant. We report separate lines based on the maximum charter (enrollment) market share that appears in our data. As expected, growth rates are faster in geographic districts that have higher eventual market shares and diminish over time.[15]

2.2 Moderator Variables

Pre-specification is important in effect heterogeneity analysis because there are usually large numbers of potential dimensions that can be studied and the odds of finding some statistically significant patterns by chance are high, i.e., the multiple comparisons problem (Athey and Imbens, 2019). It is also easy to "fish" for some statistically significant pattern. Selecting variables based on theory and prior research make this less likely as it directs analysis to dimensions where we have good reasons to expect meaningful patterns. In this section, we summarize theory and research that led us to focus on the moderators used later. This section also highlights some of the potential complexities in the interpretation of results regarding these moderators, setting the stage for the later discussion of results. We organize the moderator variables into four categories: student subgroups, market settings, charter sector characteristics, and charter-related policies.[16]

2.2.1 Student Subgroups

Race/ethnicity and parent income. These student background measures are easily the most widely studied moderators for two reasons: First, they are the most widely available in real-world data. Thus, the fact that charter school effectiveness is commonly thought to be related to student race/ethnicity and income partially reflects data availability. Second, student background is closely correlated with education outcome levels. It may be easier to raise student outcomes when they are already low (i.e., the "low hanging fruit" argument).

[13] The states in which public charter school legislation had not been passed by that time were Montana, Nebraska, North Dakota, South Dakota, and Vermont.

[14] Appendix Figure **A1** presents the trends in charter school share and charter enrollment share from spring 1991 to spring 2018.

[15] This non-linear pattern partially reflects the opening of the first charter school followed by enrollment growth in those schools, not just opening more schools.

[16] These variables were pre-specified in proposal written for this study and is available upon request.

One objective of this analysis is to try and disentangle these potential interpretations.[17]

Charter schools tend to locate in areas that are more diverse on income and race, perhaps because of heterogeneity of preferences (Glomm, Harris, and Lo, 2005) or a desire for same-race peers. While it is not obvious that this would affect test scores, it could affect high school graduation. Being closer to the preferences of students and parents might improve the utility of staying in high school relative to the alternatives like work.

These variables are captured in essentially all K-12 education data sets. The income measures are usually in the form of the percent of students eligible for free or reduced-price lunches. The underlying source for these variables in our analysis is the federal Common Core of Data (CCD).

2.2.2 Market Setting Moderators

Baseline school quality. As noted earlier, the common finding that charter schools are more effective for racial/ethnic minorities and low-income students might reflect the fact that their traditional public schools are low-performing, making improvement easier through charter schools. To test this, we use the first available baseline test scores and graduation rates (see discussion above on data sources).

Teacher supply. The importance of teachers is now widely recognized (Chetty, Friedman, and Rockoff, 2014). This is unsurprising given that education is a labor-intensive service and the vast majority of resources go to funding and supporting classroom teachers.

In the charter sector, the strongest evidence of effectiveness also comes from large cities that are attractive to young college graduates (e.g., Boston, Denver, and New Orleans). Also, charter school tend to rely on young, highly educated teachers, often from elite colleges and universities and/or teachers from outside the local area. Therefore, the variation in charter effectiveness might partially reflect the ability of charter operators to attract and retain productive teachers who fit their educational and organizational models.

Enrollment growth. We noted above how competitive responses may vary depending on funding formulas and authorizer type. District enrollment growth is another possibility. TPSs have more difficulty expanding when enrollment growth is high. While they are re-

[17] We could also describe this as a "diminishing returns" explanation but we use that term later to refer to the non-linearities charter market share effects.

quired to serve all students within their boundaries, they are less flexible, e.g., their buildings have to meet fairly specific standards and they are inclined to open in shopping malls the way that some charter schools do. Anecdotally, California school districts did not generally oppose the entry of charter schools in the 1990s and 2000s when enrollments were growing rapidly, but they have more recently as charter shares have grown and enrollments began declining. This might also imply that districts with more enrollment growth experience a smaller competitive response. In addition, authorizers might be less stringent in accepting applications from charter operators, out of desperation to provide additional capacity. Both factors suggest that we should expect smaller effects of charter entry in districts where enrollment growth is high (though the very fact of enrollment growth also leads to questions about changes in demographics that could bias our estimates).

However, our approach also has more direct implications for policy. We will learn whether various state policies are associated with improved charter market effects and whether new policies might be able to better target charter schools to those contexts where they do the most good. Our analysis also evaluates the performance of charter schools in particular states and of particular charter school authorizers. In the latter case, this is only possible with independent authorizers (in the states that have them) as the analysis requires isolating the effects of charter authorization and charter school improvement efforts from other district behaviors.

2.2.3 Charter Sector Characteristics

Charter school types. The above discussion pointed to two key differences in charter schools: whether they are managed by CMOs (versus standalone schools) and which type of charter authorizer they operate under. Based on prior evidence, we would expect that the entry of charter schools would be more positive when they are managed by CMOs.

We might also expect larger effects when the authorizer is not the district itself. School districts are unlikely to make a competitive response to schools under their own control. On the other hand, the participant effects of district-authorized schools are less well known and could potentially offset any reduced competitive response.

Diminishing marginal returns. We have written generally about the direction of charter school effects, which might imply a linear relationship between charter market share and q. However, there are reasons to expect larger effects with early charter entrants and diminishing effects and charter market share grows. First, if authorizers are picking from menu

of potential charter entrants, and they have some accurate signals of future quality then the initial charter schools may be of higher quality than subsequent entrants (Bross, Harris, and Liu, 2023).[18] Second, and similarly, the TPS competitive response seems more likely to arise with initial entrants as this is when competition first becomes an issue and is most salient. When the first charter school enters, this is likely to draw newspaper headlines and discussion about the emergence of charter schools; this type of reaction seems less likely with the second and third charter schools. (As explained later, this potential moderator requires its own analysis.)

2.2.4 State Policy Variables

Effect heterogeneity also depends on variation in the treatment itself (e.g., implementation). In the case of charter schools, this might depend on state policies that fund, restrict, and guide the work of charter operators.

Following Harris and McKenzie (2024), we also relied on assessments of state policies by charter advocacy organizations. While their views are undoubtedly biased, it is reasonable to think that charter advocates might know something about what will make their schools successful. We gathered state ratings of charter school policies published in 2014 by the Center for Education Reform (CER), the National Alliance for Public Charter School (NAPCS), and the National Association of Charter School Authorizers (NACSA). These rankings are useful for understanding regulation because these organizations are well-informed about charter laws and are likely to focus their rankings on policies that are genuinely important, at least to their charter constituents. While we do provide evidence about the relationship between each group's overall ranking and the various market outcomes, our main interest is in the information the rankings provide about the components of charter regulation. The proceeding sections provide a short description of each composite index and the index components that we included for our study.[19]

Center for Education Reform (CER). CER ranks states based on "laws that have a strong, permanent authorizing structures, equitable funding codified in law, and autonomy across state, district, and teacher rules and regulations, giving charters the freedom to do what

[18] A potential counterargument here is that, as the charter sector gains traction, more supporting structures may emerge to incubate better charter schools. This occurred in New Orleans for example, where a nonprofit began incubating charter schools once it became clear that the city was going to be opening more such schools.

[19] We selected the highest weighted components from each sub-index that are likely to be relevant for school performance, openings, and closures.

they do best." The organization's three most heavily weighted criteria are: the presence of independent (alternate) authorizers, number of schools allowed, and 100% charter funding (similar to the above "equitable funding"). The number of schools refers to enrollment caps and whether states are approving charter schools on a regular basis.[20]

National Alliance for Public Charter Schools (NAPCS). NAPCS evaluates state charter laws based on charter school laws that promote the creation of "high-quality charter schools while holding underperforming schools and authorizers accountable."[21] The five most heavily weighted components used for our study include state ratings by performance-based contract required, clear processes for renewal/nonrenewal, automatic exemption of charter schools from laws applied to traditional public schools (TPS), no caps on the growth of public charter schools, and equitable operational funding. The performance-based contract requirement component provides higher ratings to state laws that have requirements for contracts that provide academic performance expectations, operational performance expectations, and school and authorizer rights and duties. The process for renewal component examines whether there are "clear processes" for renewal, nonrenewal, and revocation decisions, including school closure and dissolution procedures. The no charter caps category determines whether there are caps on growth for public charter schools in a state. Equitable operational funding compares general operational, transportation, and other categorical funding of charter schools to TPS.

National Alliance for Public Charter Schools (NACSA). NACSA's policy ratings emphasizes policies that "facilitate the development of successful charter schools and enhance accountability for schools and authorizers alike."[22] The three most heavily weighted components include a rating for alternate authorizer, renewal standard, and default closure. The alternate authorizer ranks states based on whether they include authorizers other than school districts and whether the state can sanction poor authorizers. The renewal standard component ranks states by whether they require a strong standard that hold schools accountable for performance. A default closure policy standard means that there are minimum requirements for renewal and/or provisions for closure in extreme conditions prior to contract expiration.

Although all three industry group rankings are clearly interested in expanding charter schooling, they differ in important ways. First, NACSA, as an organization of charter autho-

[20] We note that this almost guarantees a relationship between this criterion and charter market share in the empirical analysis that follows.

[21] https://www.publiccharters.org/publications/model-law-supporting-high-quality-charter-public-schools

[22] http://www.qualitycharters.org/wp-content/uploads/2015/08/State-Policy-Analysis.pdf

rizers, is more focused on state policies that directly pertain to the authorizer roles, whereas NAPCS and CER rankings are based on a broader range of factors (e.g., funding). Second, NACSA is more focused on government accountability, while CER focuses on market accountability – "autonomy" and "freedom" for charter schools. (The NAPCS ranking is somewhere in between the two.) The NACSA approach is noteworthy given that the group represents the government-designated organizations – authorizers – who are responsible for holding schools accountable.

The participant effect is shaped by three main factors: the ability of authorizers to select high-quality charter schools from that choice set (Bross, Harris, and Liu, 2023), other supports that authorizers might provide to facilitate school improvement, and the resources available to charter schools (Harris and McKenzie, 2024). These, in turn, are shaped by state policies, which we consider to be exogenous in our analysis and are therefore labeled X_p^*.

Recall that there are three types of competitive responses. near effects, far effects, and closures. The competitive effects may be shaped by a variety of factors: local interest group politics, enrollment trends, school and district leadership, and parent responses.

State policy is obviously important here and that is no more true than with statewide charter authorizer policies. In most situations, school districts have little incentive to authorize charter schools because this creates competition with the district's existing TPS. It is therefore no surprise that the presence of an independent charter authorizer (i.e., an authorizer other than the school district itself) increases the number of charter schools that open (Harris and McKenzie, 2023). There is also some, more correlational evidence that independent authorizers tend to select charter applications that turn out to have higher value-added (Bross, Harris, and Liu, 2023; Harris and McKenzie, 2023). The presence of an independent authorizer might also change the way that districts authorize. For example, if a district believes that the state will authorize a given charter school, then the district might be more likely to recruit and/or authorize that same school itself, so that it can maintain control over it. This is an example of a far competitive response.

2.2.5 School Funding

Charter school funding deserves a separate discussion. Funding is one of the most widely studied elements in education policy and seems to affect student outcomes. We use a variety of data sources for this factor. Finally, the role of funding in the market-level effects of

charter schooling is theoretically complex.

If we were studying participant effects, then we would be most interested in the per-pupil funding charter schools receive and/or its relationship to the funding in nearby traditional public schools (the counterfactual condition) and we would hypothesize that the participant effects when charter funding is higher. But with market-level effects, we are combining participant effects and various competitive and matching effects. In the simplest case, with fixed (not weighted) per-pupil funding, higher charter funding would still increase the participant effect. It would also still increase the competitive effect, if schools are revenue maximizers.

This conclusion also holds for charter funding in more realistic scenarios. TPS revenue is usually partially buffered from enrollment losses to charter schools either because: (a) some TPS funding comes from local property taxation, irrespective of enrollment; or (b) state funding includes hold-harmless provisions. Even in the extreme, where TPS funding is completely independent of enrollment, increased charter funding would still increase the charter participant effects. There would be no competitive effects – TPS would have no revenue-based reason to respond—but the net effect of higher charter funding would still be positive.[23]

The above case would be costly because TPS revenue would remain constant even as charter funding grows with enrollment. This points to another possible funding moderator measure: average funding per pupil across TPS and charter schools. This measure also aligns with how we are constructing the outcome and other variables across sectors within local markets. But, with this measure, higher average funding means both that schools have more funding and that this funding is distributed less based on enrollment. As noted above, when TPS revenue is buffered, this drives weighted average spending upwards while reducing the competitive effect. This could paradoxically, create a negative null or even negative effect of higher average spending, especially if we have no way to control for the enrollment sensitivity of revenue.

Finally, we note that the above funding measures are subject to considerable measurement error. The above measures require accurate measurement of school funding at the school level (in the case of charter schools) and accurate accounting of how funding "passes through" school districts to charter schools, neither of which is the case in U.S. education data. This attenuates the estimated moderator role. Taken together, the above points suggest that we

[23]We are ignoring costs here. The more detached TPS funding is from revenue, the higher would be the per-pupil costs of charter schools.

should still expect charter effects to be larger when charter spending is higher, but perhaps less than one might expect.

3 Identification Strategies

We use three complementary methods to study effect heterogeneity in charter school markets. The first is the generalized difference-in-differences (GDD), which permits linear interaction heterogeneity within a fixed-effects regression. The second is the DiDCF of Gavrilova, Langørgen, and Zoutman (2025), extended here to settings with staggered treatment adoption and continuous treatment dosage by incorporating the insights of Callaway, Goodman-Bacon, and Sant'Anna (2025). The third is the causal forest with fixed effects (CFFE) of Kattenberg, Scheer, and Thiel (2023).

3.1 Notation and Definitions

In this section, we define our treament effect parameters of interest and discuss how they are identified under the GDD and Causal Forest approaches, respectively. We observe school districts (or district-grades), indexed with $i \in \{1, ..., N\}$, over time periods $t \in \{1, ..., T\}$. Our outcomes of interest, $Y_{i,t} \in \mathbb{R}$, include average high school freshman graduation rate (AFGR), standardized elementary math test scores (Math), and standardized elementary English/language arts scores (ELA). In each period, districts are observed with some continuous charter intensity $W_{i,t} \in [0, 1]$, and a set of time-invariant observable characteristics $X_i \in \mathcal{X}$.

Let $Y_{i,t}(w)$ indicate the potential outcome for district i in time period t under charter intensity w. In each time period, the observed outcome for district i is $Y_{i,t} = Y_{i,t}(W_i)$. Our treatment effect of interest is what Callaway, Goodman-Bacon, and Sant'Anna (2025) refer to as the "level treatment effect," which is defined as the difference in potential outcomes for district i if that district experienced dose w vs. if that same district had experienced zero dose in period t: $Y_{i,t}(w) - Y_{i,t}(0)$. They also define a helpful "building block" parameter:

$$ATT(w|w') = E[Y_{i,t}(w) - Y_{i,t}(0)|W = w'] \tag{1}$$

Where $ATT(w|w')$ is the average effect of receiving dose w relative to zero dose for units that actually experienced dose w'. We focus on the situation where $w = w'$ so that our target parameter is the ATT local to units that actually received dose w: $ATT(w|w)$. In our setting, charter adoption is staggered over time. Each district therefore belongs to a cohort

of districts which are first treated with a positive charter intensity in the same calendar year (or the group of districts who are never treated). We denote these timing cohorts by $G_i \in \mathcal{G}$, where $G = \infty$ for never-treated districts by convention. Also by convention, we exclude always-treated districts so that $\mathcal{G} \subseteq \{2, ..., T, \infty\}$, and $\bar{\mathcal{G}} = G \backslash \{\infty\}$ is the set of timing cohorts among ever-treated districts.

In order to account for staggered adoption of treatment, we extend the potential outcomes notation above. Denote $Y_{i,t}(g, w)$ as the potential outcome for district i in time period t subject to this district being first treated in cohort g under charter intensity w. We write $Y_{i,t}(0) = Y_{i,t}(\infty, 0)$ as the untreated potential outcome for district i. We also define the base period for cohort g as the time period before first treatment for units in cohort g, which we denote as b_g. We set $b_g = g - 1$ as the last pre-treatment period for cohort g, so that event time $h = t - b_g = 1$ corresponds to the first calendar year of treatment. Recall that each district has an associated set of observable time-invariant covariates, $X_i \in \mathcal{X}$. Some of these covariates may be measured confounders of the relationship between $W_{i,t}$ and $Y_{i,t}$, and some may not confound but may nonetheless be related to heterogeneity. Going forward, we adopt the following assumptions:

Assumption 1 (Random Sampling)**.** The observed data is drawn from an i.i.d. distribution and consists of $\{Y_{i,1}, ..., Y_{i,T}, W_{i,t}, G_i, X_i\}_{i=1}^n$.

Assumption 2 (Overlap)**.** The conditional density of treatment is bounded away from zero uniformly: there exists $\varepsilon > 0$ such that $f(W_{i,t} \mid X_i = x) \geq \varepsilon$ for all $x \in \text{supp}(X_i)$ and all $W_{i,t}$ in the interior of $[0, 1]$. This rules out regions of the covariate space in which all districts receive the same charter intensity, ensuring the causal forest has non-degenerate treatment variation at every x.

Assumption 3 (No Anticipation)**.** For all $g \in \mathcal{G}$ and $t = 1, ..., T$ with $t \leq b_g$, $Y_{i,t}(g, w) = Y_{i,t}(0)$

Assumption 4 (Conditional Parallel Trends)**.** For all $g \in \bar{\mathcal{G}}; t = 2, ..., T; w \in [0, 1]$; it must be true that $E[Y_{i,t}(0) - Y_{i,b_g}(0) | G_i = g, W_{i,t} = w, X_i = x] = E[Y_{i,t}(0) - Y_{i,b_g}(0) | G_i = \infty, W_{i,t} = 0, X_i = x]$. *In other words, it must be true that the expected counterfactual change in outcomes between period t and b_g for districts in cohort g who experienced dose w had they not been treated should be equal to the observed change in outcomes from never-treated districts, conditional on X.*

Assumptions 1 – 4 are an extension of Assumptions 1-MP, 2-MP(a), 3-MP, and PT-MP from Callaway, Goodman-Bacon, and Sant'Anna (2025) to include covariates. Only the

subset of X_i variables which are confounders are required in order to satisfy Assumption 4. The non-confounding variables in X_i are not required to obtain consistent estimates of causal parameters, but are included to further explore heterogeneity and because the set of confounders is unknown. The causal parameter of interest is the conditional average treatment effect on the treated (in levels), which we refer to as the Level-CATT (or just the CATT, because it is the only causal parameter we seek to estimate in this setting). The Level-CATT is defined for district i as the expected impact of receiving charter intensity w relative to zero charter intensity, subject to district i being in cohort g with covariates x:

$$CATT(g,t,w,x \mid g,w) \equiv \mathbb{E}[Y_{i,t}(g,w) - Y_{i,t}(0) \mid G_i = g,\ W_i = w,\ X_i = x] \tag{2}$$

Our first result is that under Assumptions 1 – 4, the Level-CATT is identified via a difference-in-differences estimator:

Lemma 1. *Under Assumptions 1 – 4:*

$$\begin{aligned} CATT(g,t,w,x \mid g,w) = \mathbb{E}[Y_{i,t} - Y_{i,b_g} \mid G_i = g,\ W_{i,t} = w,\ X_i = x] - \\ \mathbb{E}[Y_{i,t} - Y_{i,b_g} \mid G_i = \infty,\ W_{i,t} = 0,\ X_i = x] \end{aligned} \tag{3}$$

For proof, see Appendix A.

3.2 Generalized Difference-in-Differences (GDD) Identification and Estimation

The GDD identifies the CATT from within-district variation in dosage over time (for those ever treated) as well as differences between treated and untreated units (Mouganie, Ajeeb, and Hoekstra, 2020; Dow et al., 2020). This is preferable to a simple DD, in cases such as this with continuous, time-varying treatment. A simple DD would ignore most of the variation in charter market share and focus only on the point where the first charter school enters and treatment begins.

The dependent variables in our GDD analysis are test scores ($Test_{igt}$) and high school graduation (GR_{it}) in district i and grade g during year t (test scores are grade-specific; graduation is not). We include geographic unit fixed effects μ_i (usually geographic school districts) as well as state-grade-year fixed effects λ_{sgt} and district-specific linear trends $Year_i$. This implies the following models:

$$GR_{it} = \alpha + \beta_1(ChartShare_{it} \cdot H_{is}) + X_{it}\gamma + \mu_i + \lambda_{st} + Year_i + \varepsilon_{it} \quad (4)$$

$$Test_{igt} = \alpha + \beta_2(ChartShare_{igt} \cdot H_{is}) + X_{it}\gamma + \mu_i + \lambda_{sgt} + Year_i + \varepsilon_{igt} \quad (5)$$

Treatment is the continuous, time-varying charter market share, $ChartShare_{it}$. This is interacted with effect heterogeneity dimension H_{is}. None of these dimensions are time-varying. For this reason, and because of the district fixed effect, we do not enter H_{is} as a separate variable. The coefficients of interest are represented by $\boldsymbol{\beta}$.

One potential challenge with so many different dimensions of effect heterogeneity (see the theory in Section II) is that the above factors may be correlated with each other. For example, state policies may be correlated with household demographics. Also, states with more accountability might also have distinctive funding formulas. We are able to overcome this challenge because the richness of our data allows us to include essentially all of these factors at the same time to isolate the role of each.

The term X_{it} is a vector of time-varying student covariates (log enrollment, share of black, white, Hispanic, FRL, and special education students) and other district-level covariates (district revenue per student, district expenditure per student, student-teacher ratio, teacher salary, number of magnet schools, and an indicator of whether a charter law has gone into effect). As these covariates are potentially endogenous, they are only included in certain specifications and as partial tests for charter effects on the student population.

The error term ε_{igt} is clustered at the district level in the main specification. (We report results clustered at the state level in the Appendix, which only slightly reduces precision.) The estimates are weighted by the district size, so that the estimates are nationally representative; specifically, we weight by graduation rate denominator in the high school sample and by grade-level enrollment for math and ELA. In our baseline analyses, we estimate the GDD model for both the current year charter enrollment share and with a one-year lag to allow delayed effects.

Our method accounts for time-invariant unobserved differences across districts and our main identifying assumption is that there are no unobserved time-varying factors that are correlated with changes in charter market share and changes in student outcomes. We note that, even if charter market share changed based on time-invariant unobserved district characteristics that affected student outcomes, this would not introduce bias if the timing of

charter entry were conditionally random, which is possible given all the steps involved in opening a charter school (e.g., putting in an application, getting it approved, gaining access to a building, recruiting students, and hiring staff).

Still, it is worth considering specific scenarios under which this might be violated. First, charter schools are more likely to locate in districts with low contemporaneous student outcomes (Glomm, Harris, and Lo, 2005), signaling that we also might expect charter schools to also locate where *expected* future TPS performance is lower, though this is unobservable in our analysis.[24] This would lead to a downward bias in our estimates because TPS outcomes comprise the majority of the weighted average outcomes.

Alternatively, local governments might change non-charter education policies at the same time they introduce charter schools, as part of a package of reforms. Since school districts are often charter authorizers, it may be that a change in school board politics leads both to an increase in charter schools and, for example, new reading and math curricula in TPS. In this case, we might falsely attribute outcome changes to charter schools that were actually caused by other policy changes. We discuss various tests for these and related types of violations in the next section, after describing our main results.

3.3 Causal Forest Identification and Estimation

The GDD identifies the CATT within a linear parametric framework. However, there exists a large literature which considers partially linear models of conditional average treatment effects (e.g. Robinson (1988)), which can be written generally as:

$$Y_i = \alpha(X) + \tau(X)W_i + \varepsilon_i \tag{6}$$

Where the coefficient on W_i, $\tau(X)$, is an unknown function of X. Under a conditional independence condition (i.e. $\mathbb{E}[W_i\varepsilon_i|X_i] = 0$), $\tau(X)$ identifies the conditional average treatment effect (CATE)[25]. This translates simply into a generalization of the GDD framework under

[24] Relatedly, charter schools might open where districts experience idiosyncratic shocks, in which case future outcomes regress toward the mean. If charter schools prefer to locate near low-performing schools, as noted above, then this would yield an upward bias (i.e., charter schools enter because of the negative shock in existing schools, but then those outcomes bounce back in the next period). We do not see this scenario as very likely because it takes several years to create an organization that can assemble a charter application, submit the application and gain approval, hire personnel, purchase necessary capital, and recruit students. Also, with such a long-term investment, charter organizations are likely to consider multiple years of information in making their entry decisions, which reduces the chance of regression to the mean.

[25] Abadie (2003) takes this further and shows that even if W_i is endogenous, if one obtains a valid instrument then $\tau(X)$ identifies the conditional *local* average treatment effect (CLATE) within the same partially linear

our Assumptions 1 – 4, where $\tau(X)$ now identifies the Level-CATT[26]:

$$Y_{it} = \mu_i + \lambda_{st} + \tau(X_i)W_{it} + \varepsilon_{it} \tag{7}$$

It is worth noting that Equation 7 is not a structural model of potential outcomes, as interpreting it as such would require X to contain all moderators that contribute to heterogeneity (Frölich, 2007), some of which may be unobservable in practice. Nonetheless, extrapolating the result of Abadie (2003) allows us to conduct inference on the CATT without additional assumptions beyond Assumptions 1 – 4.

In order to conduct inference on the CATT, our task is to non-parametrically estimate $\tau(X_i)$ in Equation 7. We begin by briefly reviewing CATE estimation using causal forests in order to build intuition for the extensions we employ, and refer the reader to other studies which provide more detailed expositions of CATE estimation using casual forests (e.g. Knaus, Lechner, and Strittmatter (2021); Caron, Baio, and Manolopoulou (2007); Strittmatter (2023)). Next, we discuss the two methods we employ to operationalize the causal forest in DiD scenarios: the DiDCF of Gavrilova, Langørgen, and Zoutman (2025) and the CFFE of Kattenberg, Scheer, and Thiel (2023).

3.3.1 Causal Forest Overview

To discuss the basics of causal forest estimation, consider the case of a completely randomized experiment. In this case, we seek to estimate the CATE for a set of pre-specified covariates as in $\tau(X)$ in Equation 6. A *causal forest* (Wager and Athey, 2018) is built from many *causal trees* (Athey and Imbens, 2016). Causal trees group observations by recursively splitting the data into finer and finer subgroups. With each split, the causal tree aims to partition units such that the ATEs within each partition are as different as possible. Splits occur at "nodes," and the resulting partitions of the data which exist after the initial split are called "children nodes." With continuously distributed covariates, splitting can in principle continue until each unit is contained in its own partition. To prevent this level of overfitting, hyperparameters are used which define stopping rules such that a tree knows when to cease splitting along a branch. These may include (1) a minimum number of treatment/control

model without any further assumptions beyond those of a standard IV estimator.

[26]Traditionally, parallel trend assumptions like Assumption 4 only assume that counterfactual trends of *treated* units are parallel to the observed trends of control units. PTA's do not assume that the counterfactual trends of control units had they been treated are parallel to the observed outcomes of units that were actually treated. This is why we do not attempt to identify the CATE: because doing so would require a stronger PTA that stipulates the observed outcomes of treated districts are representative of the counterfactual outcomes of control districts.

units in a child node, (2) a minimum share of a parent node that must be contained in a child node, (3) imbalance penalty parameters, etc. The final partition along a branch is called a "leaf" or "terminal node," $L(x)$, and results in a small set of units defined by a similar vector of covariate values, x. Within any given leaf, the CATE is then estimated as the difference in outcomes between treated vs. untreated units in that leaf. Causal forests then provide an estimate of each unit's individualized CATE by averaging estimates across causal trees.[27] The purpose of growing many trees is to improve accuracy, as singular trees do not produce smooth estimates of CATE functions (Wager and Athey, 2018).

Athey, Tibshirani, and Wager (2019) extend the causal forest approach such that it can be applied to more general settings of causal effect estimation such as observational studies with non-randomized treatment, which they refer to as *generalized random forests* (GRF). They do this by re-framing the causal forest algorithm such that it no longer views forests as an average of causal trees, but rather as an adaptive nearest-neighbor estimator of any causal parameter that can be identified with a local moment condition. In the case of Equation 6, $\tau(X)$ is identified via the (conditional) local moment condition corresponding to conditional independence of W_i and Y_i:

$$\begin{aligned} & \mathbb{E}[W_i \varepsilon_i | X_i] = 0 \quad \forall X_i \in \mathcal{X} \\ \implies & \mathbb{E}[W_i (Y_i - \alpha(X) - W_i \tau(X_i)) | X_i] = 0 \quad \forall X_i \in \mathcal{X} \end{aligned} \tag{8}$$

Based on Equation 8, the GRF estimates of $(\hat{\alpha}(X), \hat{\tau}(X))$ are given by:

$$\big(\hat{\alpha}(X), \hat{\tau}(X)\big) \in \arg\min_{\alpha,\tau} \left\{ \Big|\Big| \sum_{i=1}^{n} \omega_i(x) [W_i (Y_i - \alpha(X) - W_i \tau(X_i))] \Big|\Big|_2 \right\} \tag{9}$$

Where $\omega_i(x)$ is the GRF-defined similarity weight which captures the relevance of the i-th observation for estimating $\hat{\alpha}(X)$ and $\hat{\tau}(X)$; which is how often observation i appears in the same terminal leaf as x across all causal trees. It is straightforward to see that Equation 9 is simply a GRF-weighted OLS estimator. The task is therefore to derive the similarity weights such that the resulting CATE estimates from Equation 9 are consistent and asymptotically normal. We will outline the GRF algorithmic approach to growing causal trees that accomplishes this.

There are two important concepts to note before explaining the GRF-training algorithm.

[27] All units will fall into some leaf within each tree, and thus will receive a treatment effect estimate from each tree.

First, GRF performs Neyman orthogonalization (i.e. local recentering) (Robinson, 1988; Chernozhukov et al., 2018; Nie and Wager, 2021) by differencing out the propensity score and conditional mean from W_i and Y_i, respectively. These propensity score functions are estimated via auxiliary regression trees with leave-one-out estimates. The purpose of this is to improve finite sample performance of the model, as GRF estimates can be biased in finite samples if the conditioning moderators in X are not fully partialled out because of the small sample size. Athey, Tibshirani, and Wager (2019) show that this reduces finite sample bias in these situations. Second, trees are estimated based on the concept of "honesty" developed by Athey and Imbens (2016). The consistency and asymptotic normality of non-parametric estimators relies on cross-training: using a different sample for model estimation than the one used for model training. The reason is that using the same sample for training and estimation introduces dependence between the structure of the model and the estimates themselves. Therefore, an honest causal tree takes the full data sample and extracts a random sub-sample to grow the tree, then further divides this sample into a training sample and an estimation sample. The training sample is used to perform splitting and define the tree structure, the estimation sample is used to populate the leaves and provide CATE estimates. Appendix B summarizes the process for growing trees and obtaining the GRF-weights.

3.3.2 DiDCF Overview

The above description of the GRF algorithm is designed for settings where treatment is credibly conditionally random, as in the case of a randomized experiment. Our study is quasi-experimental and applies difference-in-differences, which relies on the conditional parallel trends assumption stated above as Assumption 4. We therefore require an operationalization of the GRF algorithm that identifies that CATT under parallel trends rather than conditional independence. One such approach is the Difference-in-Differences Causal Forest (DiDCF) of Gavrilova, Langørgen, and Zoutman (2025). In this subsection, we will provide a brief overview of the DiDCF, which is designed for the most straightforward types of event-study: cases with a single treatment group which is affected by a binary treatment. The next section will discuss our extension of this approach to settings of staggered treatment adoption by multiple treatment groups under continuous treatment.

The target estimand of the standard DiDCF is a simpler version of the Level-CATT in Definition 2, which assumes only that all treated units are treated in the same period (and

remain treated indefinitely) and that treatment is binary:

$$\tau_t(x) = CATT(t, x \mid w) \equiv \mathbb{E}[Y_{i,t}(1) \mid W_i = 1,\ X_i = x] - \mathbb{E}[Y_{i,t}(0) \mid W_i = 1,\ X_i = x] \quad (10)$$

The DiDCF estimator proceeds in the following sequence:

Standard DiDCF Algorithm:

1. Transform the outcome variable by differencing it relative to a base year:

$$Z_{i,t} = Y_{i,t} - Y_{i,b}$$

   Conducting analysis on this transformed variable ensures that any forests trained on it are DiD-based estimators. Gavrilova, Langørgen, and Zoutman (2025) show that under a similar set of assumptions to our Assumptions 1 – 4 (and some standard regularity assumptions), that a DiD-based estimator identifies the CATT from Equation 10.

2. For each period $t \neq b$, estimate a separate (standard) causal forest (GRF) in the manner of Athey, Tibshirani, and Wager (2019). This produces an estimate of the CATT function in each period, $\hat{\tau}_t(x)$. Intuitively, this approach is equivalent to training a single causal forest on the full data, but forcing the first split to be on time period. This is necessary because not doing so would lead to cross-period contamination within leaves (i.e. units from different time periods would fall in the same leaf). If the treatment effect is not uniform across time periods, heterogeneity bias in the same spirit as that highlighted by Sun and Abraham (2021a) in the case of staggered adoption may enter the estimates. It is also important to note that this approach allows estimation of pre-treatment placebo tests of parallel trends by estimating forests in periods $t < b$, where the outcomes of not-yet-treated units are compared to the outcomes of never-treated units.

3. Predict the treatment efect for unit i in period t among only treated units, using trees not trained on unit i (the $(-i)$ notation):

$$\hat{\tau}_{i,t} = \hat{\tau}_t^{-i}(X_i) \quad \forall i \in W_i = 1$$

3.3.3 Extending DiDCF to Staggered and Continuous Treatment

Gavrilova, Langørgen, and Zoutman (2025) show that their estimator is consistent and asymptotically normal under Assumptions[28] 1 – 4. Here, we outline an approach to extend the DiDCF to a staggered adoption setting with continuous treatment.

Recall from Section 3.1 that $\mathcal{G} \subseteq \{2, ..., T, \infty\}$ is the set of cohorts of initial treatment we consider (after dropping always-treated units), and $\bar{\mathcal{G}} = G \backslash \{\infty\}$ is the set of ever-treated cohorts. Units in the same cohort have the same base period, which is the period before initial treatment, denoted b_g. Denote $h = t - b_g$ as the event-time. The primary adjustment to the DiDCF algorithm is that we first split the data into G separate sub-samples, where each subsample contains only units from cohort $g \in \bar{\mathcal{G}}$ or never-treated units (cohort $g = \infty$). A standard DiDCF is then estimated within each subsample, which yields estimates of cohort $\times$ event-time CATTs, $\hat{\tau}_{g,h}(x)$, with corresponding cohort $\times$ event-time variances, $\text{var}(\hat{\tau}_{g,h}(x))$. If Assumptions 1 – 3 in Gavrilova, Langørgen, and Zoutman (2025) hold for each cohort (i.e. parallel trends, non-extreme propensity scores, and regularity assumptions), these CATT estimates will be consistent.

It is natural to be interested in aggregating these cohort $\times$ event-time CATT estimates to the event-time level. This can be accomplished by taking a weighted average across cohorts (Callaway and Sant'Anna, 2021), where the weight on cohort g, $\phi_g = n_g / \sum_{g' \in \bar{\mathcal{G}}} n_{g'}$, is the relative frequency of treated units within each cohort:

$$\hat{\tau}_h(x) = \sum_{g=2}^{T} \phi_g \hat{\tau}_{g,h}(x) \tag{11}$$

Employing the Delta method, the variance of each event-time CATT can also be aggregated if one assumes independence of CATT estimates across cohorts:

$$\text{var}(\hat{\tau}_h(x)) = \sum_{g=2}^{T} \phi_g^2 \text{var}\big(\hat{\tau}_{g,h}(x)\big) \tag{12}$$

In our setting, we are interested in further extending the DiDCF to allow for continuous treatment dosage to estimate the Level-CATT established in Equation 2. To do this, we follow the insights of Callaway, Goodman-Bacon, and Sant'Anna (2025), particularly their

[28] Their version of the parallel trends assumption is less strict than our Assumption 4. Satisfaction of our Assumption 4 implies satisfaction of their parallel trends assumption. Nonetheless, our Assumption 4 is necessary to identify the CATT corresponding to Equation 2.

Theorem C.1. Before we establish that our staggered-adoption DiDCF design produces consistent estimates of the Level-CATT, we formally introduce the regularity assumptions needed to invoke the proof of Theorem 1 in Gavrilova, Langørgen, and Zoutman (2025):

Assumption 5 (Regularity)**.** For each cohort $g \in \bar{\mathcal{G}}$ and event time $h \geq 1$, within the cohort-g subsample $\mathcal{S}_g = \{i : G_i = g\} \cup \{i : G_i = \infty\}$, the covariates X_i are continuously distributed in the unit hypercube $(0,1)^d$, and the differenced outcome $Z_{i,g,h} = Y_{i,b_g+h} - Y_{i,b_g}$ satisfies the following conditions. (a) *Joint smoothness:* the conditional moments $\mathbb{E}[Z_{i,g,h}^k \mid X_i = x, W_{i,t} = w]$ are Lipschitz continuous jointly in (x, w) for $k = 1, 2$. (b) *Non-degenerate conditional variance:* $\text{var}[Z_{i,g,h} \mid X_i] > 0$ uniformly over $\text{supp}(X_i)$. (c) *Bounded moments:* higher-order moments of $Z_{i,g,h}$ are bounded. (d) *Independence:* observations are independent across districts. Conditions (b)–(d) correspond to Assumption 3 of Gavrilova, Langørgen, and Zoutman (2025); condition (a) strengthens their Lipschitz requirement from X_i alone to $(X_i, W_{i,t})$ jointly, as required for consistency of the continuous-treatment causal forest (Nie and Wager, 2021; Athey, Tibshirani, and Wager, 2019).

Lemma 2 (Consistency of Staggered DiDCF for Level-CATT)**.** Under Assumptions 1–5, for each cohort $g \in \bar{\mathcal{G}}$ and event time $h \geq 1$, the cohort-specific DiDCF estimator $\hat{\tau}_{g,h}(x)$ obtained by applying Algorithm 1 of Gavrilova, Langørgen, and Zoutman (2025) to the cohort-g subsample $\mathcal{S}_g = \{i : G_i = g\} \cup \{i : G_i = \infty\}$ with differenced outcome $Z_{i,g,h} \equiv Y_{i,b_g+h} - Y_{i,b_g}$ and continuous treatment $W_{i,t}$ at $t = b_g + h$, and covariates X_i, is consistent and asymptotically normal for the Level-CATT:

$$\frac{\hat{\tau}_{g,h}(x) - CATT(g,h,w,x \mid g,w)}{\sqrt{\widehat{\text{var}}(\hat{\tau}_{g,h}(x))}} \xrightarrow{d} \mathcal{N}(0,1) \quad \text{as } N \to \infty. \tag{13}$$

Consequently, the aggregated event-time estimator

$$\hat{\tau}_h(x) = \sum_{g \in \bar{\mathcal{G}}} \phi_g \, \hat{\tau}_{g,h}(x), \qquad \phi_g = \frac{n_g}{\sum_{g' \in \bar{\mathcal{G}}} n_{g'}}, \tag{14}$$

is consistent for $\tau_h(x) \equiv \sum_{g \in \bar{\mathcal{G}}} \phi_g \, CATT(g,h,w,x \mid g,w)$ as $N \to \infty$. The Delta-method variance in (12) is valid conditionally on the never-treated subsample size $n_\infty \to \infty$; see Remark 1 in the corresponding proof .

For proof, see Appendix A.

We note some implementation details regarding the estimation of staggered-adoption DiDCFs. For computational tractability (there are 210 causal forests estimated for AFGR, and 100 each for Math scores and ELA), we grow each forest with 1,000 trees. For each tree,

we hold out half of the subsample for honest estimation, and require at least 20 observations in a resulting child node for a split to occur. We use the default rule-of-thumb for determining the number of variables considered at each split point: if k is the number of covariates, select the minimum of $\sqrt{k} + 20$ or k. When training each tree, units are weighted by enrollment share for population representativeness, and we perform clustering at the district level (for AFGR) and district-grade level (for test scores). The variance of $\hat{\tau}(X)$ is constructed using the bootstrap of little bags, which is accomplished by training sets of trees and comparing CATT estimates within and across those sets (Athey, Tibshirani, and Wager, 2019).

3.3.4 CFFE Overview

In order to supplement the DiDCF operationalization of the causal forest in difference-in-differences settings, we also explore heterogeneity with a second method: the "Causal Forest Fixed Effects" (CFFE) of Kattenberg, Scheer, and Thiel (2023). The CFFE is perhaps a more natural extension of the GDD framework, as it relies on fixed effects to satisfy conditional parallel trends assumptions. In this subsection, we will provide a brief primer on the CFFE algorithm, assumptions, and identification conditions.

Let us consider the partially linear GDD model specified in Equation 7. To build intuition, let us first assume that there are two subgroups of units, $S = \{0, 1\}$, with the same treatment effect, $\tau(X_i) = \tau^s \quad \forall i \in s$. If the subgroups were known ex-ante, then Equation 7 could be estimated by simply including an indicator for a unit's group membership:

$$Y_{i,t} = \mu_i + \lambda_{s,t} + \tau^0 W_{i,t} \times \mathbb{1}\{i \in s = 0\} + \tau^1 W_{i,t} \times \mathbb{1}\{i \in s = 1\} + \varepsilon_{i,t} \tag{15}$$

An alternative, but numerically equivalent method for estimating Equation 15 is to perform orthogonalization via a double within-transformation (Wooldridge, 2010). A standard OLS regression on orthogonalized variables will produce equivalent estimates of τ^s in lieu of including fixed effects:

$$\widetilde{Y}_{i,t} = \tau^0 \widetilde{W}_{i,t} \times \mathbb{1}\{i \in s = 0\} + \tau^1 \widetilde{W}_{i,t} \times \mathbb{1}\{i \in s = 1\} + \varepsilon_{i,t} \tag{16}$$

Where $\widetilde{Y}_{i,t} = Y_{i,t} - E[Y_{i,t}|\mu_i, \lambda_{s,t}]$ and $\widetilde{W}_{i,t} = W_{i,t} - E[W_{i,t}|\mu_i, \lambda_{s,t}]$.

Kattenberg, Scheer, and Thiel (2023) propose a conditional (staggered) parallel trends assumption akin to our Assumption 4, except their comparison group is not-yet-treated units

rather than never-treated units[29]:

$$\mathbb{E}[Y_{i,t}(\infty) - Y_{i,b_g}(\infty)|G_i = g, X_i = x] = \mathbb{E}[Y_{i,t}(\infty) - Y_{i,b_g}(\infty)|G_i = g', X_i = x], \\ \forall t = 2, ..., T; \quad \forall g \in \bar{\mathcal{G}}; \quad \forall X_i \in \mathcal{X} \tag{17}$$

In a setting where Equation 17 credibly holds with the inclusion of fixed effects, μ_i and λ_{st}, in X_i, one could in principle estimate Equation 7 using the estimators of Sun and Abraham (2021a) or Borusyak, Jaravel, and Spiess (2024) by specifying subgroup membership manually. However, Equation 7 allows for subgroup membership to be an infinitely continuous function of X_i and is not assumed to be known ex-ante. Where causal forests would normally provide a method to estimate $\tau(X_i)$ from the observable covariates in the data, in this case standard causal forests fail to identify $\tau(X_i)$. The reason is because the inclusion of fixed effects in the covariate matrix introduces dependence across observations from two sources. First, all time-invariant observable covariates are absorbed in the unit fixed effects. Recall that standard causal forests use auxiliary regression trees to estimate propensity scores: $E[W_{it}|\mu_i, \lambda_{st}, X_i]$. The inclusion of fixed effects will concentrate all propensity scores around 0 and 1, which violates Assumption 2. Second, even if one residualizes globally according to Equation 16, the problem remains. Globally within-transforming over units removes the average fixed effect across all observations, but the composition of units and time periods may differ within individual tree nodes.

In order to retain fixed effect-based parallel trends while adhering to causal forest identification assumptions, Kattenberg, Scheer, and Thiel (2023) propose node-level orthogonalization. Intuitively, this is equivalent to including fixed effect dummies in the "treatment indicator" matrix and estimating a standard causal forest. Their algorithm for training causal trees can be summarized as follows:

CFFE Tree-Growing Algorithm:

For each tree $b \in B$:

1. At node $\mathcal{N}$ with observations $\{Y_{i,t}, W_{i,t}, X_i\}_{i \in \mathcal{N}}$, perform local recentering of $Y_{i,t}$ and

[29] We note that conditioning on group membership (i.e $i \in S$) as they do in their Equation (11) is equivalent to conditioning on X_i if the CATT is continuous in X_i, as we assume in our Assumption 4. We also note that this PTA is unchanged when conditioning further on dosage, W_i.

$W_{i,t}$ (here shown for $Y_{i,t}$):

$$\text{Demean by unit:} \quad \ddot{Y}^{\mathcal{N}}_{i,t} = Y^{\mathcal{N}}_{i,t} - \frac{1}{T}\sum_{t=1}^{T} Y^{\mathcal{N}}_{i,t}$$

$$\text{Demean by time:} \quad \widetilde{Y}^{\mathcal{N}}_{i,t} = \ddot{Y}^{\mathcal{N}}_{i,t} - \frac{1}{n^{\mathcal{N}}}\sum_{i\in\mathcal{N}} \ddot{Y}^{\mathcal{N}}_{i,t}$$

2. Perform standard causal forest splitting on the doubly-transformed node in the manner of Athey, Tibshirani, and Wager (2019).

3. Repeat steps (1) and (2) until stopping criteria are met.

In practice, CFFE forests are estimated similarly to the DiDCF: we estimate separate CFFEs for each time period post-treatment. The CFFE differs in that each forest contains only units in that period plus *all* pre-treatment periods, making it more efficient than the DiDCF (recall the DiDCF uses only the current period). However, this efficiency gain comes via a more strict conditional PTA. Recall, the DiDCF relied only on Assumption 4, which required that treated units be conditionally parallel only to never-treated units; the CFFE requires that all eventually-treated units be conditionally parallel to one another. Kattenberg, Scheer, and Thiel (2023) show through simulation that the CFFE achieves lower RMSE than the DiDCF when unit and time fixed effects are large relative to the treatment effect signal.

3.3.5 Comparison with the Interacted GDD

Conducting heterogeneity analysis via linear models such as the interacted DD comes with significant disadvantages that the CF-DD methods allow us to overcome. Linear models are designed to estimate average causal parameters efficiently. In the case where the average causal effect is allowed to vary continuously with covariates, the objective changes from estimating a parameter to estimating a function. Estimating such a function with linear interactions assumes that the relationship itself is linear unless the model is saturated with multiple interactions and/or polynomials. It is impossible for a saturated linear model to account for the full set of functional forms the CATT function may take on while retaining statistical power. If a researcher opts to estimate many regressions with singular heterogeneity interactions, they risk multiple hypothesis testing complications. Conversely, the CF-DD non-parametrically estimates the CATT function, producing consistent estimates while implicitly accounting for the full set of functional forms with a possibly high-dimensional vector

of covariates and avoiding multiple hypothesis testing problems.

The GDD has one important advantage over the CF: a more plausible estimate of the average causal effect (in terms of precision). The advantages in effect heterogeneity analyses are largely irrelevant if they are based on misleading heterogeneity. In ongoing work, we are trying to bring the CF more in line with the GDD, so that it provides the best of both worlds.

3.4 Policy Learning via Optimal Decision Trees

A potentially useful exercise which applies our non-parametric estimates of heterogeneous treatment effects is to understand alternative treatment assignment regimes for targeting districts towards charter school expansion. For school districts considering the decision to expand charter schools or not, what would a simple decision rule rooted in our CATT estimates look like in order to provide guidance for that question? The question we pose is: how should a social planner optimally choose which districts to expand charter schools to maximize the post-treatment outcome defined as either AFGR, Math test scores, or ELA test scores.

Formally, we seek to estimate a decision rule, $\pi \in \Pi$, that maps an individual's observed covariates to a binary decision of treatment assignment. That is, $\pi : X_i \in \mathcal{X} \rightarrow \{0, 1\}$, where $\pi(X_i) = 1$ indicates that the district should marginally expand charter school intensity and $\pi(X_i) = 0$ indicates that the district should not. Kitagawa and Tetenov (2018) specify this optimization problem as one which finds the policy that minimizes empirical regret relative to the best policy, $R(\pi)$, defined as:

$$R(\pi) = \arg\max_{\pi' \in \Pi} \{V(\pi') : \pi' \in \Pi\} - V(\pi) \tag{18}$$

Where $V(\pi) = \mathbb{E}[Y_{i,t}\big(\pi(X_i)\big) - Y_{i,t}(0)]$ is the average impact of implementing policy π relative to treating no one. Kitagawa and Tetenov (2018) solve this problem with inverse propensity score weighting, but Athey and Wager (2021) have proposed writing Equation 18 in terms of the CATT (in their case the CATE) and estimate the policy by solving:

$$\hat{\pi} \;=\; \arg\max_{\pi \in \Pi} \; \Big\{ \frac{1}{n} \sum_{i=1}^{n} \big(2\pi(X_i) - 1\big)\, \hat{\Gamma}_{i,t} \Big\} \tag{19}$$

Where $\hat{\Gamma}_{i,t}$ is the doubly robust score obtained via the augmented inverse-probability

weights (AIPW) estimator. Chernozhukov et al. (2018) shows that AIPW scores are more efficient when aggregating CATE/CATT estimates to higher-level averages. For instance, if we wanted to estimate the ATT of charter expansion via the causal forest estimators, doing so by averaging AIPW scores of the CATTs (i.e. via $\hat{\tau}_t^{ATT} = \frac{1}{n}\sum_i^n \hat{\Gamma}_{i,t}$) is more efficient than simply averaging over CATT estimates. These scores can be constructed as follows (Nie and Wager, 2021):

$$\hat{\Gamma}_{i,t} = \hat{\tau}_t(X_i) + \frac{W_{i,t} - \hat{e}(X_i)}{\hat{e}(X_i)(1-\hat{e}(X_i))}[Y_{i,t} - \hat{\mu}(X_i) - (W_{i,t} - \hat{e}(X_i))\hat{\tau}_t(X_i)] \tag{20}$$

Where $\hat{\mu}(X_i) = \mathbb{E}[Y_{i,t}|X_i = x]$ and $\hat{e}(X_i) = \mathbb{E}[W_{i,t}|X_i = x]$ come from the auxiliary forests estimated during the CF training process.

It is key to note that the solution to Equation 19 is the optimal charter expansion policy under an extremely strict set of assumptions. Namely, the solution to Equation 19 assumes that the cost to expand charter schools is equal across all districts and charter intensities. It also assumes that the social welfare function is Utilitarian and perfectly proxied by a single outcome. For this reason, we use the solution policies only to better understand the CATT function estimates from the causal forests, and do not interpret them as the true optimal treatment allocation policies.

Equation 19 can be solved easily (though inefficiently) by performing an exhaustive search over a limited class of possible policies. Athey and Wager (2021) show that doing this will yield the regret-minimizing policy within that limited class. Here, we search over the space of policies representing depth-2 and depth-3 decision trees. We compute doubly robust scores per cohort $\times$ calendar-time cell from the grf R package. For each treated district, we average scores across all post-treatment event times to obtain a single welfare score $\bar{\Gamma}_i$. We restrict the covariate matrix X to the ten moderators with the highest variable importance at $h = 1$. The policytree package (Sverdrup et al., 2020) performs the search to solve Equation 19.

4 Results

4.1 Replicating the Average Treatment Effects

We start by replicating the average treatment effects from Chen and Harris, 2023 without the interaction terms. The first column (Sample 1) of Table **??** shows the results for the overall sample. The second column (Sample 2) drops the districts whose effects initially

appear in the 5th-95th percentile (see reasoning below). Finally, the third column (Sample 3) starts with the whole sample and then shows the results dropping districts that had charter schools at the beginning of the panel, so that the estimates focus on the effects of the first charter entrant(s) due to the possibility of diminishing returns. All of the specifications assume a one-year lag in the effect from charter entry to student outcomes.

The results generally reinforce Chen and Harris, 2023's finding that the average effects are positive for each outcome.[30] All 18 estimates are positive and six are precisely estimated. The results are most consistently precise with high school graduation, though two of the six estimates are positive and significant for math. There is no clear pattern as to which samples/columns show the largest effects. The results are also uniformly positive and more precise when dropping the district-specific linear trends. Since the results with district trends are more likely to pass the parallel trends, we opt for this as our preferred method going forward while reporting.

4.2 Distribution of Effects

4.2.1 Average Treatment Effect

This study focuses on effect heterogeneity but we motivate this with a brief discussion of the average treatment effects. For the GDD, these are shown in Table **??**. These show positive effects on graduation across multiple methods, though, individually, these do not quite reach the level of statistical significance. The effects on math are nearly zero in the first two samples, but significant and large in the third sample, which restricts to the first charter entrant. Finally, for ELA, the effects are positive and mostly precise across all three specifications. (All three effects are slightly smaller than in Chen and Harris, 2023 for reasons we are still exploring.) For the heterogeneity results to follow, we will use the Sample 1 in all cases.

4.2.2 Overall Distribution Across Geographic Districts

The first set of results regarding market-level effect heterogeneity are the distributions of effects for each school district. We estimate and plot CATTs for each geographic district with the GDD (Figure 1), and we estimate CATTs for each district × event-time with the DiDCF (Figures 4–6) and CFFE (Figures 19–21) methods. These distributions aim to answer the first primary questions of our analysis: is there notable heterogeneity in charter

[30] Some slight differences exist between Table 2A and Chen and Harris, 2023. The latter paper did not use linear interpolation to clean the AFGR.

school effects, and is it estimable? For high school graduation, each method shows a wide distribution of market-level effects, with ATTs near zero. Some of these are extremely large and statistically significant, in both positive and negative directions. For the GDD method, more than 50 districts have coefficients larger than +2.0; a coefficient of that size implies that going from 0 to 10 percent charter market share (just above the national average) would increase the high school graduation rate by 20 percentage points. The distributions of district × event-time effects are much narrower with the two causal forest methods, though both approaches also estimate significant heterogeneity in both directions. The ATTs as estimated by the DiDCF and CFFE are slightly larger than the GDD (0.11 and 0.14, respectively). The range of effect estimates from the DiDCF is about [-0.75, 1], whereas the CFFE ranges from about [-0.5, 1]. Because the two causal forest methods allow us to estimate valid standard errors for district × event-time CATTs, we also denote which district × event-time estimates are statistically significant. Both causal forest methods indicate that the bulk of significant estimates are for districts that are positively impacted.

For test scores, the effect distributions are wider than we would expect from the GDD approach. For example, a larger-than-average coefficient of +1.0 means that increasing the charter market share by 10 percentage points increases annual student test scores by 0.10 s.d.. Some cities with very large market shares have generated market-level effects of this size (Harris and Larsen, 2023; Baxter, 2024)]. On the other hand, Chen and Harris, 2023 estimate that the charter participant effect is two-thirds of the total market effect, with the remainder pertaining to the various competitive effects and math effects. In that case, a market-level effect of +0.10 s.d., for a 10 percent increase in market share, would require an annual participant effect, averaged across charter schools, of 0.67 s.d.[31] The causal forest methods generally agree with the GDD regarding the range of effects, though the point estimates of the ATT are slightly different. The range of Math scores is from about [-2.5, 2] for the DiDCF, and from about [-1.75, 2] for the CFFE, with ATT estimates of -0.06 and 0.33 for DiDCF and CFFE, respectively. For ELA scores, the range of effects from the DiDCF is about [-2, 2], and about [-1, 2] from CFFE, with ATT point estimates of -0.1 from DiDCF and 0.2 from CFFE.

[31] On the other hand, Chen and Harris, 2023 estimate that the charter participant effect is two-thirds of the total market effect, with the remainder pertaining to the various competitive effects and math effects. In that case, a market-level effect of +0.10 s.d. would require an annual participant effect, averaged across charter schools, of 0.67 s.d..

4.2.3 Effect Heterogeneity by State

Districts are nested within states, which is also the institutional unit where most charter policies are made. Here we estimate equations (1) and (2) where the interaction term H_{is} is just a vector of state indicators. We also restrict to states that have at least five geographic districts contributing to the estimation. Our estimated state-specific effects on test scores vary widely from -0.83 in Oregon in ELA to +1.53 in Oklahoma in math (among those that are precisely estimated). We note that this range is narrower than the overall district-level range in Figure 1 (-3.0 to +4.10). That is, most of the variation in charter effects is within states.

The other study that has estimated state-specific effects is CREDO, Center for Research on Education Outcomes, 2023. These are shown in the last column. Also, at the bottom of each column we report the simple correlation between our estimate and the CREDO estimate, converted to the same scale as ours. We do not expect our estimates to be perfectly correlated for many reasons: (a) our time period does not match theirs; (b) we are identifying effects only from changes in market share (mostly from schools opening and closing) while CREDO's strategy incorporates essentially all charter schools; and (c) they are estimating participant effects only and we are capturing the market-level effect, which includes the participant effect but captures a broader array of mechanisms. So, we would expect a positive correlation, but one that might not be close to unity. This is what we see. The correlations range from +0.34 to +0.58 for the first two samples that focus on the overall average treatment effect. The correlations are predictably lower for Sample 3, which focus only on the subset of districts that had zero charter schools at baseline.

The appendix also shows these results in the form of maps, the first of which also shows results for high school graduation. One immediately evident pattern is that the results vary considerable within states across the maps. Only two states show consistently negative results across outcomes (Arkansas and South Carolina), while three show consistently positive effects (Illinois, Massachusetts, and Rhode Island).

4.3 Analyzing Effect Heterogeneity by District Traits

Most of the rest of the study focuses on explaining the variation across districts and states, using both the DD and CF methods. The factors are broken into five categories: student demographics (e.g., race and family income), market context (e.g., TPS baseline outcomes and teacher supply metrics), charter sector characteristics (EMO/profit share),

school finance (e.g., charter spending, TPS-charter funding gaps, and total average spending), and other charter school policies. We organize the analysis topic by topic, so that we can directly compare the results across methods and summarize the results for each set.

Interpreting the heterogeneous treatment effect estimates from the GDD and CF methods requires different approaches. The GDD coefficients we report in Table 2 represent the best *linear* unbiased estimates of the CATT along a single dimension of heterogeneity. The advantage of the GDD is the simplicity of the interpretation: point estimates and standard errors represent straightforward, linear relationships with corresponding confidence intervals that inform the researcher of the statistical significance of that relationship. However, the GDD says little about what *drives* heterogeneity[32], nor can it easily answer questions about non-linear relationships without over-saturation of the model with higher-order polynomials or multi-level interactions among moderators. Conversely, the CF methods target the full CATT function, allowing implicitly for any number of non-linearities. This results in a single (possibly noisy) estimate with valid standard errors for each unit of observation – in our case district $\times$ event-times. These individualized treatment effect estimates are implicitly derived from the model's estimation of the CATT function, which (because this function is likely to be non-linear and interactive) means that understanding how these treatment effects vary is a non-trivial task.

In order to infer (1) what moderators are driving short- and long-run heterogeneity in treatment effects and (2) how the effects vary by these moderators, we combine several metrics which individually highlight key information. Taken together, a fuller picture of the CATT function estimates from the CF methods can be pieced together. First, both CF methods allow for a simple measure of variable importance to be calculated by computing the share of total trees within each forest that split along a given variable, weighted by the depth at which the split occurred so that earlier splits are weighted more heavily. This is referred to as the "Variable Importance Factor" (VIF) and serves as our primary indicator for which moderators are driving heterogeneity. Because the DiDCF forests are trained separately for each event-time, we compute VIF matrices displayed in Figures 7 – 9 that allow us to evaluate which moderators drive heterogeneity in early periods post-treatment vs. late periods post-treatment. For the CFFE, we compute an aggregated VIF ranking for each outcome displayed in Figures 22 – 24. We interpret all further results by emphasizing

[32] A statistically significant relationship does not imply that a given moderator is driving heterogeneity. For example, two highly correlated moderators might both have independently significant point estimates, while only one is driving the effect in practice.

moderators with high VIF scores.

Following the VIFs, we report the mean values of each moderator for districts with positive CATT estimates vs. districts with negative CATT estimates, and the differences in these means. These are displayed in Tables 1 – 3 for the DiDCF and Tables 7 – 9 for the CFFE. The objective of these difference-in-means tables is to examine the "profile" of an average positively/negatively impacted district × event-time. These means may also provide some insight into the relative linearity of the relationship between a particular moderator and the CATT estimate. For example, if the mean positively-impacted district for a particular moderator is approximately equal to the mean of the average negatively-impacted district, this is evidence for a non-linear (or interactive) relationship. Conversely, if the two means are notably different, this provides evidence that there is a level difference in moderator values across positively vs. negatively impacted districts. Practically, this means that a linear approximation of the relationship between that moderator and the CATT may be of some use.

The next exercise we perform is also intended to investigate the linearity of the relationship between high-VIF moderators and the CATT estimate. Tables 4 – 6 and Tables 10 – 12 report the Best Linear Predictors (BLP) of the CATT for each outcome among the top 5 highest VIF moderators for DiDCF and CFFE, respectively. The coefficients in these tables represent the linear relationship between each moderator and the CATT and are obtained by regressing the moderators on doubly-robust scores of the CATT. Similar to the difference-in-means tables, a significant coefficient provides evidence that a linear approximation of the relationship between moderator k and the CATT ($\tau(X)$) carries useful information about $\frac{\partial \tau(X)}{\partial X_k}$.

The final two exercises aim to investigate non-linear and interactive relationships between high-VIF moderators and the CATT. First, we report event-studies for each outcome in Figures 10 – 12 and 25 – 27 for DiDCF and CFFE, respectively. These event-studies segment units by their moderator value (for the top 5 highest VIF moderators at event-time 0) such that districts falling in each quartile are plotted separately. These event-studies allow us to simultaneously assess how each moderator drives heterogeneity across its' distribution in both an overall sense and dynamically. The DiDCF event-studies also allow us to investigate the validity of the conditional parallel trends assumption discussed in Section 3 by estimating CATTs in pre-treatment periods via a placebo indicator for not-yet-treated districts.[33] Every

[33] As discussed in Section **??**, the CFFE is not well suited to investigating pre-trends because it is not

event-study (each outcome, each moderator, and each quartile of units) shows precisely zero effect in pre-treatment periods, which provides supporting evidence in favor of the conditional PTA we invoke. Second, we report shallow-depth decision trees from the policy learning exercise of Athey and Wager (2021) discussed in Section 3 in Figures 13 – 18. These trees represent the optimal treatment assignment rule among the class of depth-2 (or depth-3) rules to maximize the expected outcome in question. These assignment rules are illustrative of the interactive relationship of the CATT function estimate, and are not intended to represent the true optimal treatment assignment policy.[34]

4.3.1 Trait Set #1: Demographics

Prior research has consistently found that charter schools are more effective for disadvantaged students, broadly defined (Cohodes and Parham, 2021). The theories around this have been based partly on the lower test scores these groups typically have or, relatedly, that these groups are served by schools that are more poorly performing.

Average Freshman Graduation Rate

For AFGR, we find that demographics (or unobserved variables that are heavily correlated with them) are some of the most important drivers of charter school effects. Regarding VIF, the DiDCF lists Log Enrollment (1), Free-Reduced Lunch % (2), White % (4), Special Ed % (5), Hispanic % (7), and Black % (8) in the top 10. The CFFE similarly lists Log Enrollment (2), Hispanic % (3), White % (6), Black % (7), Special Ed % (8), and Free-Reduced Lunch % (9) in its top 10 most important variables. Notably, while Log Enrollment is the most important variable for *short run* impacts according to DiDCF, it diminishes quickly in importance when considering long run effects. Both DiDCF and CFFE agree that larger districts are impacted more positively than smaller ones: both methods' difference-in-means tables suggest that positive CATT districts are on-average larger. This is supported further by significantly positive BLP coefficients, and event-study plots which suggest that the top quartile of districts in terms of enrollment is impacted more positively than the bottom 3 quartiles.

designed to estimate pre-treatment effects via the placebo approach that the DiDCF undertakes. The CFFE is only estimated in post-treatment periods because it uses all pre-treatment information in the demeaning process.

[34]These learned policies are welfare-maximizing under a very strict set of assumptions: the social welfare function must be perfectly proxied by the outcome in question, welfare weights must be Utilitarian, and charter expansion costs must be equivalent across units. We refer the reader to Kitagawa and Tetenov (2018) for a complete discussion of these assumptions.

Both methods also agree that the share of students who are White vs. the share who are Hispanic and/or Black also matters. The difference-in-means tables for both methods suggest that districts with fewer White students and more Hispanic and/or Black students benefit more positively from charter schools. The BLP supports the direction of these relationships, but the coefficients are insignificant for each of these which points to a likely non-linear or interactive relationship. The event studies seem to confirm the non-linearity: both methods agree that districts in the bottom quartile of White % benefit the most from charter expansion (at least in the short- and medium-run). The CFFE estimates that districts in the top quartile of Hispanic % benefit more from charter expansion (DiDCF did not identify Hispanic % in the top 5 so we did not estimate event-studies for this moderator with DiDCF). Both methods also suggest that more students on Free/Reduced Price Lunch is a strong predictor of the CATT, though the relationship does seem to be likely non-linear. The DiDCF event-studies suggest that in the short run, districts with the fewest FRL students benefit less than the top 3 quartiles, but by the long run, both the bottom and top quartiles benefit roughly equally while the middle 2 quartiles benefit less. The CFFE difference-in-means table (and weakly via the BLP) corroborates this finding.

The policy trees also generally support the prior results. While the depth-2 policy tree interestingly does not split on demographics (as we will discuss later in the summary subsection), the depth-3 tree uses both FRL % and White % as decision variables. The learned policy suggests that districts with a large share of minority (and/or low socioeconomic status) students will benefit positively from charter expansion under certain conditions.

Math Test Scores

For Math scores, demographics become much less important in favor of funding metrics (to be discussed within Trait Set #3). The DiDCF (CFFE) finds Log Enrollment as the only demographic measure with a VIF score over 5%, ranking it 3rd (5th) most important. The largest districts (top quartile) appear to be negatively affected by charter expansion on-average according to the DiDCF event-study. While the BLP coefficients support this finding, the difference-in-means tables for DiDCF and CFFE do not. This finding that larger districts are impacted less positively for Math scores is notably inverse to the finding that larger districts are impacted *more* positively when considering AFGR.

Both methods also suggest that districts with positive CATTs have more Hispanic and FRL students (and fewer White students) via the difference-in-means tables, but the re-

lationship appears weak and likely non-linear when examining BLP. CFFE also identifies Special Ed % as fairly important (rank #7 but with a VIF below 5%), but seemingly non-linear/interactive with a difference-in-means across positive/negative CATT districts approximately equal and a corresponding BLP coefficient that is highly imprecise.

While the above results do not suggest that demographics are driving the heterogeneity in charter school impacts for this outcome, the learned policy trees suggest that they are nonetheless useful in recommending charter expansion within this class of learned policies. Both the depth-2 and depth-3 trees use Hispanic % and Special Ed % as decision variables, highlighting evidence that these moderators might be interactive in practice (and correlated enough with spending variables to carry useful information). Generally, the policy trees favor expanding charter school intensity if a district has fewer Hispanic and Special Ed students, which contradicts the difference-in-means tables but remains unsurprising as the BLP suggested these variables would be likely interactive/non-linear.

ELA Test Scores

The story is very similar for English test scores as it is for Math test scores. Again, Log Enrollment is the only demographic indicator with a VIF over 5% for both DiDCF and CFFE. Like with Math scores, both methods agree that the largest districts are impacted least positively (or negatively, according to DiDCF) by charter school expansion on ELA scores, while the smallest districts benefit positively in the long run according to both DiDCF and CFFE.

The policy trees again use demographics for charter school expansion recommendations despite their relatively low importance, again highlighting that these variables carry significant information when restricting policy complexity despite them not driving the heterogeneity. The depth-2 tree interestingly only splits on Special Ed %: favoring charter expansion in cases where the share of students who are special ed is below 11.6%. The depth-3 tree considers a more complex decision policy, but still splits on Hispanic % and Special Ed %. In this policy, Hispanic % is the first split, indicating it is a highly interactive moderator.

4.3.2 Trait Set #2: Market Context

Average Freshman Graduation Rate

We included three types of market context variables: region type (urban, suburban, town,

and rural), attractiveness to young teachers (proxied by if a university is located in the city), and baseline academic performance in traditional public schools. Baseline AFGR is a consistent driver of heterogeneity according to both methods: with DiDCF (CFFE) ranking it as the 3rd (1st) highest VIF moderator (and notably, when considering long run effects, Baseline AFGR increases to the highest VIF score for DiDCF as well). Urban regions also score somewhat high with CFFE (though, not DiDCF) as the 10th highest VIF with 6% of trees splitting on it. The CFFE finds that Urban school districts are more likely to have positive CATT estimates according to difference-in-means. No other market context variable carries notable information according to VIF.

Both the DiDCF and CFFE methods provide evidence that the lowest performing districts at baseline benefit the most from charter expansion, with the bottom quartile of districts having a 5pp larger treatment effect in event-time 7 than the top quartile according to DiDCF. However, though the CFFE finds that the bottom quartile benefits by as much as 10pp more than the top quartile in early- to mid-periods, that method finds that baseline AFGR does not predict long run impacts. The learned policy trees also use baseline AFGR as a decision variable. Both the depth-2 and depth-3 policy trees favor charter expansion if baseline AFGR is low.

Math Test Scores

Baseline math test scores come in as the 4th (3rd) highest VIF moderators according to DiDCF (CFFE). No other market context variables make the top 10 in VIF ranking. According to DiDCF, the bottom quartile of baseline math performance is impacted positively in the short- and medium-run, but by event-time 7 there are no lasting impacts. The middle two quartiles are negatively impacted by charter school expansion. The CFFE agrees with DiDCF that the bottom quartile of districts in baseline performance are impacted more positively than the bottom 3 quartiles, but is generally more optimistic in that CFFE finds that all quartiles are positively impacted by charter expansion in the short- and long-run.

The learned policy trees suggest that baseline math performance is highly interactive. Both the depth-2 and depth-3 trees split on baseline math scores as the initial decision variable. The depth-2 tree recommends charter expansion to districts with low baseline performance with a sufficiently low share of students who are Hispanic (again, there are likely several spending variables that are actually driving the effect – to be discussed in Trait Set #3, but are (together) correlated enough with the Hispanic student share so that the His-

panic student share is the best decision variable in a shallow decision tree), or to districts with high baseline performance with a sufficiently low share of Special Education students. The depth-3 tree again recommends charter expansion to low performing districts, or to high performing districts with high teacher salaries or sufficiently few Special Education students.

ELA Test Scores

Like with baseline math scores, we find that baseline ELA scores are highly important predictors of the CATT. Baseline ELA ranks 5th in VIF for DiDCF and 3rd for CFFE. The DiDCF estimates that the bottom quartile of baseline ELA performance benefits positively in the long run from charter expansion while the top 3 quartiles are negatively impacted. The CFFE agrees with the DiDCF that the bottom quartile of baseline ELA performance benefits the most positively, though the CFFE is again more optimistic and estimates positive effects for all 4 quartiles. Baseline ELA performance also appears as a decision variable in the depth-3 policy tree. Specifically, the depth-3 tree provides a highly similar recommendation to math scores in that charter schools should be expanded under a sufficiently low baseline performance and a sufficiently low share of Hispanic students.

4.3.3 Trait Set #3: School Funding

Average Freshman Graduation Rate

We include several different measures of school finance including Total Revenue (per-pupil), Total Spending (per-pupil), Charter Spending (per-pupil), Teacher Salaries, TPS-Charter Spending Gap, and average Student-Teacher Ratio within the district. These variables appear to drive heterogeneity in AFGR effects less than demographics. For DiDCF, Student-Teacher Ratio (6), Teacher Salary (9), Total Spending (10), and Total Revenue (11) each have VIF scores over 5% (rankings in parentheses). For CFFE, Teacher Salaries (4) and Student-Teacher Ratio (5) score over 5% as well. Overall, the results suggest that the relationship between district finance variables and the CATT is likely non-linear or interactive. The CFFE event-studies find no distinguishable pattern in effect magnitudes across quartiles of Student-Teacher Ratio, indicating an interactive relationship is likely. This is corroborated by the difference-in-means and BLPs of both DiDCF and CFFE, which find no clear level differences in Student-Teacher Ratios between positive/negative CATT districts. This is also true for the other district finance variables. While the difference-in-means tables for DiDCF and CFFE suggest that positively affected districts may spend more on-average (on both TPS's and charters) and have more revenue, the level differences are not significant

nor are the corresponding BLP coefficients.

Both the depth-2 and depth-3 policy trees leverage financing moderators in the charter expansion decision process, providing further evidence that these variables are related to the CATT interactively and non-linearly. The depth-2 tree recommends charter expansion for districts with larger Student-Teacher Ratios and lower Teacher Salaries. The depth-3 tree recommends expansion for smaller Student-Teacher Ratios if the district is already relatively high performing but with low per-pupil Total Spending.

Math Test Scores

Unlike with AFGR, district financing variables seem to be the primary drivers of heterogeneity in both math and ELA test scores. For DiDCF, Charter Spending (1), the TPS-Charter Spending Gap (2), Student-Teacher Ratio (5), and Teacher Salaries (6) each have a VIF over 5% (rankings in parentheses). The CFFE paints a similar picture with Charter Spending (1), the TPS-Charter Spending Gap (2), and Student-Teacher Ratio (4) each scoring over 5%. Notably, both methods score Charter Spending and the TPS-Charter Spending Gap as by far the most important covariates for predicting test score impacts in the short- and long-run. For instance, the DiDCF's VIF score for Charter Spending is 33% in event-time 1, and increases to 43% by event-time 7.

The relationships between these variables and the CATT are also a bit clearer for math scores than with AFGR. Both methods agree that positively impacted districts spend more on charters per-student, have larger student-teacher ratios, and pay teachers more. CFFE strongly favors districts with larger TPS-Charter spending gaps, where DiDCF weakly prefers smaller gaps (though both methods agree that the top quartile of spending gaps are impacted most positively in the short- and medium-run, and the smallest gaps are impacted least in the short- and medium-run).

ELA Test Scores

We find very similar results to math test scores when examining the impacts of school finance variables on ELA scores. For DiDCF, Charter Spending (1), the TPS-Charter spending gap (2), Teacher Salary (4), and Student-Teacher Ratio (6) each score over 5% on VIF, with Charter Spending again scoring over 30% in most periods. Per-pupil revenue also rises to the 3rd highest VIF in event-time 7, suggesting that this is a driver of long-run

charter impacts. For CFFE, Charter Spending (1), the TPS-Charter Spending Gap (2), and Student-Teacher Ratio (5) again each score highly. We come to many of the same conclusions regarding these variables' relationship to the CATT as we did with math scores. Positively impacted districts tend to spend more on charters per-student, have larger student-teacher ratios, and pay teachers more. The exception is the TPS-Charter Spending Gap, which does not exhibit a clear pattern. While the difference-in-means and BLPs for both methods suggest that larger spending gaps are associated with positive CATTs, the event-studies do not clearly show a trend.

4.3.4 Trait Set #4: State Charter Policy

We considered five policy dimensions from the charter industry groups. These results are quite mixed between the two methods and, probably not coincidentally, have very low VIFs.

In Table 4 we estimate equation (1) and (2) where H_{is} is a vector of traits, mostly household characteristics (included at the district level) and state policies (necessarily at the state level), using data taken from the baseline year to avoid endogeneity concerns. We include all of these characteristics simultaneously (each column in a single regression) because they are all correlated with one another. (the appendix shows the results where the covariates are included one at a time). Table 5 shows the placebo test for these variables. In what follows, we draw only tentative conclusions about any dimension of effect heterogeneity that fails the placebo test (especially if the placebo effect is in the same direction and magnitude as the effect estimate in Table 4). We are also more cautious about estimates that are not robust across the three samples.

The results in Table 4 simultaneously control for state policy ratings. We focus here on the NAPCS ratings, which provide the most policy categories. Again, these are: transparent charter application process ($Transpar$), performance-based contract required ($Perf$ in Table 4), clear processes for renewal/nonrenewal (Non_renew), automatic exemption of charter schools from laws applied to traditional public schools (TPS) ($Rule_exempt$), no caps on the growth of public charter schools ($Nocaps$), and equitable operational funding, i.e., higher funding for charter schools relative to TPS ($Eq_funding$).

It initially appears, in Table 4, that having a transparent application process is associated with lower charter effects, while states providing more equitable funding have better charter performance, at least in test scores. For the other policy variables, the coefficients are less consistent across samples and outcomes and/or the equivalent placebo coefficients

are of similar signs/magnitudes and sometimes precise.

For the two CF methods, these policy variables generally score low on VIF. Automatic Exemption from TPS Rules, Equitable Funding, and Clear Renewal policies seem to consistently score the highest across each method and outcome within this trait set, but never score higher than 3%. Because they have low VIFs, we omit them from BLP and event-study analyses, and they (unsurprisingly) do not appear in any of the learned policy trees. When examining difference-in-means, a few suggestive findings emerge regarding state charter policy design. First, both methods agree that districts with performance-based contracts or where charters are automatically exempt from TPS regulations are more often negatively impacted by charter expansion regarding math and ELA test scores. Second, while DiDCF generally doesn't find any significant differences across the other policy variables, the CFFE suggests that positively impacted districts on test scores more often have equitable funding agreements and/or clear renewal policies. The signs often flip for CFFE when considering AFGR: positively impacted districts more often (1) have performance-based contracts, (2) lack clear renewal policies, and (3) have no caps on charter growth.

4.3.5 Trait Set #5: Diminishing Returns

Equation (1) assumes that the effects of additional charter market share are linear, but it could be that the marginal effects depend on the baseline charter share, e.g., diminishing marginal returns (DMR). The first test for DMR comes from comparing Sample 1 (full sample) and Sample 2 (only districts with zero charter market share at baseline) in Table 2. In that case, we see some evidence of DMR with respect to test scores, but not high school graduation, i.e., the coefficients are larger for the initial charter entrants (Sample 3) than for the whole sample that leverages changes in charter entry throughout the baseline charter distribution.

A more direct test, and one that captures the entire distribution of charter entry, is to group districts on their baseline charter market share and to estimate separate effects for each subgroup. Evidence of DMR would take the form of an inverted-U. Now, we see the DMR pattern for high school graduation, but not test scores; in fact, in the latter case, the point estimates seem to grow somewhat as we move to districts with higher baseline charter shares. When we estimate without district-specific linear trends, we do see more evidence of diminishing returns for two of the three outcomes.

5 Conclusion

A common theme of the economics of education is that different educational units (teachers, schools, and so on) vary widely in their quality. A second common theme is that measurable traits in those units (teacher certification, school size, and so on) do not explain much if any of this variation. With more than one thousand school districts having at least one charter school, it is now possible to test whether that pattern holds for charter schools.

In some respects, we find the same pattern. First, do charter effects vary? Do some "bloom" and others "wilt"? Yes, as with teachers and schools, some of the effects, especially with respect to high school graduation are very large. We also see some, albeit less, variation in effects across states.

Second, can we explain that variation? Here, we see several noteworthy patterns. Districts with more low-income students benefit more from having charter schools. These results are not driven by the fact that we are controlling for so many factors at the same time in the main results (Table 4). They also hold when we estimate equation (1) with smaller blocks of variables.

Somewhat surprisingly, charter schools seem to improve geographic district outcomes when those districts were higher performing to begin with. This may be less surprising, however, when we note that some of the states with larger effects, especially Massachusetts (see Table 3), also have high baseline achievement levels; conversely, some of the states with smaller effects (Arkansas and South Carolina) have very low performance levels.

State policies may explain some of the variation in results across states. Charter policies that require less transparency in charter applications and renewal decisions are associated with more positive charter effects. One theory that could explain this is that authorizers, in states with stronger transparency, may be pressed to make their decisions more on the basis of public information, as opposed to private information that might have that is also indicative of future charter performance. Less surprising is that charter schools seem more effective, as least with respect to test scores, when state policies provide more funding to charter schools. However, as noted in section II, the role of charter funding is more complicated than the state-level policy ratings account for. We certainly expect the charter participant effects to be more positive when charter schools have more funding but, for purposes of understanding the market-level effects, we also have to be concerned with what charter entry means for

TPS in the same districts, which may influence the competitive responses. We are in the process of collecting data, at the district level, which will allow us to better understand these issues.

We see only limited evidence of diminishing marginal returns. The effects are larger for test scores when we focus only on the districts that have a first charter entry during the panel, which is suggestive of DMR. But this pattern no longer holds when we consider the full distribution of baseline charter entry. We also see little clear evidence that charter effects vary based on changes in total geographic district enrollment.

Another pattern throughout the paper is that the results differ between the high school outcomes and test outcomes. This is unsurprising for three reasons: (a) our test scores results pertain to elementary/middle schools while high school graduation pertains to a different set of schools—high schools; (b) the time span of the test score data differs from that of high school graduation, so the schools are also of different vintages; and (c) test scores and high school graduation likely have different education production functions—we do not expect schools to be equally effective in producing all outcomes. When we limit only to test scores, we see suggestive evidence that districts with declining total enrollment see more negative charter effects. We are also conducting analysis of the possibility that charter effects might display diminishing marginal returns, i.e., that the effect of charter schools is larger for initial entrants.

This work is still preliminary. In ongoing work, we are also: (a) applying shrinkage estimation techniques to understand how much of the variation is driven by measurement error; (b) collecting additional data about school types (e.g., CMO status) so we can test whether the effects are larger when certain types of schools enter; (c) collecting better data, at the district level, on charter school spending and teacher supply, so we can estimate the effects of these additional traits; and (d) carrying out additional analysis of the roles played by diminishing.

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A Appendix: Proofs

Proof. This proof derives the result in Lemma 1 which states that a difference-in-differences estimator which compares the change in outcomes of a treated group in cohort g with intensity w within local covariate group x vs. the change in outcomes from never treated units in local covariate group x identifies the Level-CATT within that local covariate group.

Step 1 (Definition). Recall Equation 2:

$$\text{Level-CATT}(g,t,w,x) = \mathbb{E}[Y_{it}(g,w) - Y_{it}(0) \mid G_i = g,\ W_i = w,\ X_i = x] \tag{S1}$$

Step 2 (Add and subtract the base-period counterfactual). Add and subtract

$\mathbb{E}[Y_{i,b_g}(0) \mid G_i = g, W_i = w, X_i = x]$ inside (S1):

$$\text{Level-CATT}(g,t,w,x) = \underbrace{\mathbb{E}\big[Y_{it}(g,w) - Y_{i,b_g}(0) \;\big|\; G_i = g,\ W_i = w,\ X_i = x\big]}_{(I)} - \underbrace{\mathbb{E}\big[Y_{it}(0) - Y_{i,b_g}(0) \;\big|\; G_i = g,\ W_i = w,\ X_i = x\big]}_{(II)}. \tag{S2}$$

Step 3 (Apply conditional parallel trends to term (II)). Plug in Assumption 4 into term (II),

$$(II) = \mathbb{E}\big[Y_{it}(0) - Y_{i,b_g}(0) \;\big|\; G_i = \infty,\ W_i = 0,\ X_i = x\big] \tag{S3}$$

Step 4 (Replace potential outcomes with observables). We handle terms (I) and (II) separately.

Term (I): treated districts. For districts with $G_i = g$ and post-treatment time $t \geq b_g$:

- *Post-period:* By consistency (SUTVA), $Y_{it} = Y_{it}(G_i, W_i) = Y_{it}(g,w)$.
- *Base period:* Since $b_g < g$ (i.e., b_g is strictly pre-treatment for cohort g), Assumption 3 gives $Y_{i,b_g}(g,w) = Y_{i,b_g}(0)$, and by consistency $Y_{i,b_g} = Y_{i,b_g}(G_i, W_i) = Y_{i,b_g}(g,w) = Y_{i,b_g}(0)$.

Therefore,

$$(I) = \mathbb{E}\big[Y_{it} - Y_{i,b_g} \;\big|\; G_i = g,\ W_i = w,\ X_i = x\big] \tag{S4a}$$

Term (II): never-treated districts. For districts with $G_i = \infty$ and $W_i = 0$, consistency gives $Y_{it} = Y_{it}(\infty, 0) \equiv Y_{it}(0)$ for all t. Therefore,

$$\mathbb{E}\big[Y_{it}(0) - Y_{i,b_g}(0) \;\big|\; G_i = \infty,\ W_i = 0,\ X_i = x\big] = \mathbb{E}\big[Y_{it} - Y_{i,b_g} \;\big|\; G_i = \infty,\ W_i = 0,\ X_i = x\big]. \tag{S4b}$$

Conclusion. Under Assumptions 1 – 4, we can obtain the difference-in-differences estimator from Lemma 1.

□

Proof of Lemma 2

Proof. We show (i) that the identification equation of Lemma 1 places the Level-CATT in the form of a causal-forest estimand with continuous treatment; (ii) that existing consistency results for continuous-treatment GRF apply within each cohort subsample; and (iii) that the

cohort-specific estimates aggregate consistently.

Notation refresh: Let cohorts be $g \in \bar{\mathcal{G}}$, event times $h \geq 1$, and set $t = b_g + h$ so that $Z_{i,g,h} = Y_{i,t} - Y_{i,b_g}$.

Step 1 (Identification target and relationship to CGBS Theorem C.1). By Lemma 1, under Assumptions 1–4,

$$CATT(g, h, w, x \mid g, w) = \mathbb{E}[Z_{i,g,h} \mid G_i = g,\ W_{i,t} = w,\ X_i = x] - \mathbb{E}[Z_{i,g,h} \mid G_i = \infty,\ W_{i,t} = 0,\ X_i = x]. \tag{21}$$

Within the cohort-g subsample $\mathcal{S}_g$ at event time $h = t - b_g$, we identify the dose variable W_i with the post-treatment charter intensity $W_{i,t}$ of district i at calendar time $t = b_g + h$. For cohort-g districts ($G_i = g$), this is the actual treatment dose received. For never-treated districts ($G_i = \infty$), $W_{i,t} = 0$ by definition.

Equation (21) is the conditional-on-X_i analog of Callaway, Goodman-Bacon, and Sant'Anna (2025) Theorem C.1, which establishes $ATT(g, t, d \mid g, d)$ via the same DiD comparison but under unconditional parallel trends (their Assumption PT-MP) and with the comparison group consisting of all units with $W_t = 0$, not only never-treated units. Our Assumption 4 is strictly stronger than their PT-MP in two respects: it conditions on $X_i = x$ throughout, and it restricts the comparison group to $G_i = \infty$ rather than all currently-untreated units, thereby avoiding the "forbidden comparisons" of Sun and Abraham (2021a). The differenced outcome $Z_{i,g,h}$ corresponds to $Y_t - Y_{g-1}$ in their notation, with $g - 1 \equiv b_g$.

Step 2 (Consistency via continuous-treatment causal forest). Within subsample $\mathcal{S}_g$, equation (21) shows that the Level-CATT equals the causal-forest estimand applied to $Z_{i,g,h}$ with *continuous* treatment $W_{i,t} \in [0, 1]$ and covariates X_i. Specifically, by Assumption 4, the conditional distribution of $Z_{i,g,h}(0)$ given X_i is the same for treated ($G_i = g$) and never-treated ($G_i = \infty$) districts, so the causal-forest contrast in $Z_{i,g,h}$ between units with $W_{i,t} = w > 0$ and units with $W_{i,t} = 0$ identifies $CATT(g, h, w, x \mid g, w)$—precisely the derivation in Gavrilova, Langørgen, and Zoutman (2025) equations (3)–(5).

Note that Assumption 4 holds for each w in the support of $W_{i,t}$, so the conditional parallel trends condition on differenced outcomes $Z_{i,g,h}(0)$ holds uniformly over the dose range. This delivers the uniform conditional moment condition required by the continuous-treatment GRF consistency result.

Gavrilova, Langørgen, and Zoutman (2025)'s Proposition 1 is stated for binary $W_i \in$

$\{0,1\}$ and invokes Wager and Athey (2018) Theorem 11. In our setting, $W_{i,t}$ is continuous. We therefore invoke two complementary results. First, Theorem 6 of Athey, Tibshirani, and Wager (2019) establishes consistency and asymptotic normality of the GRF estimator under continuous treatment, with the honesty and subsampling conditions ensured by the algorithm. Second, Theorem 1 of Nie and Wager (2021) provides the rate of convergence for the same estimand under the R-learner reduction. Both results require: (a) Lipschitz continuity of the conditional mean and treatment effect functions jointly in (x, w); (b) bounded conditional moments; and (c) overlap in the sense that the conditional density $f(W_{i,t} \mid X_i)$ is bounded away from zero. Our Assumption 5(a)–(c) delivers conditions (a) and (b); our Assumption 2 delivers condition (c). The cohort-specific DiDCF, which (i) differences outcomes relative to b_g via $Z_{i,g,h}$, (ii) estimates a separate causal forest for each event time $h \neq 0$, and (iii) extracts predictions only for treated districts, is Algorithm 1 of Gavrilova, Langørgen, and Zoutman (2025) applied within $\mathcal{S}_g$. By Nie and Wager (2021) Theorem 1 and Athey, Tibshirani, and Wager (2019) Theorem 6 (applied to $Z_{i,g,h}$ as the outcome and $W_{i,t}$ as the continuous treatment), $\hat{\tau}_{g,h}(x)$ is consistent and asymptotically normal for $CATT(g, h, w, x \mid g, w)$ as stated in (13).

Step 3 (Aggregation; fixed G). Here we treat $G \equiv |\bar{\mathcal{G}}|$, the number of cohorts, as fixed as $N \to \infty$; extensions to growing G require additional rate conditions on $\min_g n_g \to \infty$.

Since each treated district belongs to exactly one cohort, the *treated-unit* components of $\{\mathcal{S}_g\}_{g \in \bar{\mathcal{G}}}$ are disjoint. However, the never-treated units ($G_i = \infty$) appear in every $\mathcal{S}_g$, so $\hat{\tau}_{g,h}(x)$ and $\hat{\tau}_{g',h}(x)$ are not fully independent for $g \neq g'$. As established in Remark 1, the cross-cohort covariance terms are $O(1/n_\infty)$ under the maintained covariance assumption, and are asymptotically negligible relative to the leading nonparametric variance terms $O(n_g^{-2/(d+2)})$ whenever the rate condition $n_\infty = \omega(n_g^{2/(d+2)})$ holds for each g. Under this condition (which holds in our application as discussed in Remark 1) consistency of

$$\hat{\tau}_h(x) = \sum_{g \in \bar{\mathcal{G}}} \phi_g \hat{\tau}_{g,h}(x)$$

for $\tau_h(x) = \sum_{g \in \bar{\mathcal{G}}} \phi_g \, CATT(g, h, w, x \mid g, w)$ follows by the continuous mapping theorem, and the Delta-method variance in (12) is an asymptotically valid approximation, as in Gavrilova, Langørgen, and Zoutman (2025) Section 3. □

Remark 1 (Cross-cohort dependence and the rate condition for diagonal variance)**.** *Because $\mathcal{S}_g$ and $\mathcal{S}_{g'}$ ($g \neq g'$) share the never-treated units ($G_i = \infty$), the cohort-specific estimators*

$\hat{\tau}_{g,h}(x)$ and $\hat{\tau}_{g',h}(x)$ are not independent. The full variance of the aggregated estimator is

$$\mathrm{Var}(\hat{\tau}_h(x)) = \sum_{g \in \bar{\mathcal{G}}} \phi_g^2 \, \mathrm{Var}(\hat{\tau}_{g,h}(x)) \; + \; 2 \sum_{\substack{g,g' \in \bar{\mathcal{G}} \\ g<g'}} \phi_g \, \phi_{g'} \, \mathrm{Cov}(\hat{\tau}_{g,h}(x), \, \hat{\tau}_{g',h}(x))$$

Rate condition. *In the nonparametric GRF setting with d-dimensional covariates, the pointwise variance of each cohort-specific estimator shrinks at the nonparametric rate $\mathrm{Var}(\hat{\tau}_{g,h}(x)) = O(n_g^{-2/(d+2)})$ (Athey, Tibshirani, and Wager, 2019). The diagonal formula (12) is an asymptotically valid approximation for $\mathrm{Var}(\hat{\tau}_h(x))$ if and only if the cross-cohort covariance terms are asymptotically negligible relative to these leading terms.*

Cross-cohort covariance (maintained assumption). *We assume that the shared never-treated units contribute $O(1/n_\infty)$ covariance to each estimator pair: $\mathrm{Cov}(\hat{\tau}_{g,h}(x), \hat{\tau}_{g',h}(x)) = O(1/n_\infty)$ for all $g \neq g'$, where $n_\infty = |\{i : G_i = \infty\}|$. This bound holds for kernel-smoothed nonparametric estimators and is assumed to hold for the GRF adaptive-weighting scheme; a formal derivation for GRF would require tracking the dependence structure of the forest weights across subsamples and is left as an open theoretical problem.*

Negligibility condition. *Under this assumption, the cross-cohort covariance terms are negligible relative to the diagonal variance terms if and only if:*

$$n_\infty^{-1} = o\big(n_g^{-2/(d+2)}\big) \quad \text{for each } g \in \bar{\mathcal{G}}$$

equivalently $n_\infty = \omega(n_g^{2/(d+2)})$. In particular, n_∞ must grow faster *than $n_g^{2/(d+2)}$. This condition is increasingly mild as d grows (for large d, $2/(d+2) \to 0$ and any $n_\infty \to \infty$ suffices), and that can be verified empirically given the number of moderators d and the cohort and never-treated sample sizes in the data.*

When this rate condition holds as it does in our application (since never-treated districts constitute a substantial fraction of the sample and n_∞ grows proportionally to N), the diagonal formula (12) is asymptotically valid. Following Gavrilova, Langørgen, and Zoutman (2025) Section 3, we proceed under this condition and treat (12) as the operative variance approximation; a finite-sample correction would substitute the full expression above, estimating the covariance terms from the shared never-treated subsample.

B Appendix: Technical Summary of GRF Estimation Procedure

A generalized random forest grows B causal trees indexed by $b = 1, ..., B$. Each tree is grown by the following steps:

1. From the full data sample, draw a random subsample without replacement.

2. Further divide the subsample into a training sample, $\mathcal{J}_1$, and an estimation sample, $\mathcal{J}_2$, which are non-overlapping partitions.

3. With training sample $\mathcal{J}_1$, perform greedy and recursive splitting by grid searching over the covariate space and selecting the single covariate, $x_k \in X$, and value, $x_k = c$, that maximizes the squared difference in treatment effects across the resulting child nodes. The objective function is given as:

$$\max_{x_k, c} \quad N_{C_1} N_{C_2} (\hat{\tau}_{C_1} - \hat{\tau}_{C_2})^2$$

   Where C_1 and C_2 denote the resulting child nodes such that N_{C_1} and N_{C_2} are the number of units in each resulting child node and $\hat{\tau}_{C_1}$ and $\hat{\tau}_{C_2}$ are the resulting ATEs within each child node. For computational tractability (i.e. in order to avoid estimating linear regressions for each potential split), Athey, Tibshirani, and Wager (2019) show that this process reduces to splitting on a pseudo-outcome corresponding to a first-order gradient-based approximation of the subsample ATEs.

   (a) Denote a parent node, P, as any node that is currently being assessed for an optimal split. Compute $\hat{\alpha}_P$ and $\hat{\tau}_P$ by solving Equation 9, which can be alternatively expressed as:

$$\sum_{i=1}^{N_P} \omega_i(x)(Y_i - \alpha(x) - \tau(x)W_i) = 0$$

   (b) Also compute an estimate of the gradient of the expectation (once, with all of the data) of $\psi_{\alpha(x),\tau(x)} = Y_i - \alpha(x) - \tau(x)W_i$, denoted $A_p = \nabla \mathbb{E}[\psi_{\hat{\alpha}(x)_P, \hat{\tau}(x)_P} | X_i \in P]$:

$$A_P = \frac{1}{|\{i : X_i \in P\}|} \sum_{\{i: X_i \in P\}} \nabla \psi_{\hat{\alpha}(x)_P, \hat{\tau}(x)_P}$$

   (c) Compute the pseudo-outcomes as:

$$\rho_i = -\xi' A_P^{-1} \psi_{\hat{\alpha}(x)_P, \hat{\tau}(x)_P}$$

Where ξ is simply a vector that picks out the $\tau(x)$ coordinate from the $\big(\alpha(x), \tau(x)\big)$ vector.

(d) Finally, use the peudo-outcome in a standard classification and decision tree (CART), which just selects the split that maximizes the sum of the average influence functions between the resulting child nodes:

$$\widetilde{\Delta}(C_1, C_2) = \sum_{j=1}^{2} \frac{1}{|\{i : X_i \in C_j\}|} \Big(\sum_{|\{i: X_i \in C_j\}|} \rho_i \Big)^2$$

Intuitively, the influence function captures how much the treatment effect in the child node is likely to differ from the treatment effect estimate in the parent node. In the simplest case of a randomized experiment where the above process refers to OLS-based identification (without a constant, for simplicity), this first-order approximation looks like:

$$\hat{\tau}_{C_j} \approx \hat{\tau}_P - \frac{1}{\sum_{i \in C_j} W_i^2} \sum_{i \in C_j} W_i (Y_i - W_i \hat{\tau}_P)$$

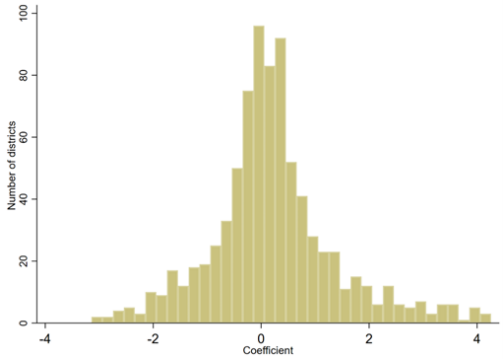
(e) With the two child nodes, repeat steps (a) – (d) on each, treating each node as a new parent node until a full tree, $\mathcal{T}$, is trained.

4. Within tree $\mathcal{T}$, collect all observations in the estimation subsample $\mathcal{J}_2$ that fall in the same leaf as x, $L_b(x)$. Compute the GRF-weights as the share of total trees where unit i falls into the same leaf as x:

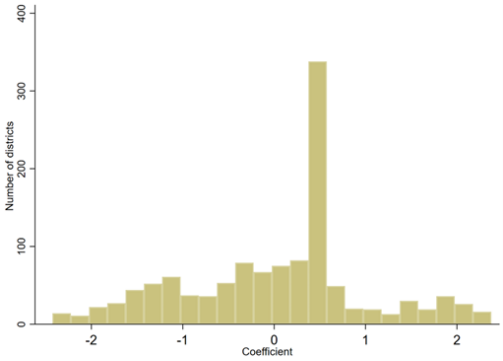
$$\omega_i = \frac{1}{B} \sum_{b=1}^{B} \frac{1}{|L_b(x)|} \mathbb{1}\{X_i \in L_b(x)\}$$

C Appendix: Tables and Figures

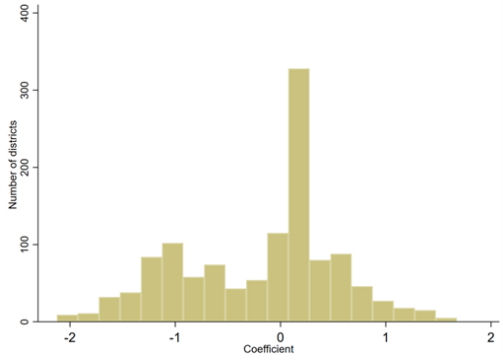
C.1 GDD Result Tables and Figures



(a) Graduation Rate



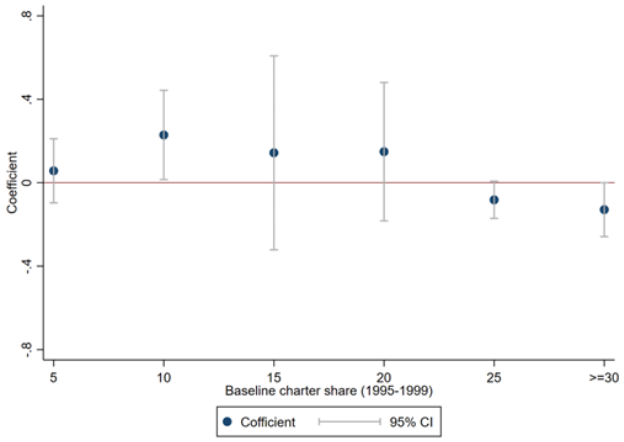
(b) Math Achievement



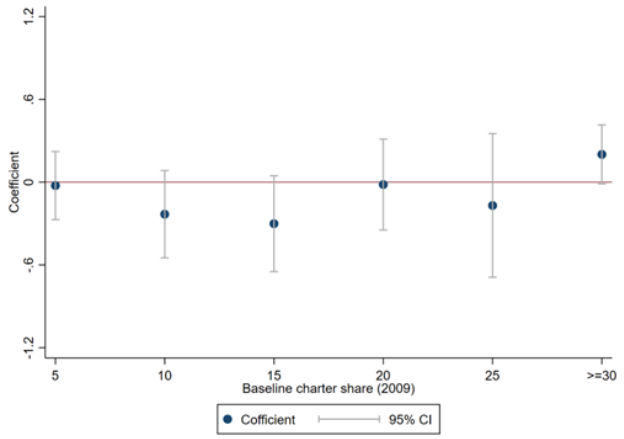
(c) ELA Achievement

Figure 1: Histogram Plots of Coefficient Distributions

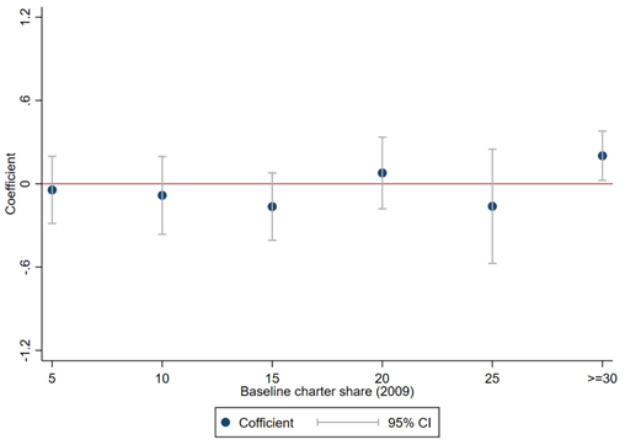
Notes: This figure presents histogram plots of estimates for the charter effects for each treated district. Specifically, for each treated district, we use the non-charter districts in the same state as the comparison group to perform the GDD analysis. The coefficients below the 5th percentile and above the 95th percentile are not plotted here.



(a) Graduation Rate



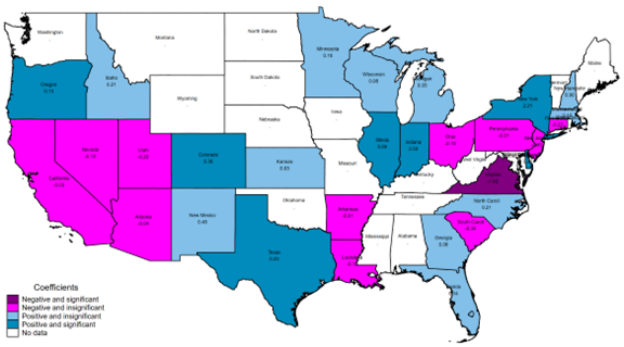
(b) Math Achievement



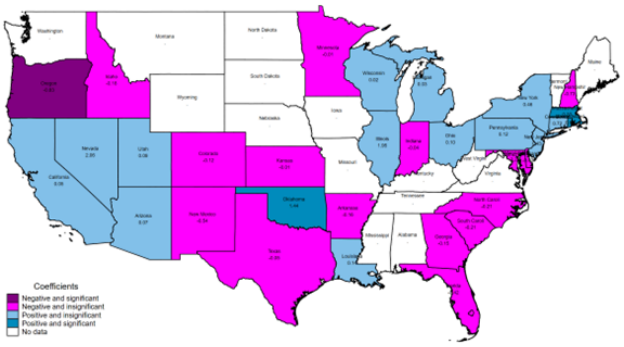
(c) ELA Achievement

Figure 2: Diminishing Returns

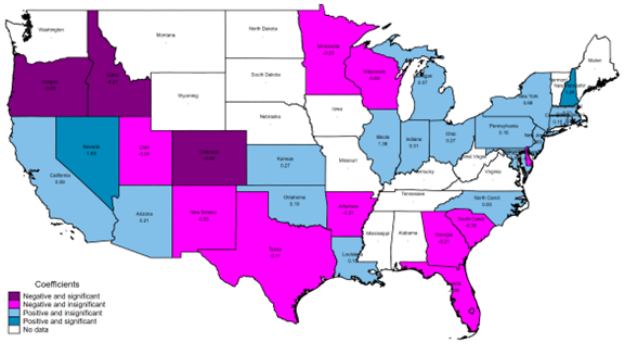
Notes: This figure presents the coefficient of the interaction of charter share with baseline charter share groups. The baseline charter share is the charter share in the first period of the sample, but for graduation sample, we use the max charter share of 1995-1999 as the baseline charter share as only a few districts have charter schools in 1995.



(a) Graduation Rate



(b) Math Achievement



(c) ELA Achievement

Figure 3: Charter Effects Map (include linear time trend)

Notes: This figure presents the effect map for full sample with district time trend included.

Table 1 Summary Statistics

Variable\Sample	Graduation rate			Test Score		
	Sample 1	Sample 2	Sample 3	Sample 1	Sample 2	Sample 3
Graduation rate	0.75	0.76	0.75	N.A.	N.A.	
Math	N.A.	N.A.	N.A.	0.00	0.01	0.18
ELA	N.A.	N.A.	N.A.	-0.01	0.01	0.21
Enrollment	64,837	45,567	52,310	59,901	38,322	11,357
White	0.63	0.64	0.64	0.52	0.55	0.64
Black	0.16	0.16	0.16	0.17	0.16	0.12
Hispanic	0.15	0.15	0.14	0.25	0.24	0.18
FRL	0.30	0.30	0.30	0.53	0.52	0.47
Special education	0.12	0.12	0.12	0.13	0.13	0.13
Urban	0.31	0.29	0.29	0.29	0.28	0.14
Suburb	0.38	0.37	0.38	0.43	0.40	0.43
Town	0.13	0.14	0.13	0.11	0.12	0.16
Rural	0.19	0.20	0.20	0.18	0.20	0.27
Revenue per student	9,105	9,067	9,104	12,636	12,807	13,097
Expenditure per student	9,279	9,214	9,274	12,631	12,781	13,034
Student-teacher ratio	16.87	16.86	16.75	16.77	16.80	16.24
Teacher salary	80,514	80,130	79,808	102,166	103,407	102,890
No. magnet school	1.26	1.30	0.72	4.84	3.72	0.61
Observation	135,735	132,514	135,272	464,436	451,298	414,181
N (district)	9,239	8,994	9,208	9,976	9,684	8,951

Notes: This table presents weighted means of outcome variables (graduation rate, math, and ELA) and control variables. graduation rate sample is weighted by the first-period enrollment. Sample (1) is the full sample, Sample (2) drops charter districts that coefficients above 95 percentile and below 5 percentile, and Sample (3) drops the always-charter districts. Data source: National Longitudinal School Database.

Table 2 Effects of charter entry on student outcomes

	Sample 1		Sample 2		Sample 3	
	(1)	(2)	(3)	(4)	(5)	(6)
	Panel A: Graduation rate					
Charter share	0.063*	0.058*	0.060*	0.053*	0.046	0.040
	[0.032]	[0.030]	[0.033]	[0.030]	[0.039]	[0.036]
R-squared	0.946	0.946	0.941	0.942	0.945	0.945
Observations	135,137	135,137	131,939	131,939	134,676	134,676
	Panel B: Math					
Charter share	0.040	0.041	0.009	0.014	0.123*	0.120*
	[0.052]	[0.052]	[0.054]	[0.053]	[0.064]	[0.065]
R-squared	0.888	0.889	0.887	0.887	0.865	0.865
Observations	464,436	464,436	451,298	451,298	414,181	414,181
	Panel C: ELA					
Charter share	0.036	0.037	0.004	0.008	0.107	0.101
	[0.045]	[0.044]	[0.048]	[0.048]	[0.068]	[0.069]
R-squared	0.912	0.912	0.911	0.911	0.891	0.891
Observations	464,436	464,436	451,298	451,298	414,181	414,181
District FE	Yes	Yes	Yes	Yes	Yes	Yes
State-by-year	Yes	Yes	Yes	Yes	Yes	Yes
District control	No	Yes	No	Yes	No	Yes
District trend	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table shows estimates of the effects of charter entry on student outcomes using different samples. Sample 1 is the full sample; Sample 2 drops charter districts that coefficients above 95 percentile and below 5 percentile, and Sample 3 drops the always-charter districts. Charter share in the data set range from 0 to 1. District control includes log of enrollment, share of black, white, Hispanic, FRL, special education, revenue per student, expenditure per student, student-teacher ratio, teacher salary, number of magnet schools, and an indicator of charter law effectiveness. Regressions are weighted by the first-period high school enrollment for graduation rate and grade-level enrollment for Math and ELA. Standard errors are clustered at the district level.

Table 3 - State-by-state treatment effects using various methods and comparing with CREDO

State	Sample 1		Sample 2		Sample 3		CREDO (2023)	
	Math	ELA	Math	ELA	Math	ELA	Math	ELA
AL	.	.	.	.	.	.	.	.
AK	0.02	0.53	0.2	0.02	.	.	.	.
AZ	0.07	0.21	0.11	0.18	0.13	0.18	0.18**	3.51**
AR	-0.16	-0.31	-0.30*	-0.63**	-0.16	-0.32	-0.18	0.53
CA	0.05	0.09	0.03	0.07	0.31***	0.20*	0.70**	1.93**
CO	-0.12	-0.69**	0.4	-0.52*	1.97***	1.32***	2.28*	2.63*
CT	0.72	0.16	2.17**	1.40***	.	.	.	.
DE	-0.53	-0.13	-1.72	-2.75**	.	.	.	.
DC	.	.	.	.	.	.	5.61**	0.88**
FL	-0.42	-0.06	-0.51	0.12	1.98***	0.93***	-0.18**	1.40**
GA	-0.15	-0.21	0.04	0.03	0.25	-0.63	.	.
HI	.	.	.	.	.	.	.	.
ID	-0.18	-0.21*	-0.25*	-0.19	0.87*	-0.34	1.40*	2.98*
IL	1.95	1.36	-0.27	0.04	.	.	8.42**	7.02**
IN	-0.04	0.31	0.37	0.12	-0.07	0.13	-4.74	0.53
IA	.	.	.	.	3.17***	10.69***	.	.
KS	-0.01	0.27	0.13	0.47***	.	.	.	.
KY	.	.	.	.	.	.	.	.
LA	0.14	0.18	0.13	0.25	-0.1	0.17	1.05	0.53
ME	.	.	.	.	.	.	.	.
MD	-2.28	1.48	.	.	.	.	6.49**	6.49**
MA	0.30*	0.21	0.17	0.15	.	.	7.19**	7.19**
MI	0.03	0.07	0.09	0.1	0.47	0.46	4.21**	6.32**
MN	-0.01	-0.2	-0.39	-0.44**	-0.44	-0.47	1.40**	3.68**
MS	.	.	.	.	.	.	9.82*	6.84*
MO	.	.	.	.	.	.	.	.
MT	.	.	.	.	.	.	.	.
NE	.	.	.	.	.	.	0.53	1.4
NV	2.06	1.63*	-1.37	0.11	.	.	.	.
NH	-0.73	1.24*	-1.00***	0.64	-0.99	0.57	.	.
NJ	0.43	0.09	0.64*	0.35	0.59***	0.72	5.61*	5.79*
NM	-0.54	-0.53	-0.27	0.16	-4.52**	-4.83***	1.23	1.93
NY	0.46	0.68	0.03	-0.04	.	.	12.81**	13.16**
NC	-0.21	0	-0.26	-0.05	-0.08	0.15	-2.98**	2.28**
ND	.	.	.	.	.	.	.	.
OH	0.1	0.27	0.13	0.16	-0.72**	-0.09	-6.67**	-0.70**
OK	1.44***	0.19	1.47***	0.2	.	.	.	.
OR	-0.83***	-0.68***	-0.85***	-0.64***	-0.29	-0.34*	-5.61*	-3.16*
PA	0.12	0.16	0.14	0.18	-1.12**	-0.74	-3.68	-0.18
RI	0.93**	0.69	0.81**	0.62	.	.	15.44**	15.79**
SC	-0.21	-0.36	-0.26	-0.43	-0.89	-0.46	-8.25**	-1.40**
SD	.	.	.	.	.	.	.	.
TN	.	.	.	.	.	.	6.84**	5.96**
TX	-0.05	-0.11	-0.23	-0.24*	-0.05	0.12	-0.88**	4.21**
UT	0.09	-0.04	-0.03	-0.16	-0.1	0.3	-2.46	-0.35
VT	.	.	.	.	.	.	.	.
VA	.	.	.	.	.	.	.	.
WA	.	.	.	.	.	.	6.84	4.56
WV	.	.	.	.	.	.	.	.
WI	0.02	-0.06	-0.01	-0.08	0.19	0.06	2.81	2.63
WY	.	.	.	.	.	.	.	.
Corr.	0.34	0.58	0.44	0.47	0.25	0.16	-	-

Notes: This table compares our estimates with that of CREDO (2023)'s Figure 1.10 and Figure 1.11. Corr is the correlation with CREDO (2023). The coefficients of CREDO (2023) are divided by 5.7 to adjust from learning days to effect size.

Table 4 Effect heterogeneity analysis (include district time trend)

	Graduation rate			Math			ELA		
	Sample 1	Sample 2	Sample 3	Sample 1	Sample 2	Sample 3	Sample 1	Sample 2	Sample 3
Charter*Black	-0.035	-0.032	-0.042	0.584	0.771**	0.748*	0.533*	0.477	0.766
	[0.160]	[0.149]	[0.162]	[0.372]	[0.352]	[0.446]	[0.299]	[0.314]	[0.500]
Charter*Hispanic	-0.053	-0.095	-0.063	0.381	-0.002	0.608	0.319	0.008	0.309
	[0.234]	[0.228]	[0.257]	[0.546]	[0.514]	[0.399]	[0.325]	[0.361]	[0.504]
Charter*FRL	0.431**	0.430**	0.391*	1.102**	1.102**	0.255	0.340	0.452	0.239
	[0.209]	[0.199]	[0.232]	[0.452]	[0.474]	[0.440]	[0.323]	[0.355]	[0.475]
Charter*Performance	0.364**	0.354**	0.323**	1.049***	0.954***	0.455***	0.515***	0.472***	0.090
	[0.160]	[0.151]	[0.162]	[0.109]	[0.109]	[0.128]	[0.073]	[0.068]	[0.117]
Charter* Nocaps	0.196	0.220	0.204	-0.045	0.118	0.655	-0.059	0.026	0.929
	[0.289]	[0.288]	[0.298]	[0.318]	[0.365]	[0.555]	[0.255]	[0.306]	[0.624]
Charter* Transpar	-0.102	-0.127	-0.060	-0.798*	-0.816*	-0.701	-0.678*	-0.380	-0.540
	[0.261]	[0.257]	[0.291]	[0.480]	[0.495]	[0.593]	[0.403]	[0.425]	[0.513]
Charter* Perf	0.334	0.343	0.304	0.162	-0.444	-0.411	-0.154	-0.400	-0.513
	[0.227]	[0.219]	[0.331]	[0.478]	[0.416]	[0.545]	[0.320]	[0.327]	[0.571]
Charter* Non_renew	0.466	0.439	0.421	0.121	-0.238	-0.486	-0.065	-0.332	-1.150**
	[0.331]	[0.324]	[0.326]	[0.609]	[0.557]	[0.560]	[0.440]	[0.476]	[0.585]
Charter*Exam	-0.151	-0.121	-0.122	0.201	0.219	-0.374	0.247	0.311	-0.124
	[0.121]	[0.114]	[0.178]	[0.350]	[0.358]	[0.402]	[0.269]	[0.289]	[0.439]
Charter* Eq_funding	0.025	0.040	-0.055	0.358	0.390	0.534**	0.171	0.270	0.480*
	[0.138]	[0.136]	[0.172]	[0.246]	[0.249]	[0.255]	[0.173]	[0.182]	[0.277]
Charter	-0.822**	-0.836**	-0.760**	-1.003*	-0.548	-0.055	-0.170	-0.184	0.131
	[0.366]	[0.361]	[0.382]	[0.521]	[0.481]	[0.547]	[0.371]	[0.400]	[0.534]
R-squared	0.945	0.940	0.943	0.885	0.884	0.858	0.909	0.908	0.886
Observations	155,723	152,445	155,247	525,502	511,874	474,731	525,502	511,874	474,731
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
District Trend	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State by year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
District control	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table presents the effect heterogeneity analysis. *Nocaps* refers to No Caps on the growth of charter schools in a state; $Transpar$ refers to Transparent Charter Application, Review, and Decision-making Processes; $Perf$ refers to Performance-based Charter School Contracts Required; Non_renew refers to Clear Processes for Renewal, Nonrenewal, and Revocation Decisions; $Exam$ refers to Automatic Exemptions from Many State and District Laws and Regulations; $Eq_funding$ refers to Equitable Access to Capital Funding and Facilities. These NAPCS sub categories rangs from 0 to 1 and higher values indicate a better policy.

C.2 DiDCF Tables and Figures

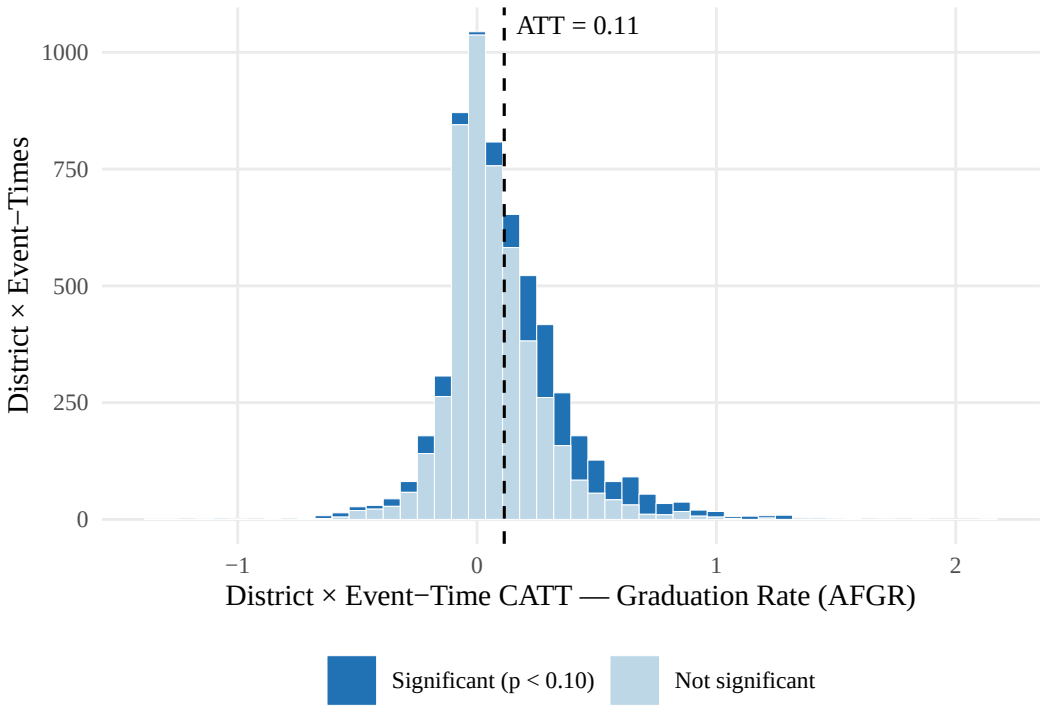


Figure 4: DiDCF CATT Distribution: Graduation Rate (AFGR)

Notes: Distribution of cohort × event-time Level-CATT estimates from the DiDCF applied to graduation rates (AFGR; 15 entry cohorts, 1995–2009). Bars colored by statistical significance ($p < 0.10$, two-sided); vertical line marks the weighted ATT.

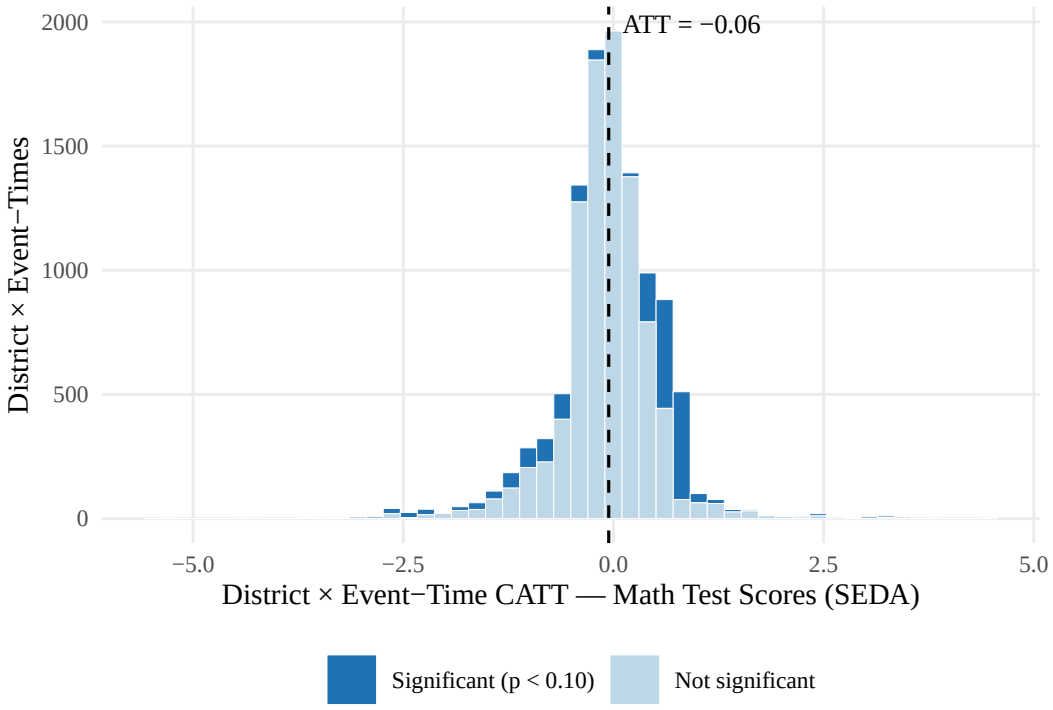


Figure 5: DiDCF CATT Distribution: Math Scores

Notes: Distribution of cohort × event-time Level-CATT estimates from the DiDCF applied to math test scores (10 entry cohorts, 2009–2018). Bars colored by statistical significance ($p < 0.10$, two-sided); vertical line marks the weighted ATT.

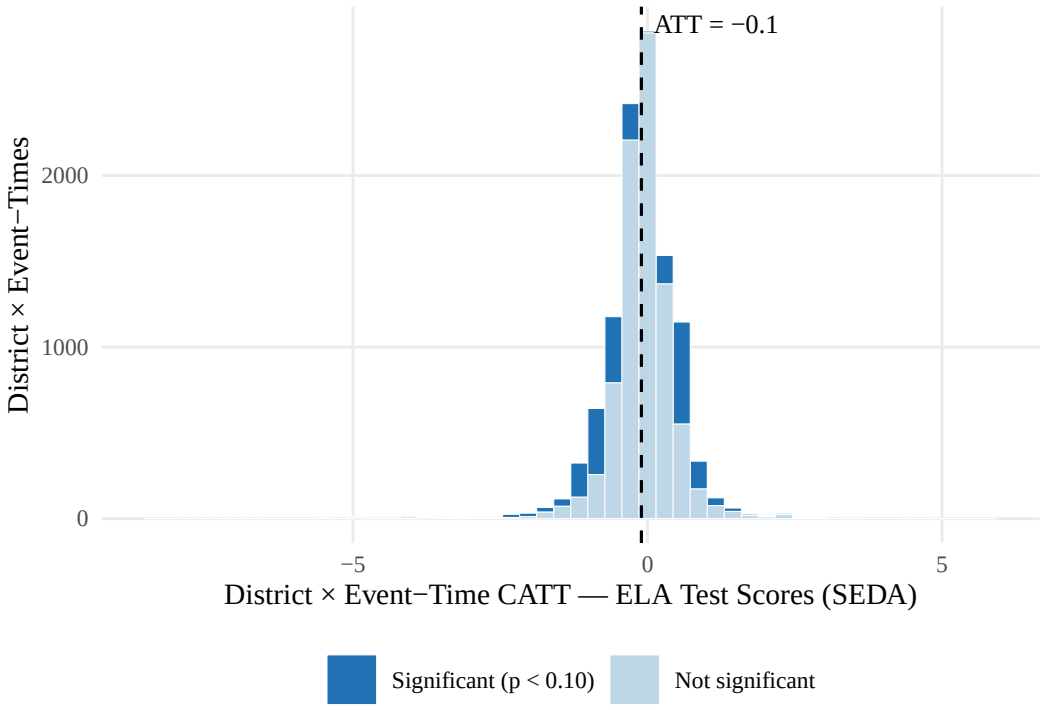


Figure 6: DiDCF CATT Distribution: ELA Scores

Notes: Distribution of cohort × event-time Level-CATT estimates from the DiDCF applied to ELA test scores (10 entry cohorts, 2009–2018). Bars colored by statistical significance ($p < 0.10$, two-sided); vertical line marks the weighted ATT.

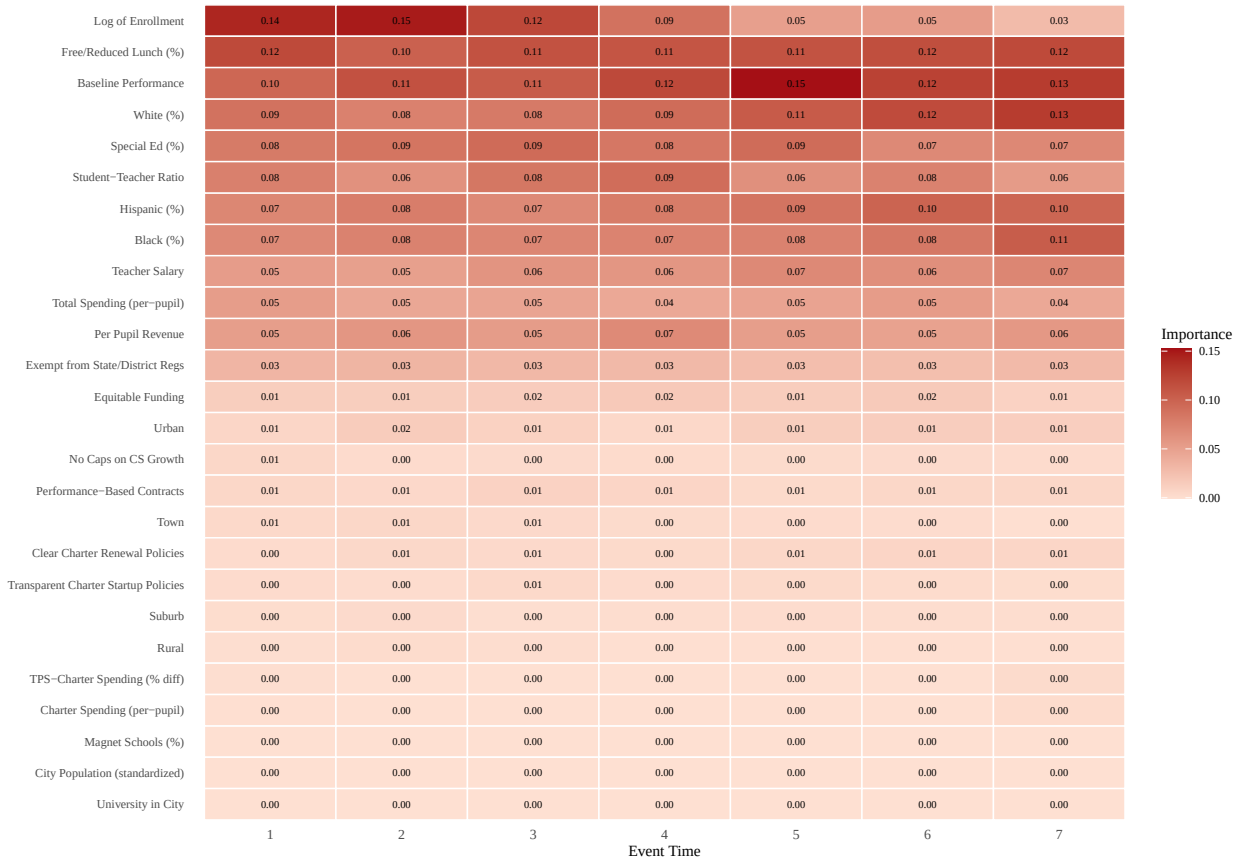


Figure 7: DiDCF Variable Importance Factors: Graduation Rate (AFGR)

Notes: Heatmap of variable importance factors (VIF) across all covariates and post-treatment event-times ($h = 1, \ldots, 7$) from the DiDCF applied to graduation rates. Cell values show the share of splits on each covariate at each event-time. Grey cells indicate covariates absent from the data.

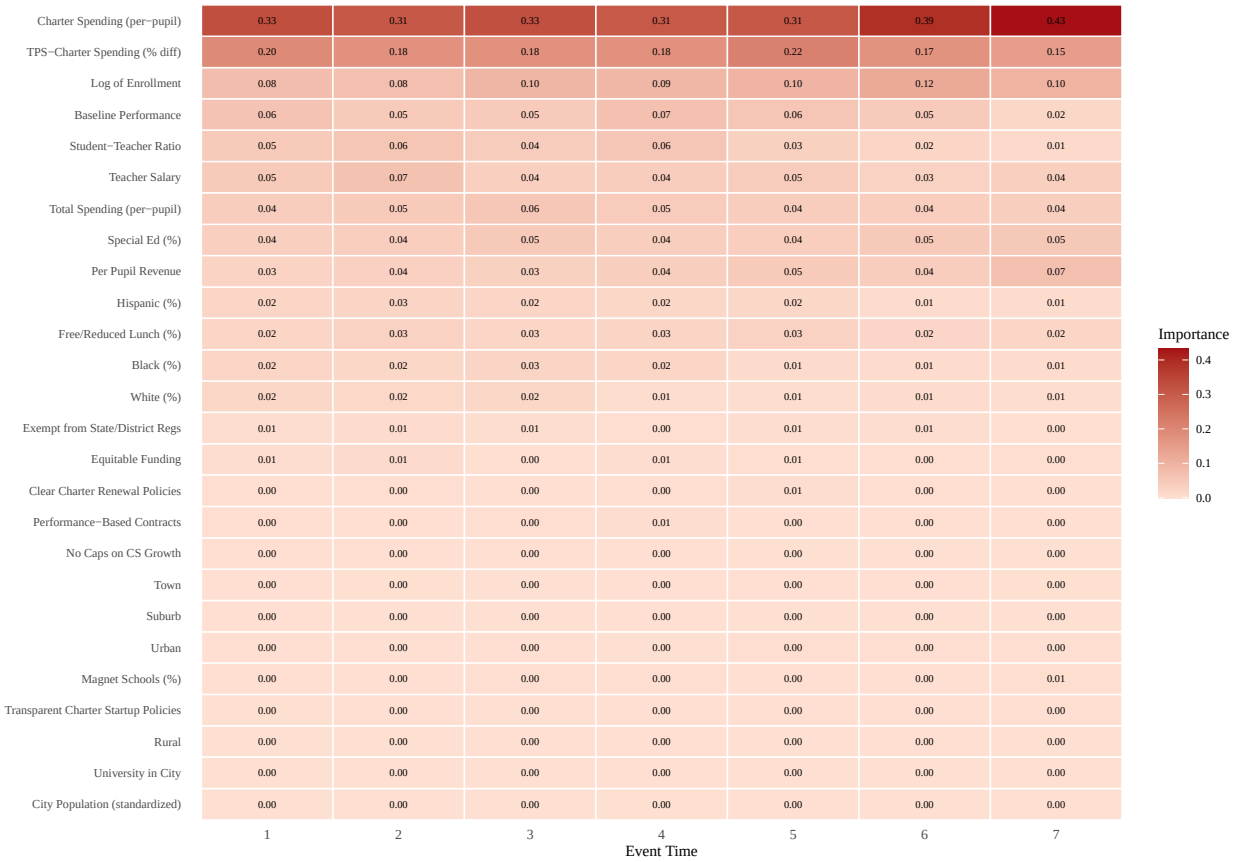


Figure 8: DiDCF Variable Importance Factors: Math Scores

Notes: Heatmap of variable importance factors (VIF) across all covariates and post-treatment event-times ($h = 1, \ldots, 7$) from the DiDCF applied to math test scores. Cell values show the share of splits on each covariate at each event-time. Grey cells indicate covariates absent from the data.

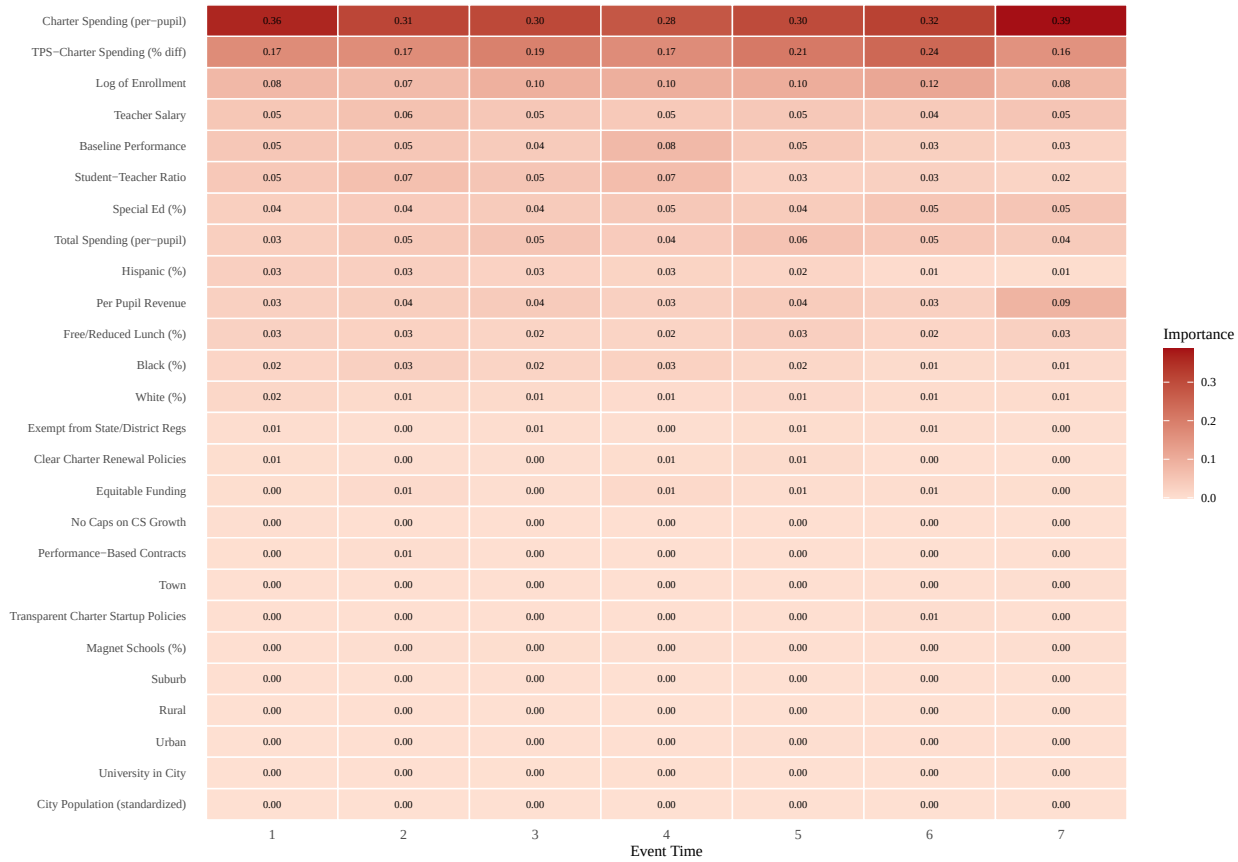


Figure 9: DiDCF Variable Importance Factors: ELA Scores

Notes: Heatmap of variable importance factors (VIF) across all covariates and post-treatment event-times ($h = 1, \ldots, 7$) from the DiDCF applied to ELA test scores. Cell values show the share of splits on each covariate at each event-time. Grey cells indicate covariates absent from the data.

Table 1: DiDCF Differences-in-Means: Graduation Rate (AFGR)

Variable	Mean (pos. CATT)	Mean (neg. CATT)	Difference	p-value
Log of Enrollment	8.81	8.44	0.36	0.00
Free/Reduced Lunch (%)	0.23	0.24	-0.01	0.71
White (%)	0.70	0.75	-0.05	0.02
Special Ed (%)	0.11	0.11	0.00	0.91
Student-Teacher Ratio	17.96	18.11	-0.15	0.56
Hispanic (%)	0.12	0.10	0.03	0.05
Black (%)	0.12	0.11	0.02	0.29
Teacher Salary	62251.95	62963.95	-712.00	0.53
Total Spending (per-pupil)	6323.11	6197.47	125.64	0.29
Per Pupil Revenue	6326.44	6225.70	100.73	0.41
Exempt from State/District Regs	0.62	0.61	0.01	0.56
Equitable Funding	0.48	0.51	-0.04	0.16
Urban	0.27	0.22	0.05	0.17
No Caps on CS Growth	0.79	0.77	0.02	0.11
Performance-Based Contracts	0.52	0.51	0.01	0.38
Town	0.21	0.20	0.01	0.80
Clear Charter Renewal Policies	0.62	0.62	-0.01	0.70
Transparent Charter Startup Policies	0.51	0.51	0.00	0.75
Suburb	0.31	0.29	0.02	0.53
Rural	0.21	0.29	-0.08	0.03
TPS-Charter Spending (% diff)	0.11	0.09	0.01	0.45
Charter Spending (per-pupil)	5761.73	5220.59	541.13	0.24
Magnet Schools (%)	0.00	0.00	0.00	NaN
City Population (standardized)	0.01	0.00	0.01	0.66
University in City	-0.01	0.00	-0.01	0.44

Notes: For each moderator, the table reports the mean value among districts with positive CATT estimates (Mean^{+}), the mean among districts with negative CATT estimates (Mean^{-}), their difference, and the associated p-value of the difference. Rows sorted by variable importance factor.

Table 2: DiDCF Differences-in-Means: Math Scores

Variable	Mean (pos. CATT)	Mean (neg. CATT)	Difference	p-value
Charter Spending (per-pupil)	7086.51	5548.66	1537.85	0.00
TPS-Charter Spending (% diff)	0.08	0.12	-0.04	0.00
Log of Enrollment	8.48	8.53	-0.05	0.33
Student-Teacher Ratio	16.74	16.51	0.24	0.10
Teacher Salary	96462.04	95910.79	551.25	0.54
Total Spending (per-pupil)	11859.84	12199.01	-339.17	0.03
Special Ed (%)	0.13	0.13	0.00	0.15
Per Pupil Revenue	11768.45	11989.21	-220.76	0.13
Hispanic (%)	0.23	0.19	0.04	0.00
Free/Reduced Lunch (%)	0.51	0.48	0.03	0.00
Black (%)	0.16	0.15	0.01	0.26
White (%)	0.55	0.60	-0.05	0.00
Exempt from State/District Regs	0.64	0.63	0.00	0.71
Equitable Funding	0.54	0.53	0.01	0.62
Clear Charter Renewal Policies	0.63	0.64	-0.01	0.31
Performance-Based Contracts	0.52	0.54	-0.02	0.02
No Caps on CS Growth	0.82	0.81	0.01	0.40
Town	0.23	0.19	0.04	0.03
Suburb	0.36	0.35	0.01	0.64
Urban	0.16	0.19	-0.03	0.05
Magnet Schools (%)	0.26	0.40	-0.14	0.06
Transparent Charter Startup Policies	0.52	0.54	-0.02	0.01
Rural	0.24	0.26	-0.02	0.37
University in City	0.01	0.01	0.00	0.98
City Population (standardized)	0.01	0.00	0.01	0.15

Notes: For each moderator, the table reports the mean value among districts with positive CATT estimates (Mean^{+}), the mean among districts with negative CATT estimates (Mean^{-}), their difference, and the associated p-value of the difference. Rows sorted by variable importance factor.

Table 3: DiDCF Differences-in-Means: ELA Scores

Variable	Mean (pos. CATT)	Mean (neg. CATT)	Difference	p-value
Charter Spending (per-pupil)	6482.16	6082.63	399.53	0.10
TPS-Charter Spending (% diff)	0.12	0.09	0.03	0.01
Log of Enrollment	8.48	8.53	-0.05	0.35
Teacher Salary	95071.27	97027.67	-1956.40	0.03
Student-Teacher Ratio	16.31	16.86	-0.55	0.00
Special Ed (%)	0.13	0.13	0.00	0.97
Total Spending (per-pupil)	12053.32	12033.79	19.53	0.90
Hispanic (%)	0.22	0.21	0.01	0.56
Per Pupil Revenue	11955.85	11833.22	122.62	0.41
Free/Reduced Lunch (%)	0.51	0.49	0.02	0.01
Black (%)	0.17	0.15	0.02	0.03
White (%)	0.57	0.59	-0.02	0.08
Exempt from State/District Regs	0.62	0.65	-0.03	0.00
Clear Charter Renewal Policies	0.63	0.64	0.00	0.55
Equitable Funding	0.51	0.56	-0.04	0.00
No Caps on CS Growth	0.81	0.81	0.00	0.95
Performance-Based Contracts	0.53	0.53	0.00	0.95
Town	0.22	0.20	0.02	0.35
Transparent Charter Startup Policies	0.53	0.54	-0.01	0.32
Magnet Schools (%)	0.27	0.39	-0.12	0.10
Suburb	0.37	0.35	0.02	0.23
Rural	0.25	0.25	0.00	1.00
Urban	0.16	0.20	-0.04	0.01
University in City	0.00	0.01	-0.01	0.28
City Population (standardized)	0.00	0.00	0.00	0.53

Notes: For each moderator, the table reports the mean value among districts with positive CATT estimates (Mean^+), the mean among districts with negative CATT estimates (Mean^-), their difference, and the associated p-value of the difference. Rows sorted by variable importance factor.

Table 4: DiDCF Best Linear Predictor: Graduation Rate (AFGR)

	Estimate	Std. Error	p-value
(Intercept)	-0.15	0.08	0.06
Hispanic (%)	0.03	0.05	0.59
Free/Reduced Lunch (%)	-0.02	0.03	0.64
White (%)	-0.06	0.04	0.14
Special Ed (%)	0.04	0.08	0.62
Log of Enrollment	0.02	0.01	0.00
Student-Teacher Ratio	0.00	0.00	0.84

Notes: Best linear predictor (BLP) regression of cohort-weighted CATT estimates on district moderators, applied to graduation rates. Each row reports a separate OLS coefficient from regressing the mean post-treatment CATT on the listed covariate. Estimates above zero indicate that districts with higher values of the covariate experience larger positive effects of charter entry.

Table 5: DiDCF Best Linear Predictor: Math Scores

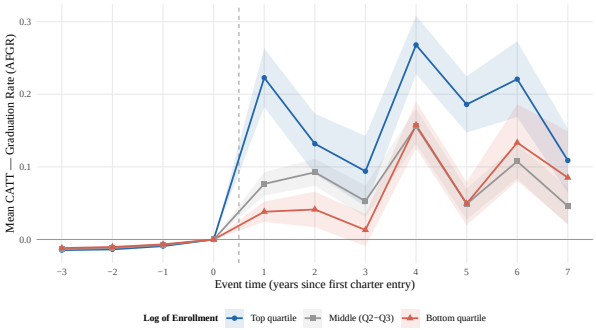
	Estimate	Std. Error	p-value
(Intercept)	-0.07	0.19	0.73
Log of Enrollment	-0.02	0.01	0.05
Total Spending (per-pupil)	0.00	0.00	0.10
Student-Teacher Ratio	0.00	0.01	0.63
Teacher Salary	0.00	0.00	0.54
TPS-Charter Spending (% diff)	0.03	0.04	0.48
Charter Spending (per-pupil)	0.00	0.00	0.00

Notes: Best linear predictor (BLP) regression of cohort-weighted CATT estimates on district moderators, applied to math test scores. Each row reports a separate OLS coefficient from regressing the mean post-treatment CATT on the listed covariate.

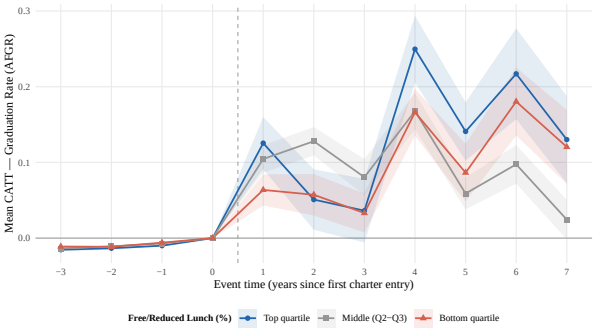
Table 6: DiDCF Best Linear Predictor: ELA Scores

	Estimate	Std. Error	p-value
(Intercept)	0.17	0.20	0.40
Special Ed (%)	-0.77	0.31	0.01
Log of Enrollment	-0.03	0.01	0.03
Student-Teacher Ratio	-0.01	0.01	0.08
Teacher Salary	0.00	0.00	0.26
TPS-Charter Spending (% diff)	0.08	0.04	0.04
Charter Spending (per-pupil)	0.00	0.00	0.00

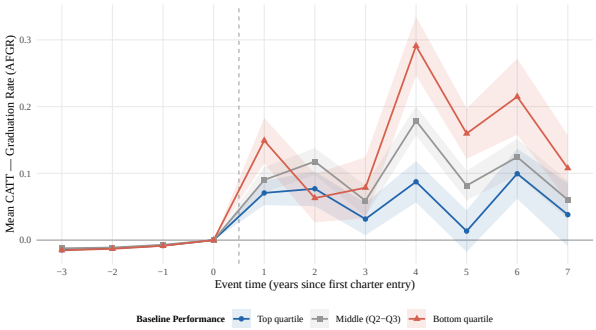
Notes: Best linear predictor (BLP) regression of cohort-weighted CATT estimates on district moderators, applied to ELA test scores. Each row reports a separate OLS coefficient from regressing the mean post-treatment CATT on the listed covariate.



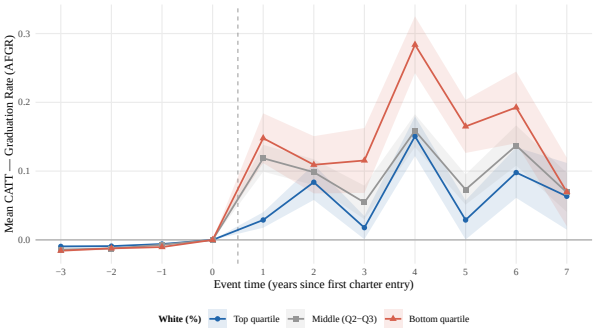
(a) Log Enrollment



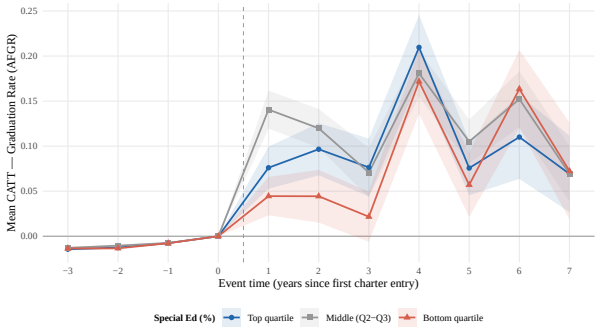
(b) Percent Free/Reduced Lunch



(c) Baseline Graduation Rate



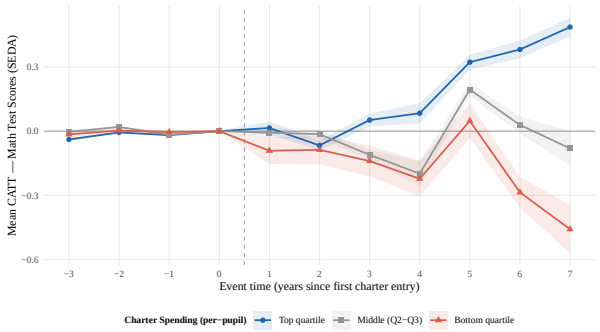
(d) Percent White



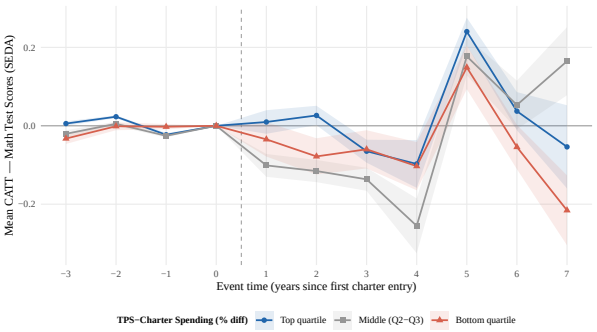
(e) Percent Special Education

Figure 10: DiDCF Quartile Event Studies: Graduation Rate (AFGR)

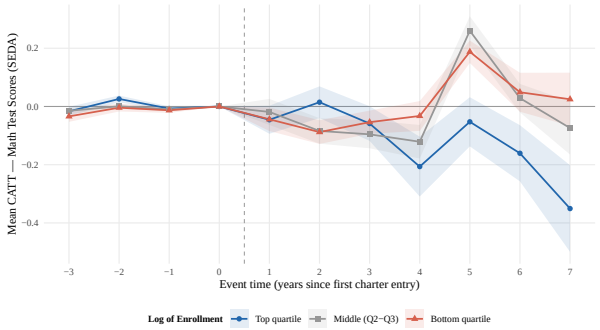
Notes: Each panel shows cohort-weighted ATT estimates by event-time, separately for districts in the top quartile (blue), middle two quartiles (grey), and bottom quartile (red) of the named moderator. Panels correspond to the five highest-ranked covariates by VIF at $h = 1$ for graduation rates.



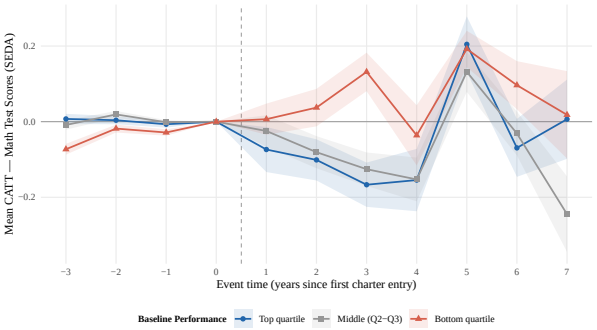
(a) Charter Spending Per-Pupil



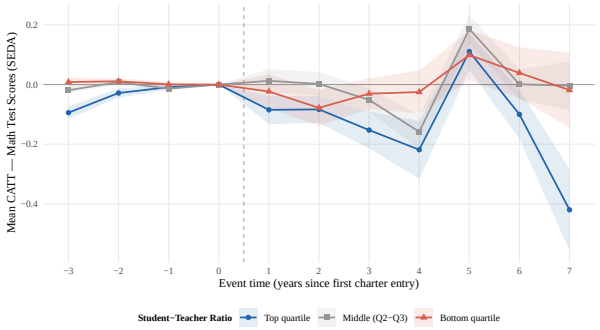
(b) TPS vs. Charter Spending Gap



(c) Log Enrollment



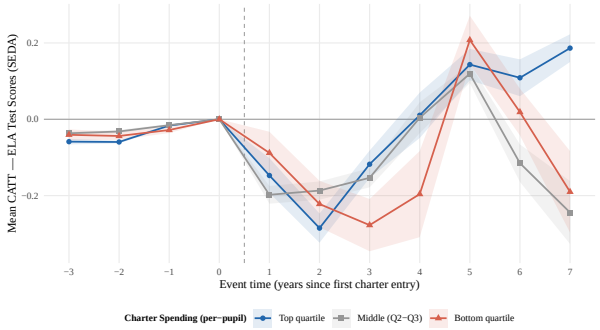
(d) Baseline Math Score



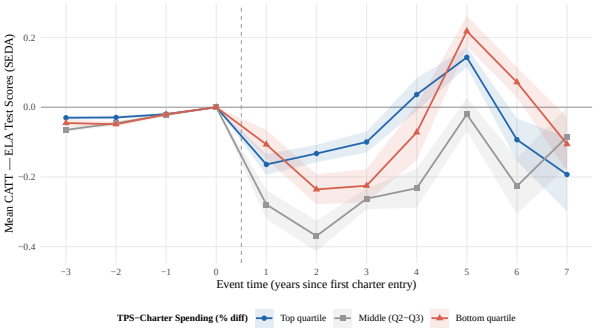
(e) Student-Teacher Ratio

Figure 11: DiDCF Quartile Event Studies: Math Scores

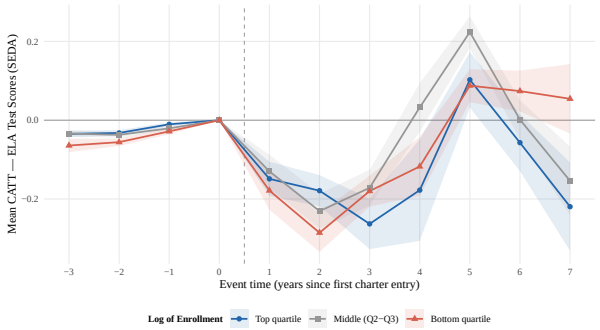
Notes: Each panel shows cohort-weighted ATT estimates by event-time, separately for districts in the top quartile (blue), middle two quartiles (grey), and bottom quartile (red) of the named moderator. Panels correspond to the five highest-ranked covariates by VIF at $h = 1$ for math scores.



(a) Charter Spending Per-Pupil



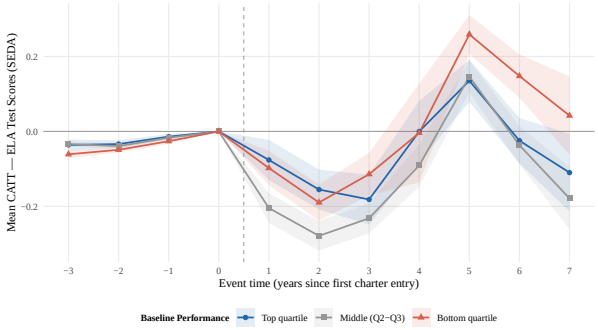
(b) TPS vs. Charter Spending Gap



(c) Log Enrollment



(d) Teacher Salary



(e) Baseline ELA Score

Figure 12: DiDCF Quartile Event Studies: ELA Scores

Notes: Each panel shows cohort-weighted ATT estimates by event-time, separately for districts in the top quartile (blue), middle two quartiles (grey), and bottom quartile (red) of the named moderator. Panels correspond to the five highest-ranked covariates by VIF at $h = 1$ for ELA scores.

Figures 13–18 display optimal policy trees estimated via the Athey and Wager (2021) framework applied to DiDCF doubly robust scores. Each outcome is shown at depth 2 (four terminal leaves) and depth 3 (eight terminal leaves). Blue leaves recommend expanding charter market share; red leaves recommend against expansion. Internal nodes display the splitting variable and threshold; "Yes" (condition satisfied, left branch) and "No" (condition not satisfied, right branch) labels accompany each edge.

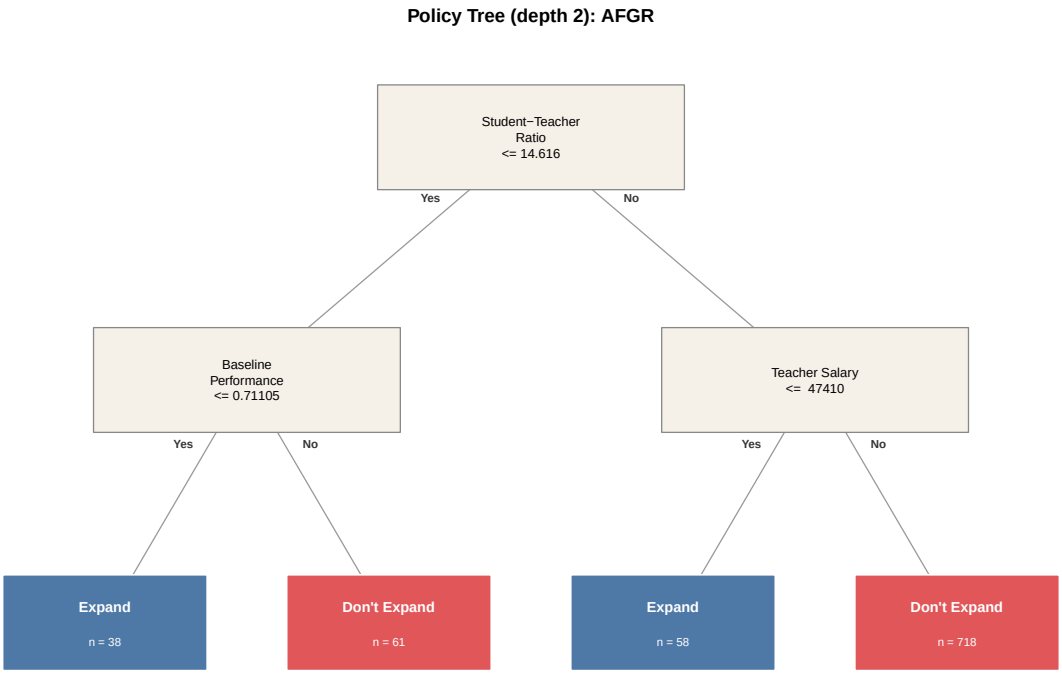


Figure 13: Optimal Policy Tree (Depth 2): Graduation Rate (AFGR)

Notes: Depth-2 optimal policy tree estimated via the Athey and Wager (2021) framework applied to DiDCF doubly robust scores for graduation rates (AFGR, 15 entry cohorts 1995–2009). Each internal node shows the split variable and threshold; leaf colors indicate the recommended policy action (blue = expand charter market share; red = do not expand). The sample consists of treated districts with non-missing DR scores averaged across post-treatment event times, restricted to the ten highest-VIF moderators at $h = 1$.

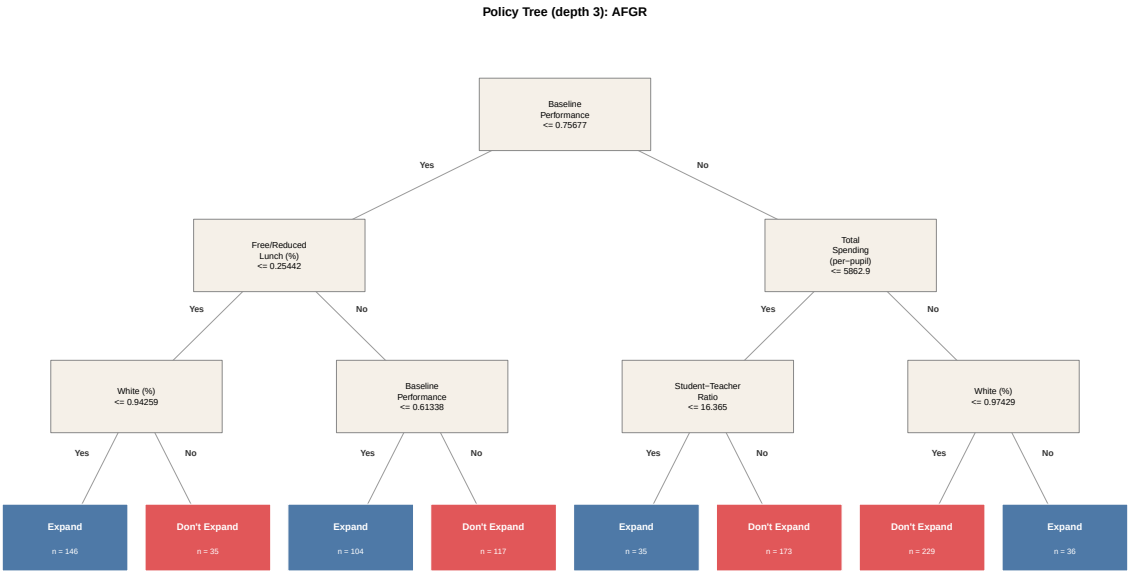


Figure 14: Optimal Policy Tree (Depth 3): Graduation Rate (AFGR)

Notes: Depth-3 optimal policy tree for graduation rates. See notes to fig. 13.

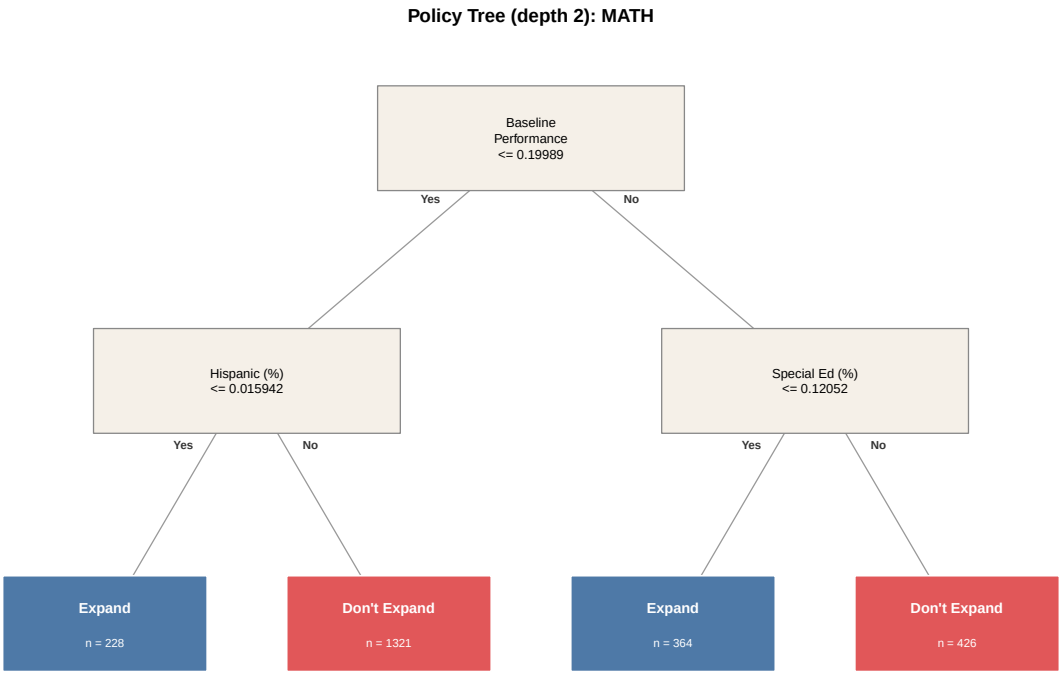


Figure 15: Optimal Policy Tree (Depth 2): Math Scores

Notes: Depth-2 optimal policy tree applied to DiDCF doubly robust scores for math test scores (10 entry cohorts, 2009–2018). See notes to fig. 13.

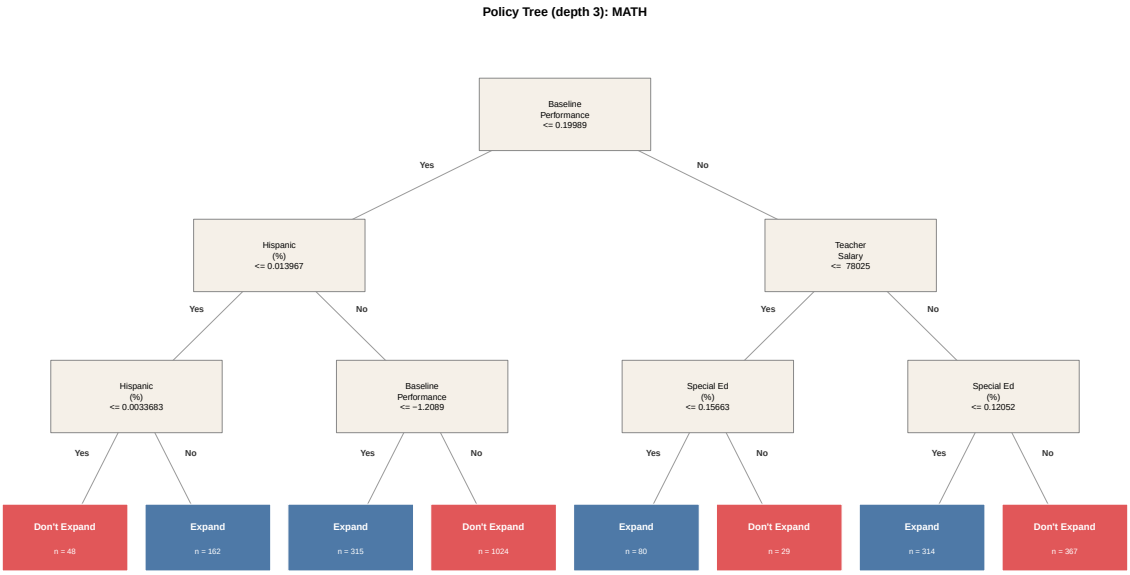


Figure 16: Optimal Policy Tree (Depth 3): Math Scores

Notes: Depth-3 optimal policy tree for math test scores. See notes to fig. 13.

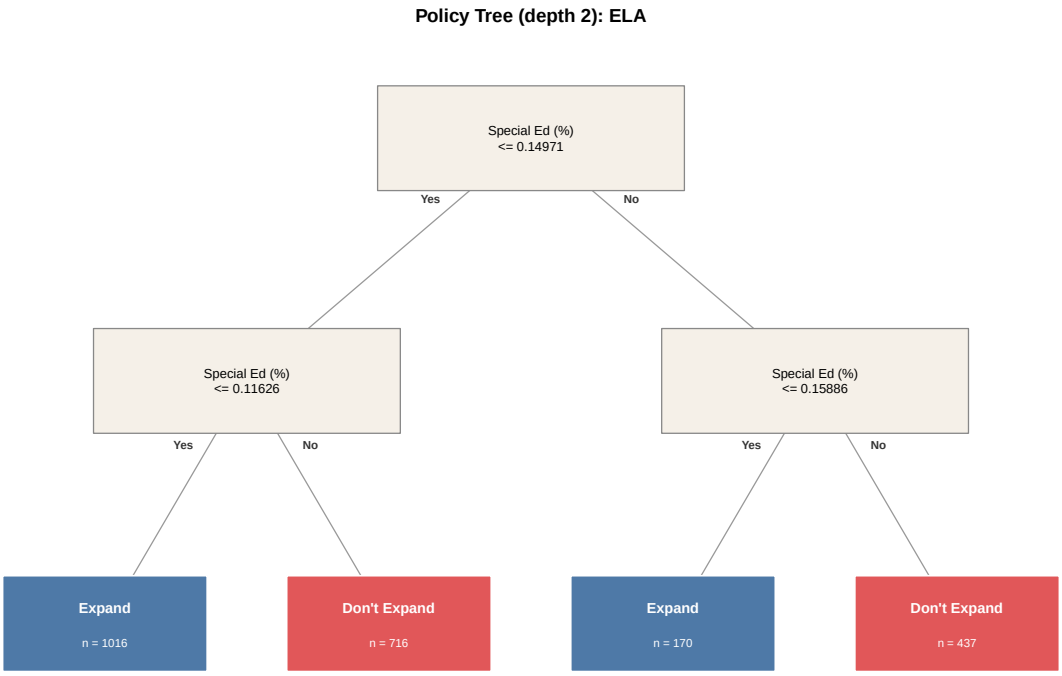


Figure 17: Optimal Policy Tree (Depth 2): ELA Scores

Notes: Depth-2 optimal policy tree applied to DiDCF doubly robust scores for ELA test scores (10 entry cohorts, 2009–2018). See notes to fig. 13.

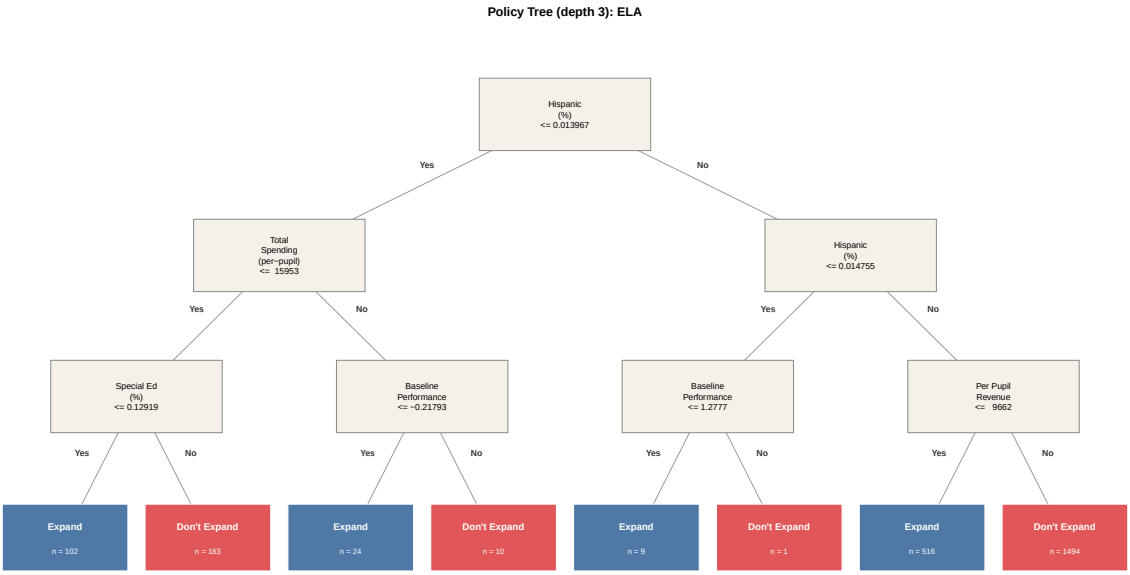


Figure 18: Optimal Policy Tree (Depth 3): ELA Scores

Notes: Depth-3 optimal policy tree for ELA test scores. See notes to fig. 13.

C.3 CFFE Results

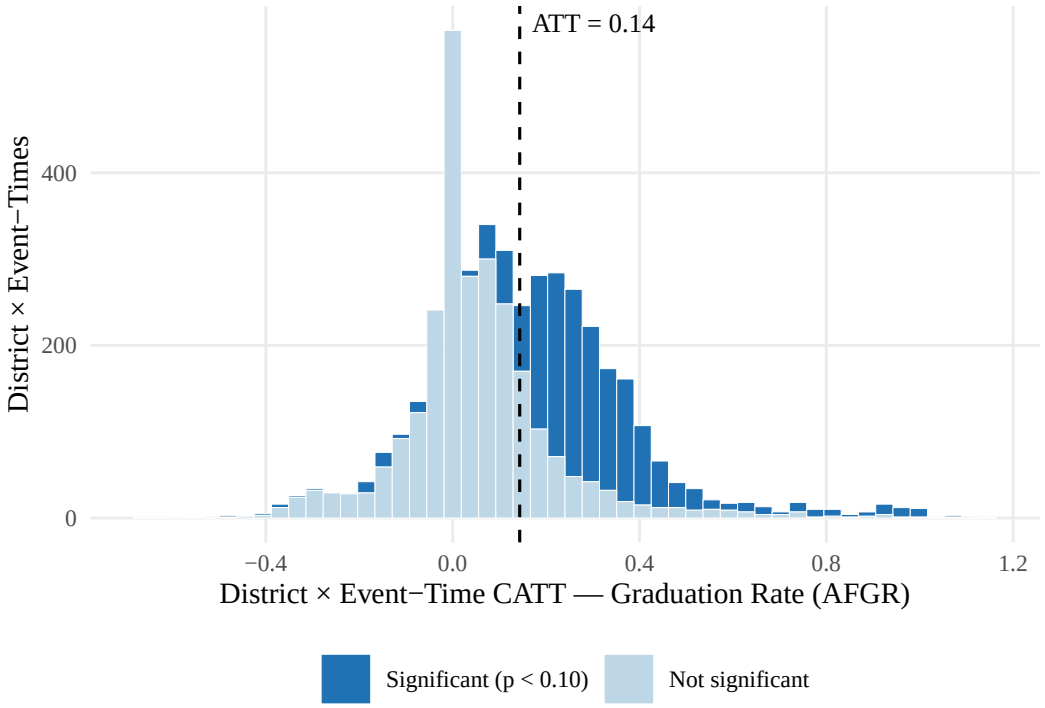


Figure 19: CFFE CATT Distribution: Graduation Rate (AFGR)

Notes: Distribution of Level-CATT estimates from the CFFE applied to graduation rates (15 entry cohorts, 1995–2009). Bars colored by statistical significance ($p < 0.10$, two-sided); vertical line marks the weighted ATT.

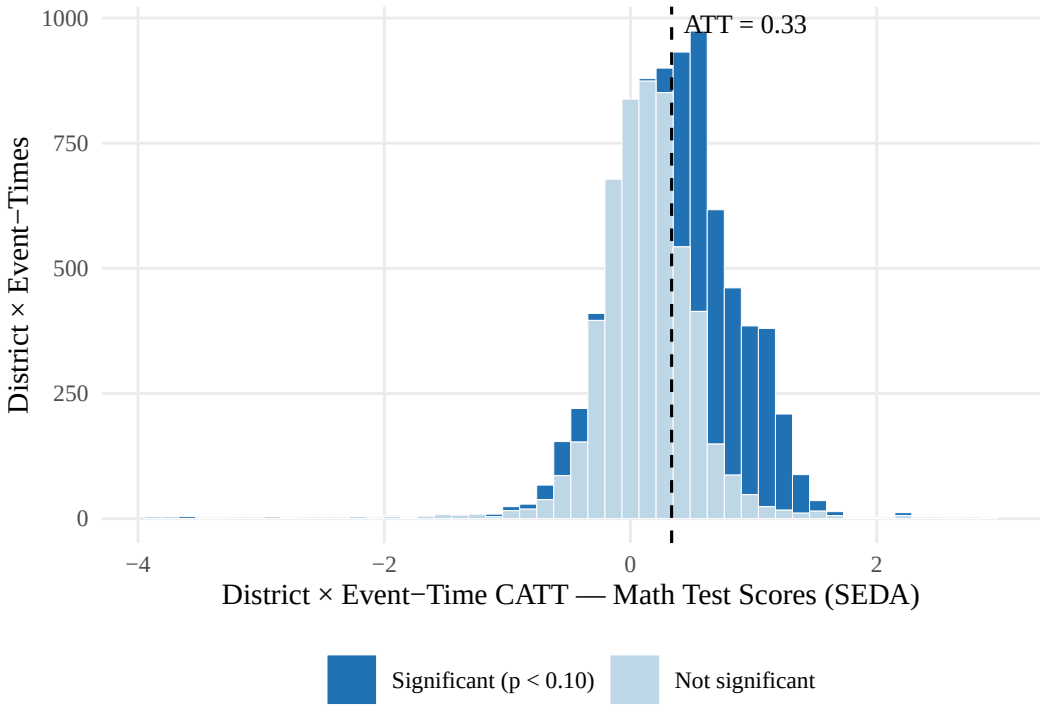


Figure 20: CFFE CATT Distribution: Math Scores

Notes: Distribution of Level-CATT estimates from the CFFE applied to math test scores (10 entry cohorts, 2009–2018). Bars colored by statistical significance ($p < 0.10$, two-sided); vertical line marks the weighted ATT.

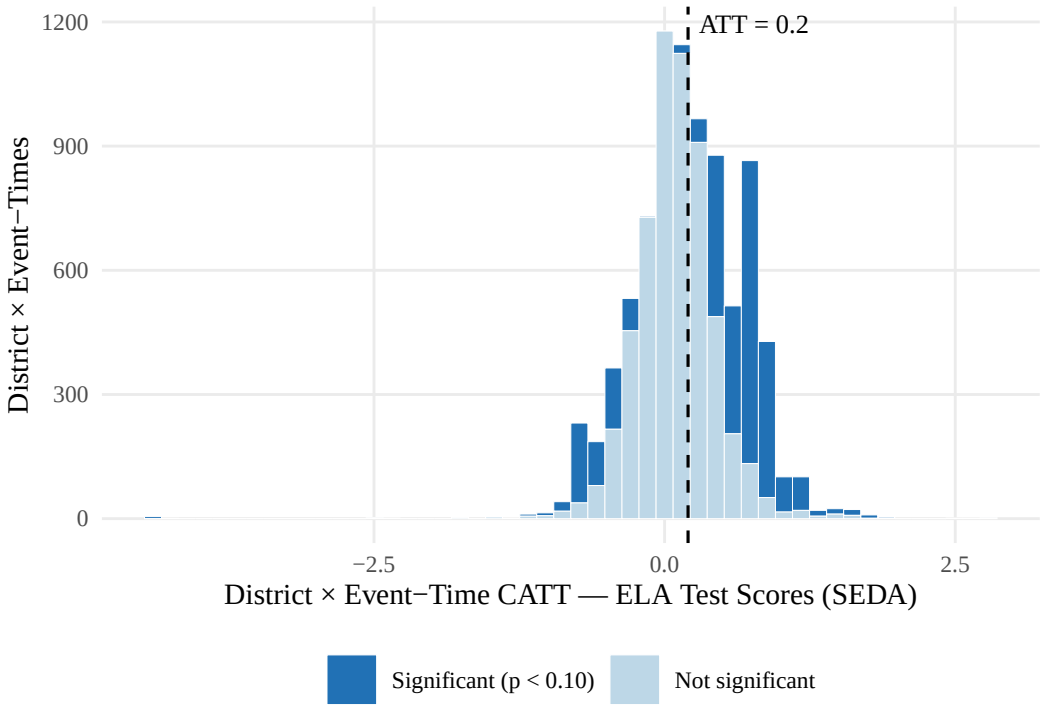


Figure 21: CFFE CATT Distribution: ELA Scores

Notes: Distribution of Level-CATT estimates from the CFFE applied to ELA test scores (10 entry cohorts, 2009–2018). Bars colored by statistical significance ($p < 0.10$, two-sided); vertical line marks the weighted ATT.

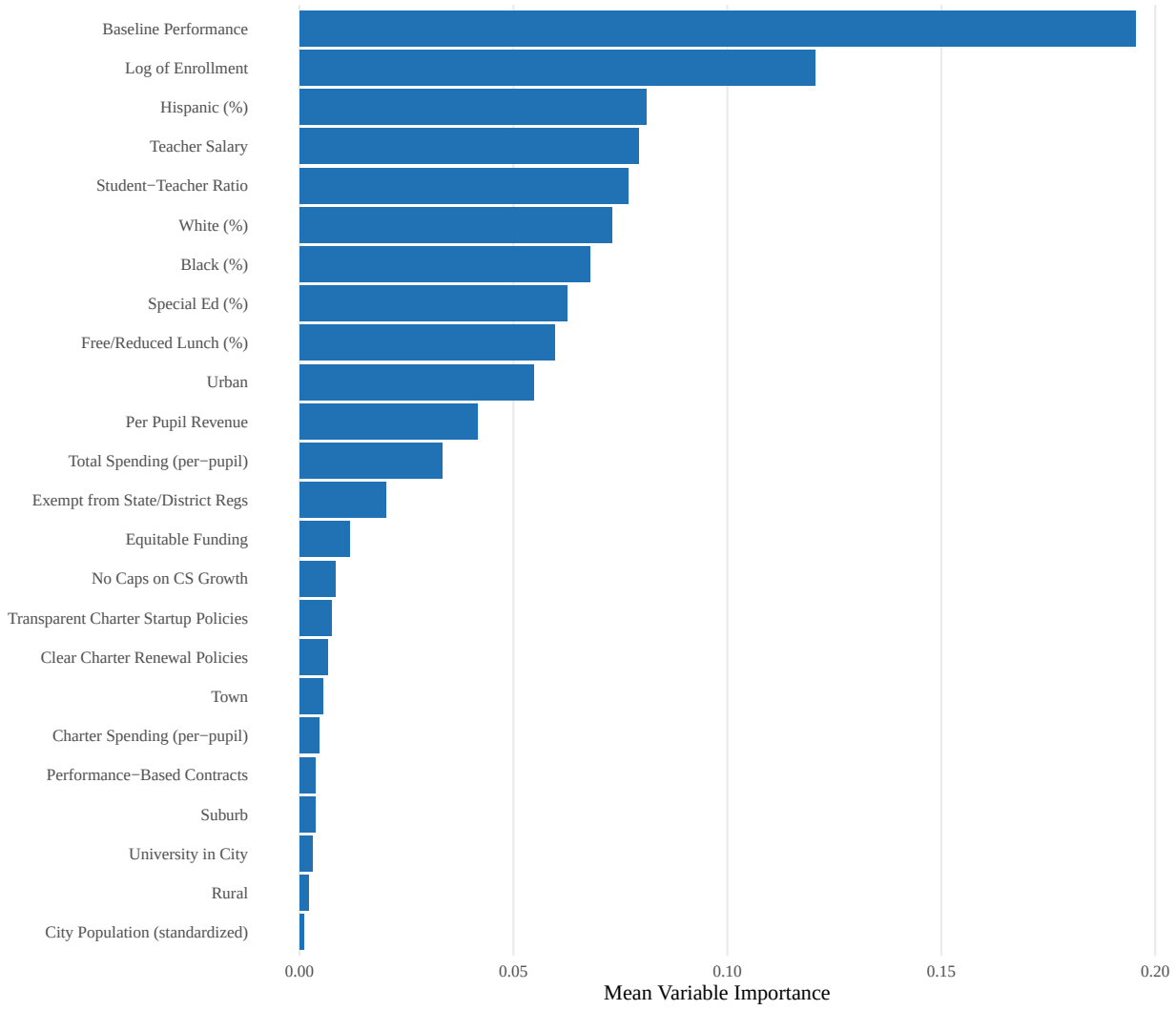


Figure 22: CFFE Variable Importance Factors: Graduation Rate (AFGR)

Notes: Bar chart of variable importance factors (VIF) averaged across all cohort-period forests ($h = 1, \ldots, 7$) from the CFFE applied to graduation rates. Bars show the mean share of splits on each covariate across all estimated models, sorted by importance.

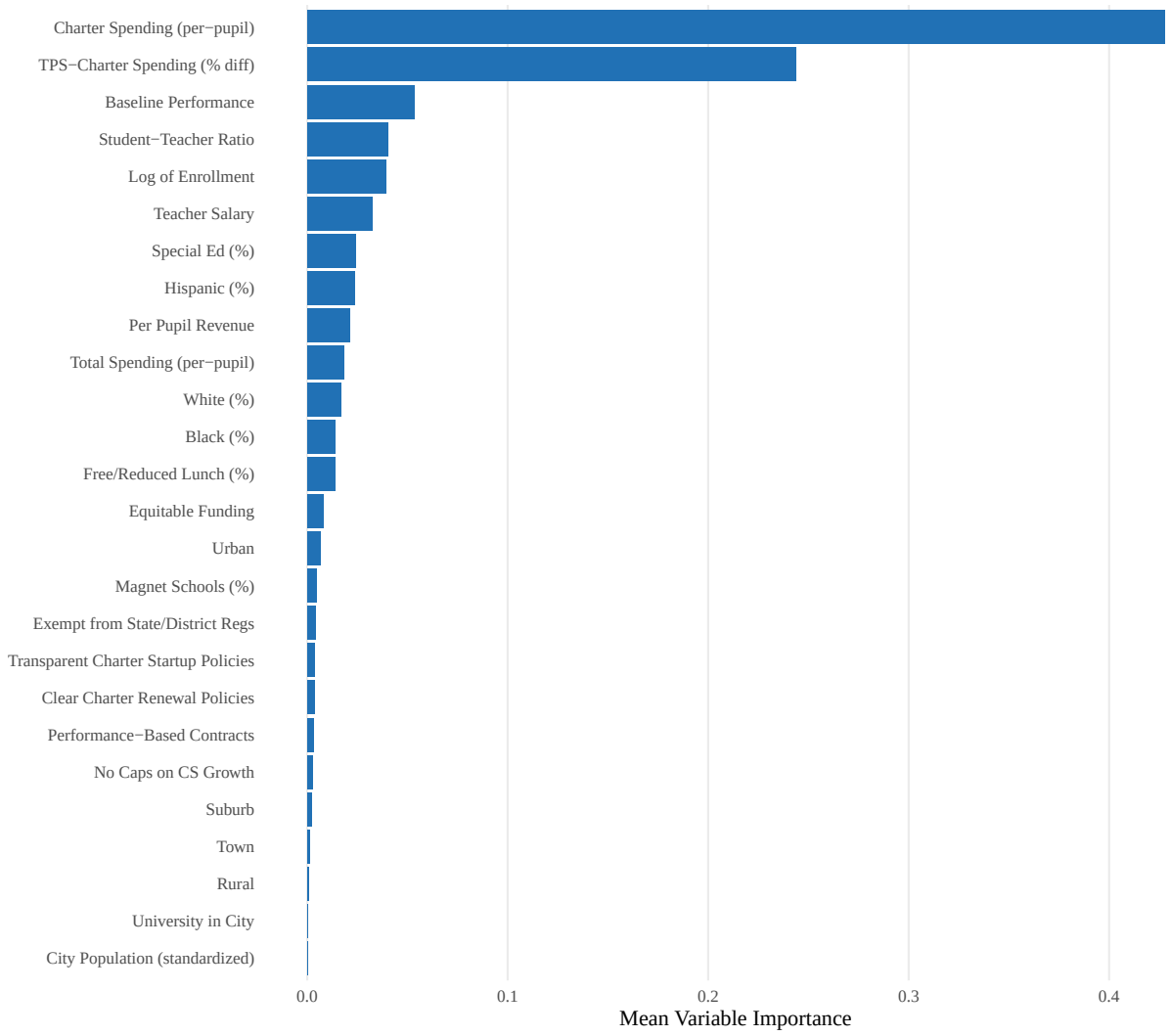


Figure 23: CFFE Variable Importance Factors: Math Scores

Notes: Bar chart of variable importance factors (VIF) averaged across all cohort-period forests from the CFFE applied to math test scores. Bars show the mean share of splits on each covariate across all estimated models, sorted by importance.

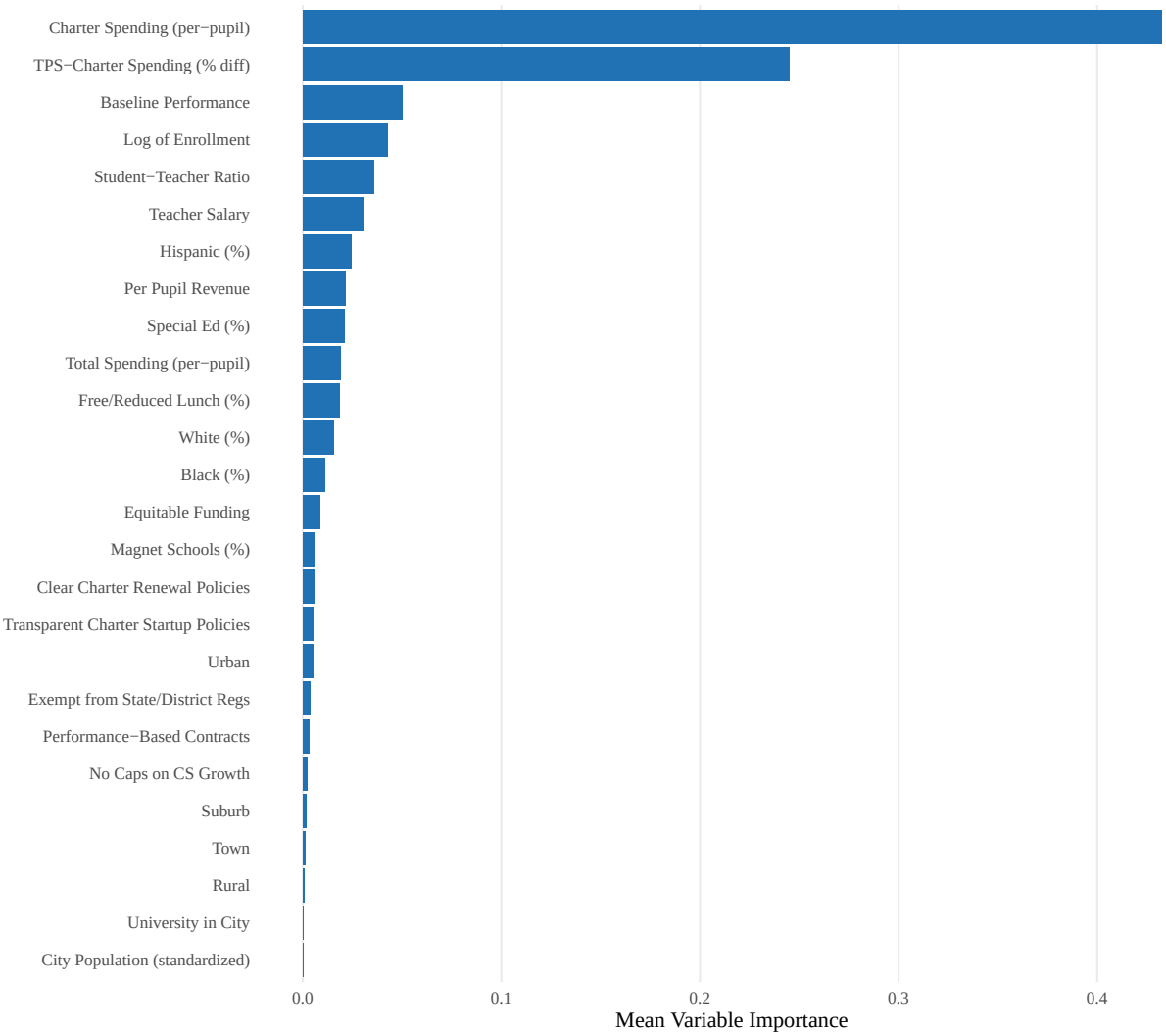


Figure 24: CFFE Variable Importance Factors: ELA Scores

Notes: Bar chart of variable importance factors (VIF) averaged across all cohort-period forests from the CFFE applied to ELA test scores. Bars show the mean share of splits on each covariate across all estimated models, sorted by importance.

Table 7: CFFE Differences-in-Means: Graduation Rate (AFGR)

Variable	Mean (pos. CATT)	Mean (neg. CATT)	Difference	p-value
Log of Enrollment	8.89	8.35	0.54	0.00
Student-Teacher Ratio	17.89	18.44	-0.56	0.02
Hispanic (%)	0.13	0.10	0.03	0.04
White (%)	0.69	0.76	-0.07	0.00
Special Ed (%)	0.11	0.10	0.01	0.04
Black (%)	0.13	0.09	0.04	0.00
Teacher Salary	62509.09	62447.71	61.38	0.95
Free/Reduced Lunch (%)	0.24	0.22	0.02	0.13
Urban	0.29	0.21	0.08	0.02
Per Pupil Revenue	6335.81	6180.64	155.17	0.23
Total Spending (per-pupil)	6334.06	6127.45	206.61	0.11
Exempt from State/District Regs	0.62	0.61	0.01	0.69
Equitable Funding	0.48	0.52	-0.03	0.18
No Caps on CS Growth	0.81	0.74	0.06	0.00
Clear Charter Renewal Policies	0.61	0.64	-0.03	0.04
Transparent Charter Startup Policies	0.51	0.51	0.00	0.85
Town	0.20	0.22	-0.02	0.47
Performance-Based Contracts	0.53	0.50	0.03	0.05
Suburb	0.31	0.28	0.03	0.41
University in City	-0.01	0.00	-0.01	0.24
Charter Spending (per-pupil)	5839.33	5146.45	692.88	0.13
Rural	0.20	0.29	-0.08	0.02
City Population (standardized)	0.00	0.01	-0.01	0.50
Magnet Schools (%)	0.00	0.00	0.00	NaN
TPS-Charter Spending (% diff)	0.11	0.09	0.02	0.26

Notes: For each moderator, the table reports the mean value among districts with positive CATT estimates (Mean^+), the mean among districts with negative CATT estimates (Mean^-), their difference, and the associated p-value of the difference. Rows sorted by variable importance factor.

Table 8: CFFE Differences-in-Means: Math Scores

Variable	Mean (pos. CATT)	Mean (neg. CATT)	Difference	p-value
Charter Spending (per-pupil)	7222.94	3124.30	4098.64	0.00
TPS-Charter Spending (% diff)	0.13	0.01	0.12	0.00
Student-Teacher Ratio	16.88	15.90	0.99	0.00
Teacher Salary	97124.32	94525.17	2599.15	0.03
Special Ed (%)	0.13	0.13	-0.01	0.00
Per Pupil Revenue	11883.99	12077.39	-193.40	0.37
Log of Enrollment	8.54	8.42	0.12	0.07
White (%)	0.57	0.62	-0.04	0.01
Hispanic (%)	0.23	0.16	0.07	0.00
Total Spending (per-pupil)	12025.03	12284.49	-259.46	0.26
Black (%)	0.14	0.18	-0.03	0.01
Free/Reduced Lunch (%)	0.50	0.48	0.02	0.15
Equitable Funding	0.54	0.48	0.06	0.00
Performance-Based Contracts	0.52	0.55	-0.03	0.00
Exempt from State/District Regs	0.63	0.67	-0.04	0.00
Urban	0.20	0.16	0.03	0.14
Transparent Charter Startup Policies	0.53	0.53	-0.01	0.54
Clear Charter Renewal Policies	0.64	0.62	0.02	0.10
Magnet Schools (%)	0.30	0.59	-0.30	0.09
No Caps on CS Growth	0.81	0.82	0.00	0.64
Suburb	0.36	0.32	0.04	0.09
Town	0.22	0.18	0.03	0.13
Rural	0.23	0.33	-0.11	0.00
University in City	0.01	0.00	0.01	0.06
City Population (standardized)	0.00	0.00	0.00	0.47

Notes: For each moderator, the table reports the mean value among districts with positive CATT estimates (Mean^+), the mean among districts with negative CATT estimates (Mean^-), their difference, and the associated p-value of the difference. Rows sorted by variable importance factor.

Table 9: CFFE Differences-in-Means: ELA Scores

Variable	Mean (pos. CATT)	Mean (neg. CATT)	Difference	p-value
Charter Spending (per-pupil)	6694.03	5783.86	910.17	0.00
TPS-Charter Spending (% diff)	0.11	0.07	0.04	0.00
Student-Teacher Ratio	17.01	15.84	1.17	0.00
Teacher Salary	98001.34	92871.58	5129.77	0.00
Per Pupil Revenue	11872.70	12051.74	-179.04	0.38
Log of Enrollment	8.54	8.46	0.09	0.14
Hispanic (%)	0.23	0.18	0.05	0.00
Special Ed (%)	0.13	0.13	0.00	0.01
White (%)	0.57	0.60	-0.02	0.11
Total Spending (per-pupil)	11955.79	12399.10	-443.31	0.04
Free/Reduced Lunch (%)	0.50	0.48	0.02	0.15
Black (%)	0.14	0.18	-0.05	0.00
Equitable Funding	0.55	0.49	0.06	0.00
Clear Charter Renewal Policies	0.65	0.59	0.06	0.00
Performance-Based Contracts	0.50	0.58	-0.07	0.00
Exempt from State/District Regs	0.63	0.65	-0.01	0.18
Urban	0.20	0.16	0.04	0.04
Transparent Charter Startup Policies	0.53	0.53	0.00	0.72
Magnet Schools (%)	0.38	0.28	0.10	0.27
Town	0.21	0.20	0.02	0.40
Suburb	0.36	0.36	0.00	0.97
Rural	0.23	0.29	-0.06	0.01
No Caps on CS Growth	0.81	0.82	-0.01	0.27
University in City	0.01	0.00	0.02	0.04
City Population (standardized)	0.00	0.00	0.01	0.08

Notes: For each moderator, the table reports the mean value among districts with positive CATT estimates (Mean^+), the mean among districts with negative CATT estimates (Mean^-), their difference, and the associated p-value of the difference. Rows sorted by variable importance factor.

Table 10: CFFE Best Linear Predictor: Graduation Rate (AFGR)

	Estimate	Std. Error	p-value
(Intercept)	-0.37	0.10	0.00
Black (%)	-0.03	0.06	0.65
Hispanic (%)	0.07	0.06	0.25
White (%)	0.04	0.05	0.51
Special Ed (%)	0.01	0.12	0.93
Log of Enrollment	0.04	0.01	0.00
Student-Teacher Ratio	0.00	0.00	0.94

Notes: Best linear predictor (BLP) regression of CFFE CATT estimates on district moderators, applied to graduation rates. Each row reports a separate OLS coefficient from regressing mean post-treatment CATT on the listed covariate. Estimates above zero indicate that districts with higher values of the covariate experience larger positive effects of charter entry.

Table 11: CFFE Best Linear Predictor: Math Scores

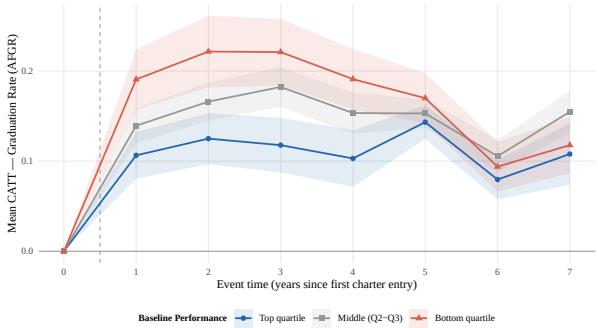
	Estimate	Std. Error	p-value
(Intercept)	0.01	0.22	0.98
Special Ed (%)	-0.11	0.33	0.74
Per Pupil Revenue	0.00	0.00	0.32
Student-Teacher Ratio	0.02	0.01	0.00
Teacher Salary	0.00	0.00	0.42
TPS-Charter Spending (% diff)	0.13	0.04	0.00
Charter Spending (per-pupil)	0.00	0.00	0.00

Notes: Best linear predictor (BLP) regression of CFFE CATT estimates on district moderators, applied to math test scores. Each row reports a separate OLS coefficient from regressing mean post-treatment CATT on the listed covariate.

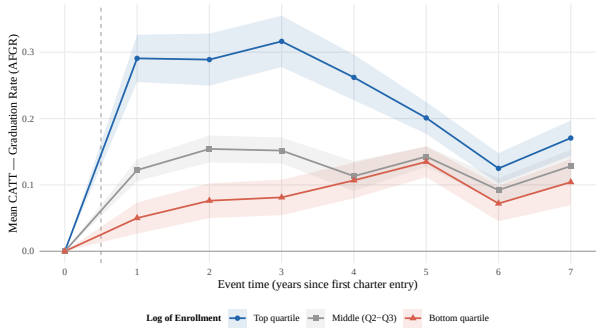
Table 12: CFFE Best Linear Predictor: ELA Scores

	Estimate	Std. Error	p-value
(Intercept)	0.45	0.18	0.01
Log of Enrollment	-0.02	0.01	0.05
Per Pupil Revenue	0.00	0.00	0.65
Student-Teacher Ratio	0.01	0.00	0.25
Teacher Salary	0.00	0.00	0.01
TPS-Charter Spending (% diff)	0.10	0.04	0.01
Charter Spending (per-pupil)	0.00	0.00	0.00

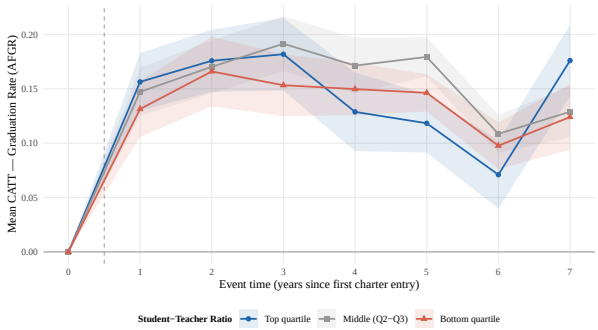
Notes: Best linear predictor (BLP) regression of CFFE CATT estimates on district moderators, applied to ELA test scores. Each row reports a separate OLS coefficient from regressing mean post-treatment CATT on the listed covariate.



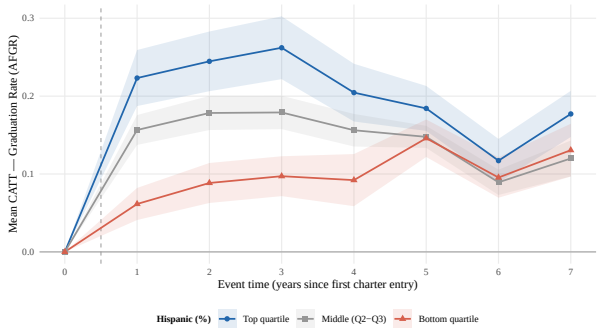
(a) Baseline Graduation Rate



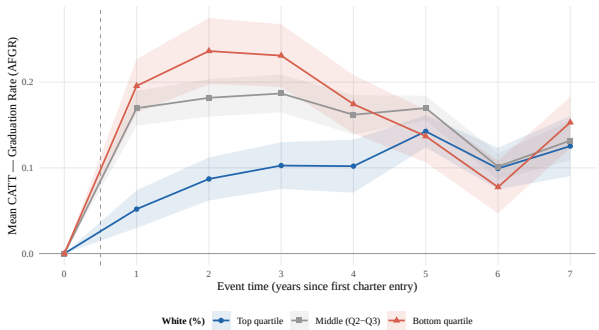
(b) Log Enrollment



(c) Student-Teacher Ratio



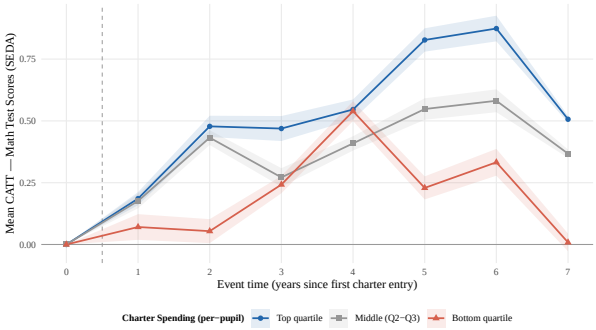
(d) Percent Hispanic



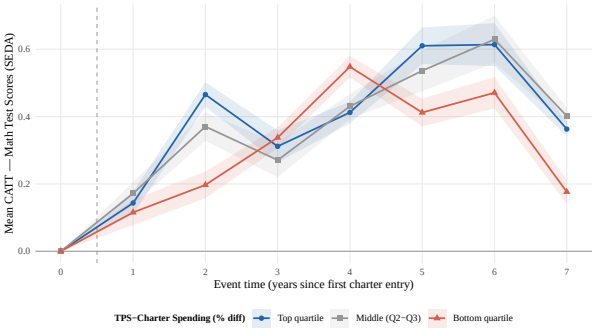
(e) Percent White

Figure 25: CFFE Quartile Event Studies: Graduation Rate (AFGR)

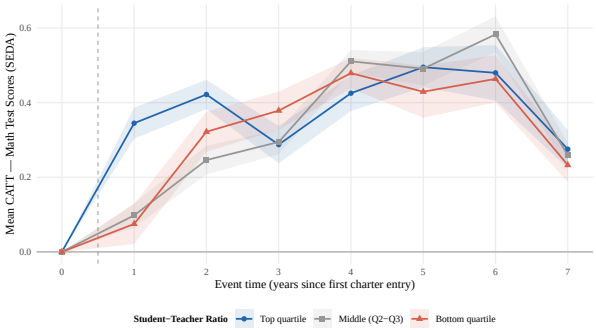
Notes: Each panel shows cohort-weighted ATT estimates by event-time from the CFFE, separately for districts in the top quartile (blue), middle two quartiles (grey), and bottom quartile (red) of the named moderator. Panels correspond to the five highest-ranked covariates by VIF (averaged across $h = 1, \ldots, 7$) for graduation rates.



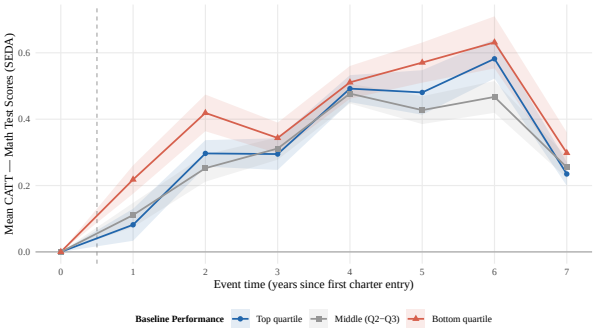
(a) Charter Spending Per-Pupil



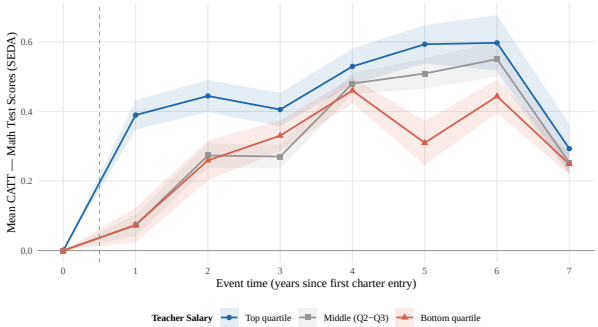
(b) TPS vs. Charter Spending Gap



(c) Student-Teacher Ratio



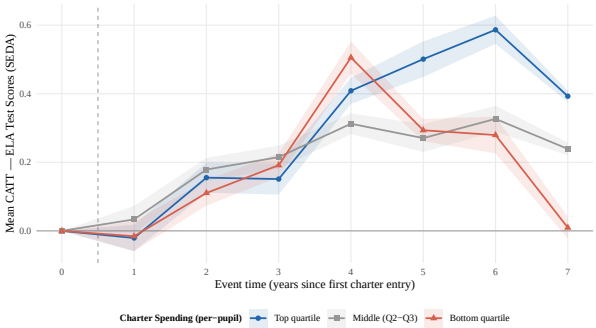
(d) Baseline Math Score



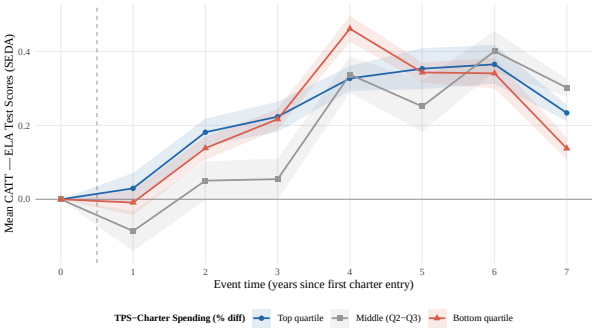
(e) Teacher Salary

Figure 26: CFFE Quartile Event Studies: Math Scores

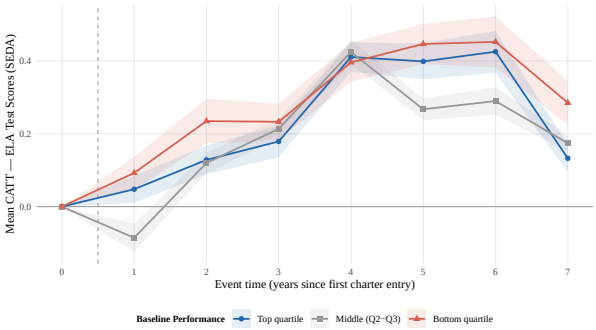
Notes: Each panel shows cohort-weighted ATT estimates by event-time from the CFFE, separately for districts in the top quartile (blue), middle two quartiles (grey), and bottom quartile (red) of the named moderator. Panels correspond to the five highest-ranked covariates by VIF (averaged across $h = 1, \ldots, 7$) for math test scores.



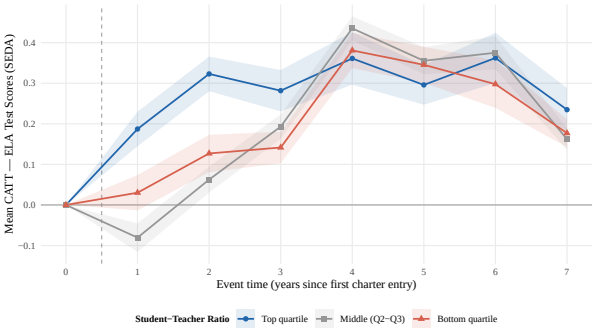
(a) Charter Spending Per-Pupil



(b) TPS vs. Charter Spending Gap



(c) Baseline ELA Score



(d) Student-Teacher Ratio



(e) Teacher Salary

Figure 27: CFFE Quartile Event Studies: ELA Scores

Notes: Each panel shows cohort-weighted ATT estimates by event-time from the CFFE, separately for districts in the top quartile (blue), middle two quartiles (grey), and bottom quartile (red) of the named moderator. Panels correspond to the five highest-ranked covariates by VIF (averaged across $h = 1, \ldots, 7$) for ELA test scores.